



SHZVG Type Oil-Immersed Vacuum On-Load Tap-Changer

Technical Data

HM 0.154.7201-8/27/2025

Important Note!

Read carefully before use.

Keep for future reference.



Shanghai Huaming Power Equipment Co., Ltd.

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Manufacturer

Shanghai Huaming Power Equipment Co., Ltd.

Address: No. 977 Tongpu Road, Putuo District, Shanghai, China

Postal Code: 200333

Tel: +86 21 5270 8966 (Main line)

Fax: +86 21 5270 3385

Website: www.huaming.com

Email: Order@huaming.com, export@huaming.com (after-sale service)

For further information or a copy of this manual, please contact us.



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1 Overview

The SHZVG Type Oil-Immersed Vacuum On-Load Tap-Changer (hereinafter referred to as "tap-changer") is a combined on-load tap-changer used for oil-immersed regulating transformers. It consists of a diverter switch and a tap selector. Both of them are immersed in the transformer tank. The diverter switch is equipped with an independent oil compartment, and the tap selector shares the transformer tank with transformer windings. The tap-changer is installed on the transformer box cover with head flange in bell-type installation method, and driven by a motor drive unit (MDU). It is connected to the MDU through a head gearbox, driving shaft, and bevel gearbox, enabling local or remote control operation.

There are two connection types for the tap-changer, including three-phase star-connected neutral point and single-phase arbitrary connections. Three single-phase tap-changers can achieve any connection and voltage regulation position for three-phase transformers.

The basic connection methods are shown in Figure 1-1:

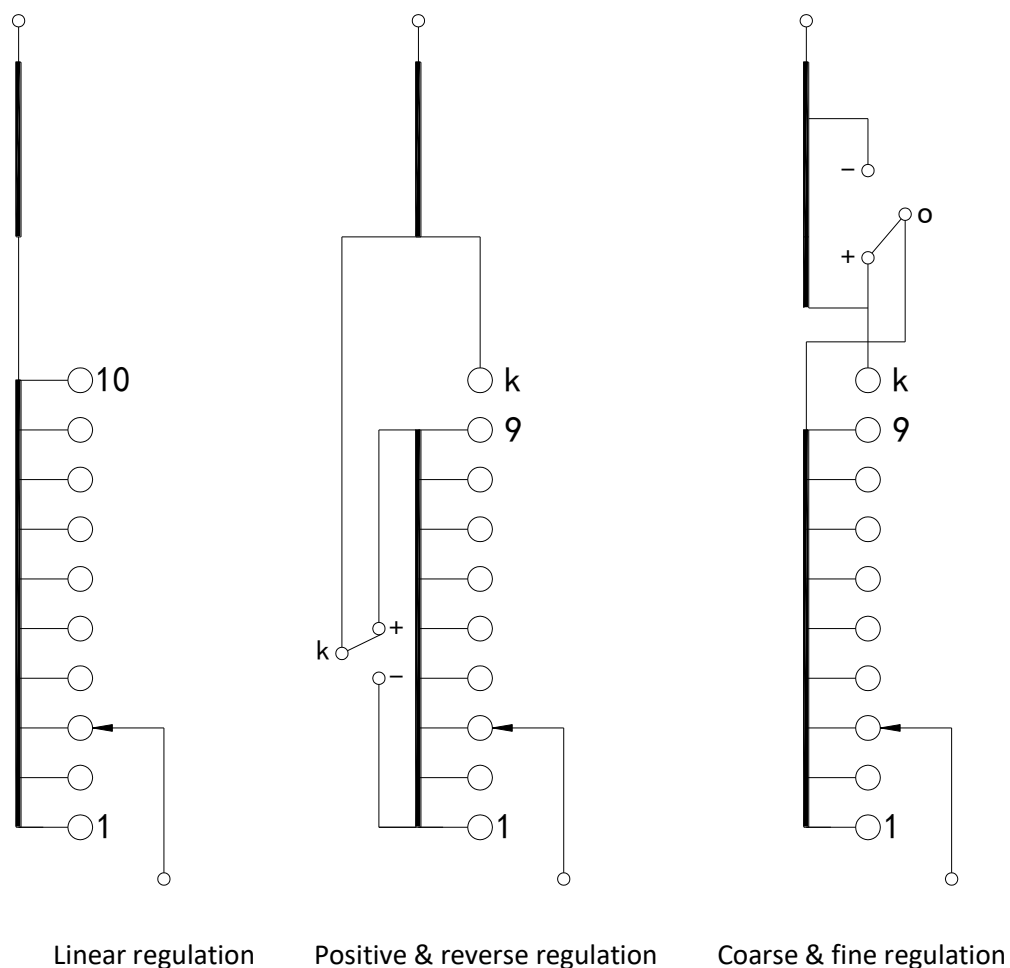


Figure 1-1 Basic connection methods of the transformer regulating winding

2 Technical Specifications

The tap-changer complies with the standards of GB/T 10230.1-2019 and IEC 60214-2014. For its technical parameters, see Table 2-1.

Table 2-1 Technical parameters of the tap-changer

No.	Model		SHZVG III 1300Y SHZVG I 1300	SHZVG III 1500Y SHZVG I 1500	SHZVG I 2000	SHZVG I 2400	SHZVG I 3000	SHZVG I 3600
1	Maximum rated through current (A)		1,300	1,500	2,000	2,400	3,000	3,600
2	Rated frequency (Hz)		50 or 60					
3	Connection type		Three-phase Y connection, single-phase arbitrary connection					
4	Maximum rated step voltage (V)		6,000					
5	Rated step capacity (kVA)		6,000		9,600		14,400	
6	Short-circuit withstand capability (kA)	Thermal stability (3 s)	15	20	24	30	32	40
		Dynamic stability (peak)	37.5	50	60	75	80	100
7	Maximum tap positions		18 without change-over selector, 35 with change-over selector					
8	Ground insulation level (kV)	Highest equipment voltage	72.5	126	170	252	300	363
		Rated withstand voltage (50 Hz, 1 min)	140	230	325	460	480	510
		Rated lightning impulse withstand voltage (1.2/50 μ s)	350	550	750	1,050	1,100	1,175
9	Tap selector		It is classified into five specifications based on insulation level: B, C, D, DE and E.					
10	Mechanical lifespan		1.5 million					
11	Maintenance-free operations		300,000					
12	Switch oil compartment	Operating pressure	0.03 MPa					
		Sealing performance	0.1 MPa/24 h leak-free					



No.	Model		SHZVG III 1300Y SHZVG I 1300	SHZVG III 1500Y SHZVG I 1500	SHZVG I 2000	SHZVG I 2400	SHZVG I 3000	SHZVG I 3600
		Overpressure protection	Rupture disk bursts at $0.3 \pm 20\%$ MPa					
		Protective relay	Set flow rate: 2.5 m/s $\pm$ 10%					
13	MDU matched		SHM series or CMA7					

3 Model Description

3.1 Model designation Method

The tap-changer has various specifications due to different combinations of phase number, maximum rated through current, highest equipment voltage, insulation grade and connection method of the selector.

Therefore, in model of the tap-changer, the above-mentioned performance parameters should be clearly indicated. Detailed explanations of the code symbols are shown in Figure 3-1.

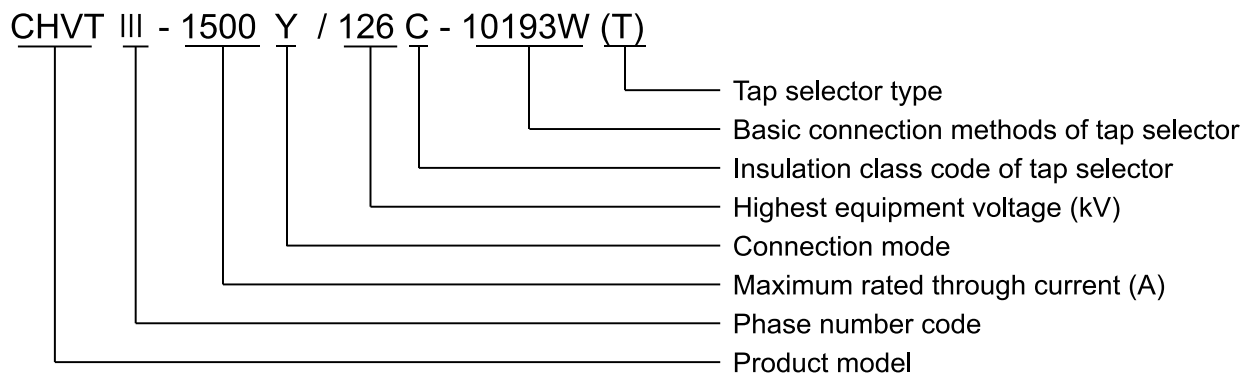


Figure 3-1 Model description of the tap-changer

3.2 Basic connection methods of tap selector

Various tap selector configurations are available to accommodate different voltage regulation ranges and winding connection methods. Specifications of the tap selector include: the number of distributed contacts, the number of operating positions, the number of intermediate positions, and the form of the change-over selector. The parameter expression in the tap selector model is shown in Figure 3-2.

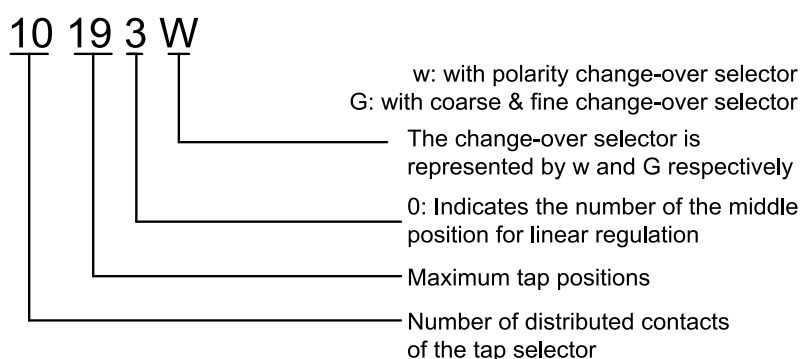


Figure 3-2 Basic connection method explanation of the tap selector

3.3 Basic wiring diagram of tap selector

Different tap winding configurations of the transformer regulating coil correspond to different specifications of tap selector basic wiring diagrams. Figure 3-3 shows the common wiring diagrams. Special designs are available for custom requirements.

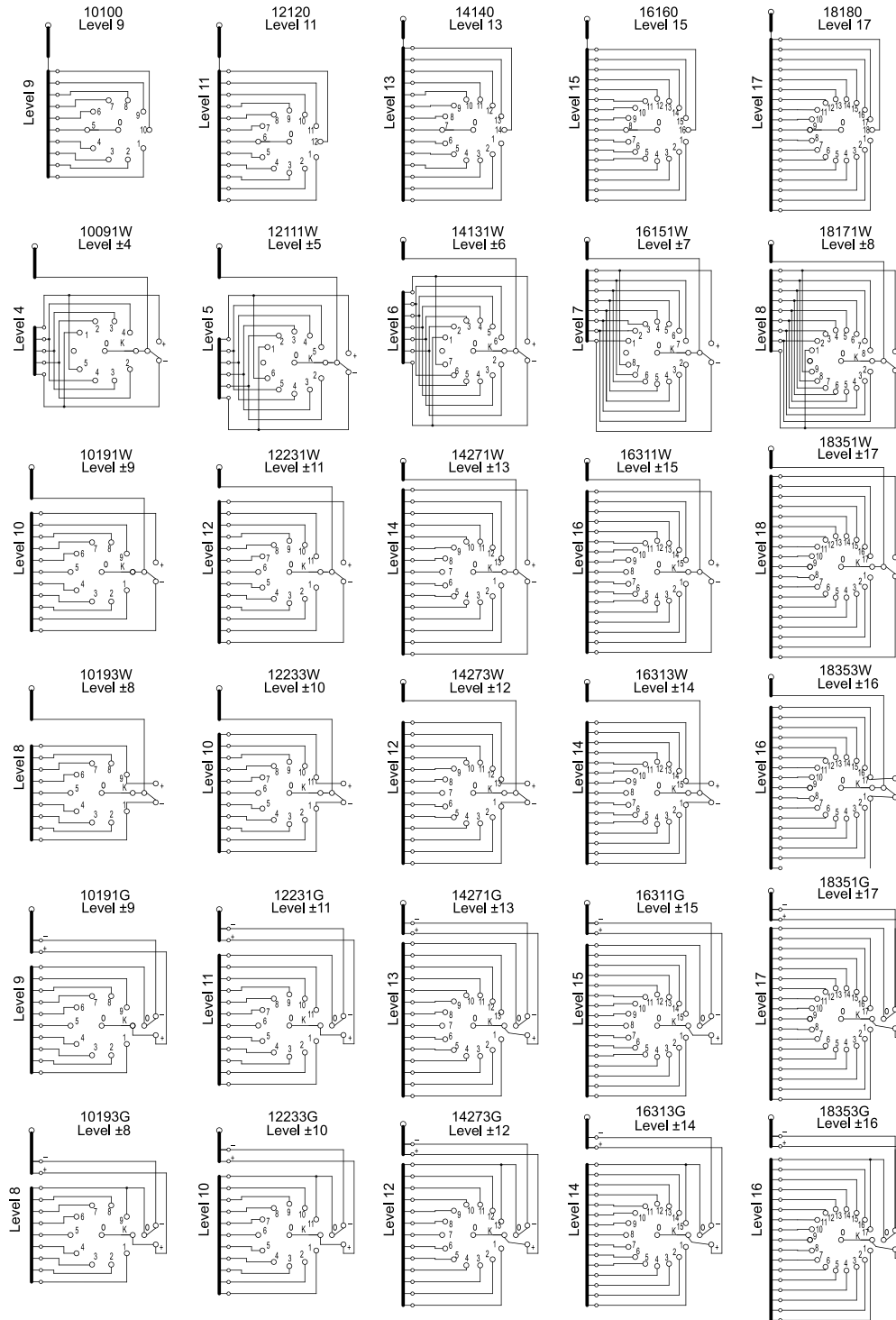


Figure 3-3 Basic wiring schematic diagram of the tap selector

4 Technical Performance Parameters

4.1 Through current

Rated current through I_u : The current flowing from the tap-changer to the external circuit. It can be transferred from one tap to another by the tap-changer under the relevant step voltage. The tap-changer continuously carries the current when requirements in this section are met.

Maximum rated through current I_{um} : The maximum rated through current designed for the tap-changer. It serves as reference current in relevant tests.

4.2 Step voltage

Rated step voltage U_i : The maximum allowable voltage between two adjacent tap terminals connected to the transformer for each rated through current.

Maximum rated step voltage U_{im} : The maximum value of rated step voltage designed for the tap-changer. The maximum rated step voltage of the SHZVG type tap-changer is 6,000 V.

4.3 Breaking capacity

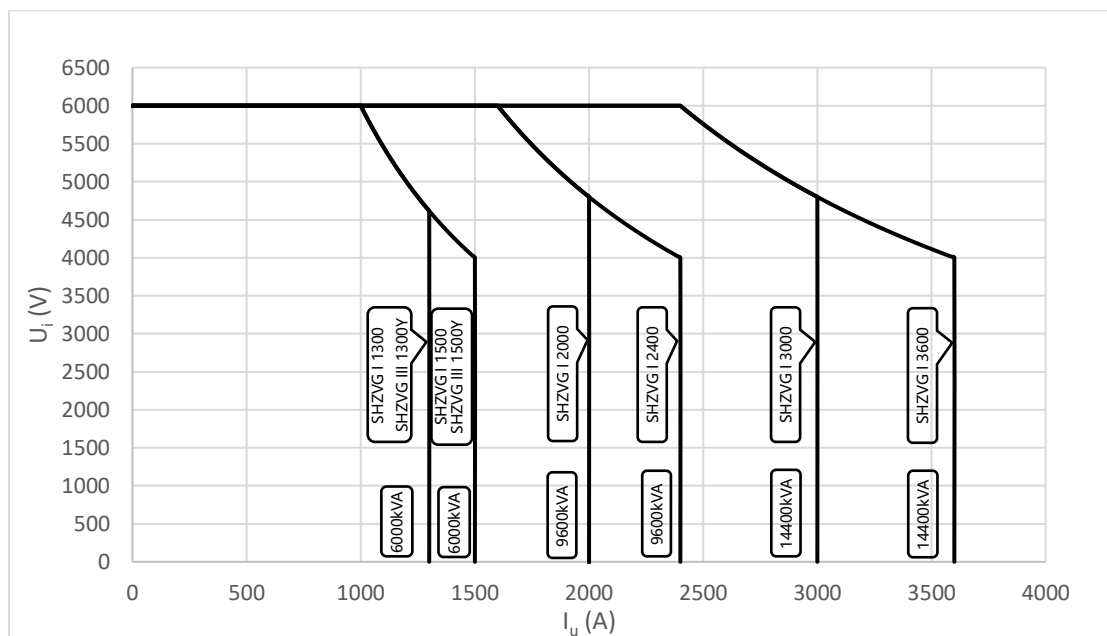


Figure 4-1 Rated step capacity of the tap-changer



According to the standards of IEC 60214-1:2014 and GB/T 10230.1-2019, the tap-changer should break 40 times at twice the maximum rated through current and relevant rated step voltage. Breaking capacity:

$$P_{st,max}=2P_{stN} \approx 2I_{um} \times U_{stN}$$

P_{stN} : rated step capacity

I_{um} : maximum rated through current (A)

U_{stN} : relevant rated step voltage

4.4 Electrical life of vacuum interrupter

Electrical life of the vacuum interrupter is 500,000 operations.

4.5 Short-circuit withstand capability

According to the standards of IEC 60214-1:2014 and GB/T 10230.1-2019, all types of continuous-carrying contacts of the tap changer should be able to withstand a short-circuit current impact for 2 s ($\pm 10\%$) without any phenomena such as welding, deformation or mechanical damage. The initial peak current for each instance should be 2.5 times ($\pm 5\%$) of the rated short-circuit current's square root value. For detailed parameters regarding the tap-change short-circuit withstand capability, refer to Table 2-1.

4.6 Operating conditions of the tap-changer

1. When the tap-changer is used in an oil environment, its temperature range is within -25°C to $+100^{\circ}\text{C}$.
2. The storage environment temperature of the tap-changer should within 40°C to -25°C . The humidity condition should not exceed 85%.

According to the standard design, the operating environment temperature of the tap-changer should not exceed 40°C and not be lower than -25°C .

If temperature in the operating environment is higher than 40°C or lower than -25°C , specify when placing the order. Materials and accessories of the tap-changer will be specially treated according to the order requirements to meet requirements of the operation environment.

3. After the tap-changer is installed on the transformer, the verticality error between it and the ground is 2%.
4. Application site of the tap-changer should be free from severe dust and other explosive or corrosive gases.

NOTICE If the tap-changer or MDU needs to operate under more harsh environmental conditions, contact technical department of Huaming Company for confirmation.

4.7 Ground insulation level of the tap-changer

The ground insulation level of the tap-changer, that is, the insulation between tap-changer live parts and grounded components, is determined by the test values of 1-minute AC power-frequency withstand voltage and the impulse voltage. Its value depends on the tap winding position of the connected transformer, voltage regulating range, winding connection method, structural arrangement, and rated voltage of the transformer winding. It is defined by the insulation level of the transformer regulating winding to the ground.

Ground insulation level of the tap-changer should be selected from standard values specified in IEC 60214-1:2014 and GB/T 10230.1-2019 (see Table 4-2) based on the highest equipment operating voltage of the tap-changer. This allows minimum insulation values to cover the entire application range.

Table 4-1 Abbreviation for insulation level

LI	Full-wave lightning impulse (kV, 1.2/50 μ s)
LIC	Chopped-wave lightning impulse (kV, 1.2/50/3 μ s)
AC	Applied voltage (kV, 50 Hz, 1 min)
SI	Operational impulse voltage (kV, 250/2,500 μ s)

Table 4-2 Ground insulation level (kV)

Highest equipment voltage U_m	LI	LIC	AC	SI
72.5	350	385	140	-
126	550	605	230	460
170	750	825	325	620
252	1,050	1,155	460	850
300	1,100	1,210	480	890
363	1,175	1,300	510	950

4.8 Internal insulation level of the tap-changer

Internal insulation levels of the tap-changer are classified into four grades: B, C, D, DE, and E. Tap-changers with different insulation levels have different withstand voltages for: step-to-step, max-min tap positions, phase-to-phase. Level B is the lowest withstand capability, while the Level DE is the highest one. Figure 4-2 Figure 4-2 Basic wiring diagrams and insulation symbols shows potential distribution of each part after the tap-changer is connected to the transformer regulating winding. Table 4-3 presents the power frequency impulse withstand voltage of the tap-changer at each corresponding insulation distance.

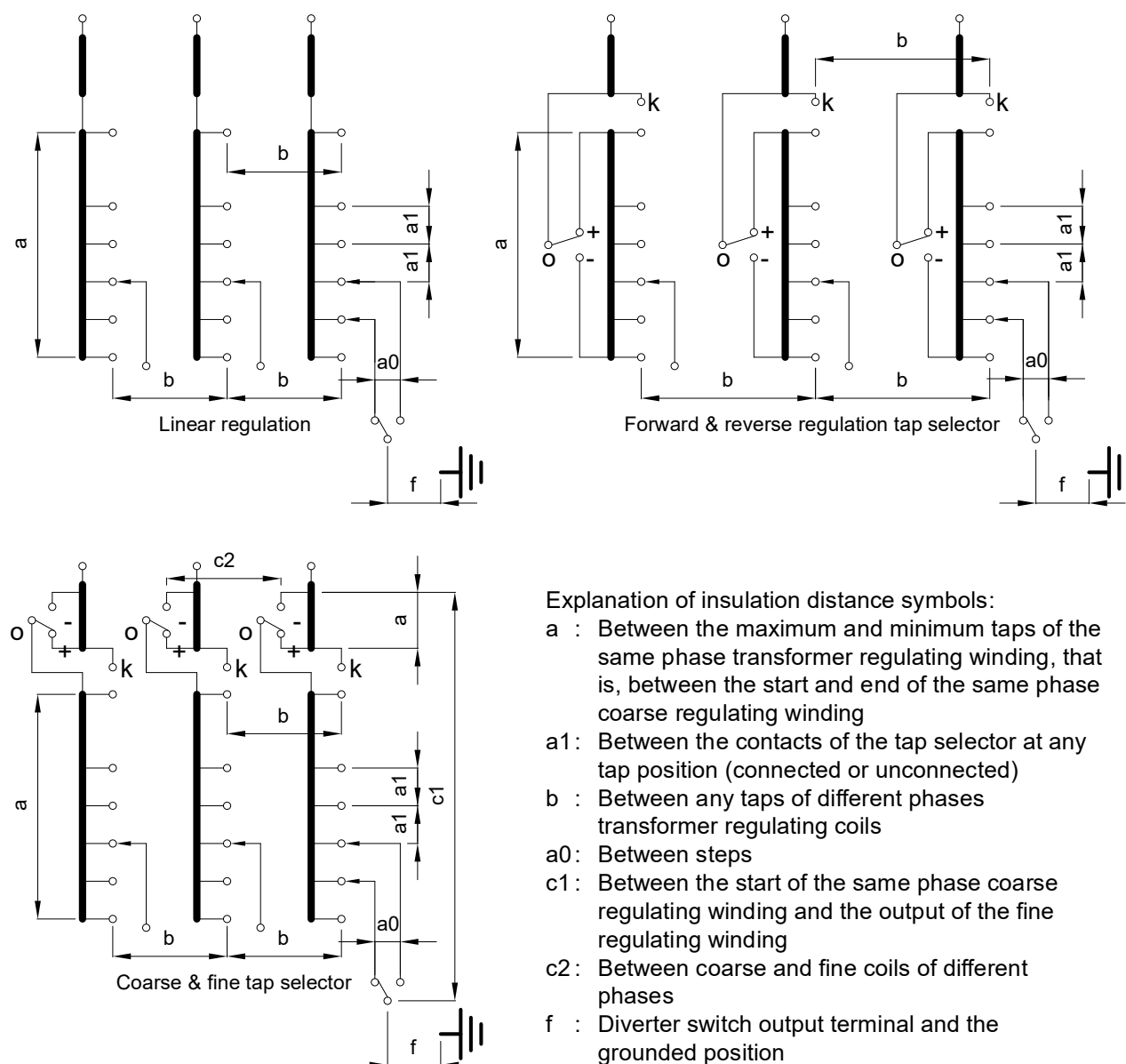


Figure 4-2 Basic wiring diagrams and insulation symbols

Table 4-3 Internal insulation level of the tap changer

(Unit: kV)

Insulation Distance Symbol		Level B Tap Selector	Level C Tap Selector	Level D Tap Selector	Level DE Tap Selector	Level E Tap Selector
a	LI	265	365	460	550	670
	LIC	295	385	540	605	735
	AC	50	82	105	120	180
	SI	175	230	320	360	435
b	LI	265	365	460	550	670
	LIC	295	385	540	605	735
	AC	50	82	146	160	180
	SI	175	230	320	360	435
a0	LI	90				
	AC	20				
a1	LI	150				
	LIC	165				
	AC	30				
	SI	100				
c1	LI	500	550	590	660	820
	LIC	535	600	650	725	900
	AC	145	180	225	230	250
	SI	315	355	385	430	535
c2	LI	500	550	590	660	820
	LIC	545	605	650	725	900
	AC	145	195	225	250	280
	SI	325	360	385	430	535

NOTICE The internal insulation distance of a0 is ZnO protection, with the insulation level: 1.2/50 μ s; 50 to 70 kV, 100% response under 70 kV.



4.9 Installation method of the tap-changer

The SHZVG type tap-changer is installed on transformer tank cover using a head flange. Therefore, there should be a mounting flange on the transformer. For its detailed dimensions, see The SHZVG type tap-changer is used for bell-type transformers. The support flange of the tap-changer only serves as a temporary support when the transformer is wired. After bell tank cover of the transformer is installed, fix the tap-changer to the transformer mounting flange. The left-right arrangement structure of the head flange for SHZVG type tap-changer is shown in (and for details).

5 Special Design of the Tap Changer

5.1 Potential connection of transformer regulating winding

In a high-voltage on-load tap-changing transformer with large regulation ranges, when the tap-changer change-over selector is operating, the transformer regulating winding will be momentarily electrically disconnected from the main winding and be in a "floating" state. At this time, the transformer regulating winding will obtain a new potential determined by the coupling capacitance to ground (C_e) and the coupling capacitance to adjacent windings (C_w) (For details, see Figure 5-2). Usually, this potential is different from the potential of the transformer regulating winding before the change-over selector operates. The potential difference between the two is called the offset voltage. This offset voltage appears as a recovery voltage across contact break of the change-over selector. When the offset voltage reaches a certain critical value, it may cause discharge on the change-over selector, accompanied by lots of gas. This is very serious. Therefore, if the offset voltage exceeds a certain limit, potential connection of the transformer regulating winding must be considered to prevent discharge during operation of the change-over selector.

For SHZVG type tap-changer, the allowable recovery voltage value on contacts of the change-over selector is 35 kV. When it is above this limit, connect a tie-in resistor with a fixed value permanently to the transformer regulating winding (For details, see Figure 5-1). For installation position and dimensions of the tie-in resistor for the SHZVG type tap-changer, see

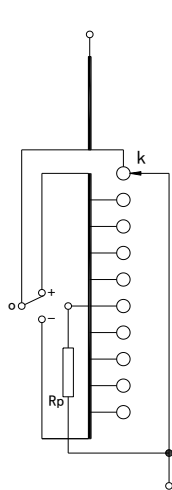


Figure 5-1 Connection method of the tie-in resistor

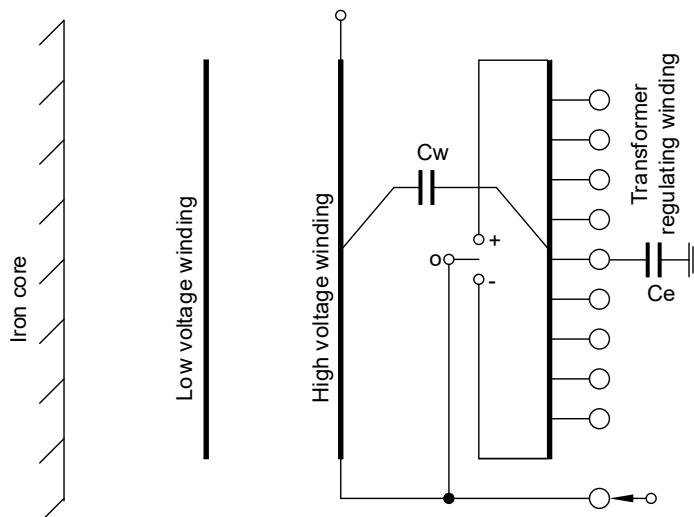


Figure 5-2 Arrangement of positive & reverse regulation windings of double-winding transformer

To allow Huaming to clearly determine load of the change-over selector and the size and number of potential resistance elements, etc., when selecting type of tap-changer, specify following parameters of the transformer:

1. All performance parameters of the transformer: rated capacity, rated voltage, transformer regulating range, connection mode of windings, and insulation level
2. Relative arrangement mode of windings: relative position of the transformer regulating winding and adjacent windings or winding components
3. Voltages at both ends adjacent to the transformer regulating winding and the electrical position of this coil in the transformer windings

4. Capacitance between the transformer regulating winding and adjacent winding components
5. Capacitance between the transformer regulating winding and the ground or adjacent grounded windings (if any)
6. Impulse load across half of the transformer regulation winding
7. Working voltage and power frequency test voltage of the transformer regulating winding across half of it.

5.2 Two-phase or single-phase tap-changer

A tap-changer can be designed as one MDU (or possibly three MDUs) driving three single-phase tap-changers or a combination of one two-phase tap-changers and one single-phase tap-changer. These configurations are employed for voltage regulation in delta-connected transformers or at locations other than the neutral point.

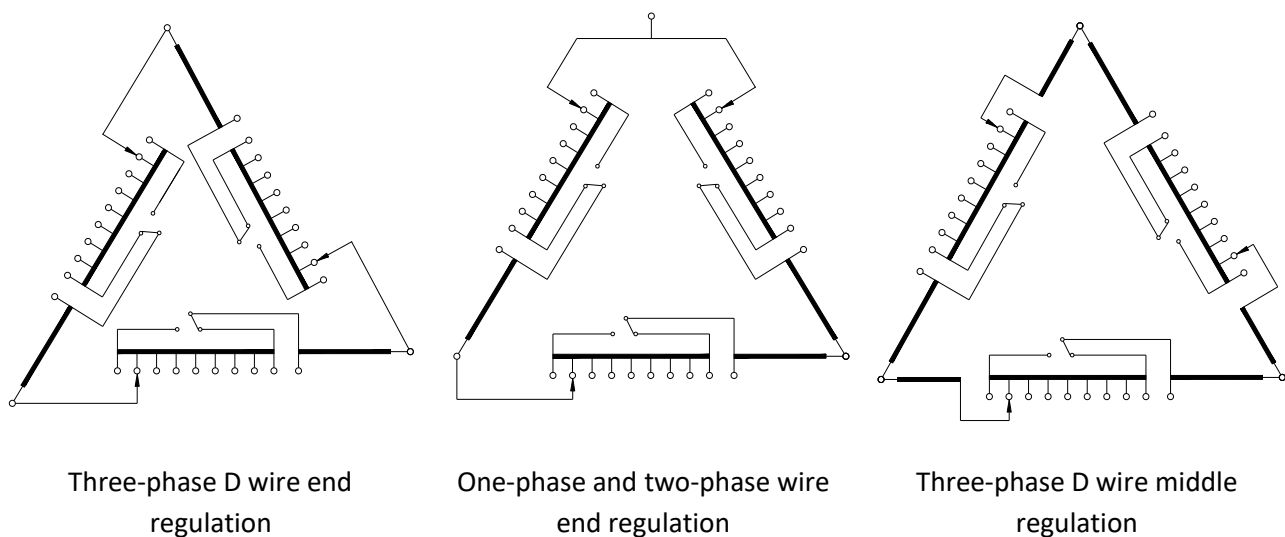


Figure 5-3 Basic connection methods of the transformer winding for D connection

5.3 Structural selection

The types of tap selectors are represented by T and L respectively.

T represents drum type tap selectors. They can be used when all the following three conditions are met:

1. The number of distributed contacts is not above 14.
2. The three-phase rated current is not above 1,300 A.
3. The insulation level is not above D.

L represents cage type tap selectors. They can be used when any of the following conditions are met:

1. Number of distributed contacts is above 14.
2. The three-phase rated current is above 1,300 A.
3. The insulation level is above D.



6 Motor Drive Unit (MDU)

The tap-changer is driven by a motor drive unit (MDU). You should select the appropriate MDU based on the tap-changer model. Its technical parameters are listed in Table 6-1.

Table 6-1 Technical parameters of the MDU

Motor drive unit (MDU)		SHM-D	CMA7	
Item				
Motor	Rated power (W)	750	750	1,100
	Rated voltage (V)	200 to 240 (AC)	Three-phase 380 (AC)	
	Rated current (A)	4	2.0	2.8
	Frequency (Hz)	50 or 60	50 or 60	
	Rotational speed (r/min)	330 direct drive	1,400	
Output shaft rotational torque (Nm)		35 (static torque)	18	26
Rotations number of tap changeover driving shaft per step (revolutions)		33	33	
Rotations number of tap changeover driving handle per step (revolutions)		33	33	
Tap changeover electric operation time per step (s)		About 6	About 5	
Maximum operation position number		107	107	
Voltage of control circuit and heater (V)		220 (AC)	220 (AC)	
Power consumption of heater		100	50	
Insulation test of auxiliary circuit (kV/50 Hz, 1 min)		2	2	
Weight (kg)		80	90	
Protection level		IP66	IP66	
Mechanical life of MDU (ten-thousand cycles)		≥200	≥80	

7 Operation Controllers of the Tap Changer

7.1 SHM-KX remote digital controller

The SHM-KX remote digital controller (hereinafter referred to as SHM-KX) can be used in conjunction with electric operating mechanisms such as SHM-D, A7/A9, and ZDK to control the tap-changer manually and automatically. SHM-KX is an intelligent product with independent intellectual property rights launched by our company.

SHM-KX adopts a 5.1-inch 1024*720 high-resolution screen. It is equipped with a domestically produced main control chip, has a storage capacity of 8G and a memory of 2G. It can be linked with the electric mechanism in multiple ways and also supports multiple modes of information interaction with the background.

Technical parameters of SHM-KX remote digital controller (partial):	
Working Voltage: 85 to 264 V (AC), 50/60 HZ; 100 to 250 V (DC)	
Power Consumption: About 7 W	
Maximum display position number: 107	
Working environment: indoor 0 °C to 50 °C; Relative humidity < RH95% without condensation	

7.2 HMC-3C type on-load tap-changer position display

The HMC-3C type on-load tap-changer position display can work together with the MDU remote position indication circuit as the remote indicator of the on-load tap-changer gear position. At the same time, it also has the "1-N", "Stop", and "N-1" functions of the on-load tap-changer, outputs BCD code position signals. In addition, it is equipped with a remote control indicator light.

Technical parameters of HMC-3C display:	
Working Voltage: 220 V (AC)	Working Frequency: 50 Hz
Maximum display position number: 107	Working Temperature: -10 °C to +40 °C
NOTICE For other power supply technical requirements of the above-mentioned controllers, specify clearly when placing an order.	



8 Tap-Changer Accessories

8.1 Bevel gearbox

The bevel gearbox is used to connect the horizontal shaft of the tap-changer and the vertical shaft of the MDU, transmitting drive torque from the MDU to the tap-changer. For outline dimensions of the bevel gearbox, see relay

The protective relay is a protective device used for oil-immersed tap-changers. When an internal fault of the tap-changer causes oil decomposition to generate gas or oil flow, the relay contacts activate, to connect the specified control circuit, and issue a signal in time or automatically disconnect the transformer.

The protective relay is installed on the connecting pipeline between the oil pipe at head of the tap-changer and the conservator. During installation, make arrow on the relay face the conservator side. Huaming Company provides protective relays of various models for you to choose for tap-changers. For detailed installation dimensions of the protective relay, see (If you have special requirements, contact technical department of our company)

8.2 Pressure relief device (PRD)

The pressure relief device (PRD) and rupture disk are safety protection devices for oil-immersed on-load tap-changer. When there is an accident inside the tap-changer, oil in the oil compartment will be gasified, generating lots of gas. This makes pressure in the oil compartment rise sharply. If this pressure is not released in time, oil tap-changer of the tap-changer will deform or even burst. Therefore, installing a PRD can avoid the expansion of the accident.

The PRD is initially installed as a safety protective device on the transformer tank. It is an automatic sealing relief valve. When overpressure occurs, the valve cover opens, and the overpressure is immediately released. After that, it closes again. It can be reused and keep the liquid loss during each action to a minimum level.

The rupture disk creates a weak point on top cover of the oil compartment. Once pressure in the oil compartment exceeds the set value, it bursts the rupture disk to release overpressure from the oil compartment. By this way, the oil compartment is protected against damage.

The PRD is a low-energy fault protection device, while the rupture disk is a high-energy protection device. Since tap-changer accidents are mostly high-energy, it is recommended that the tap-changer adopts a rupture membrane as a safety protection device or a double safety protection device with the rupture disk as the main and the PRD as the auxiliary. The PRD is an optional accessory for the tap-changer. Specify your exact configuration when placing an order as per your needs.

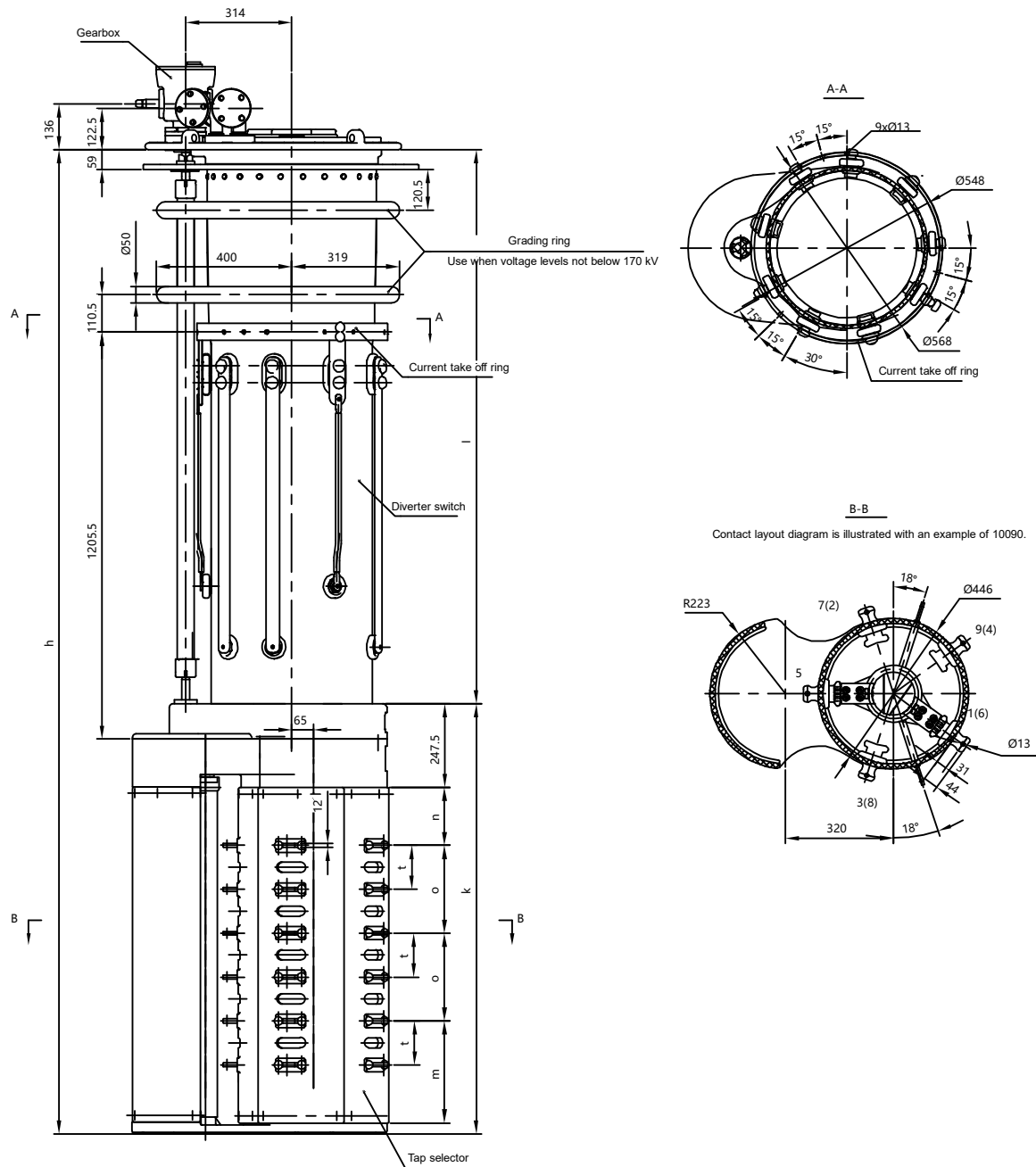
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9.1 Outline dimension drawing of SHZVG III 1300A linear regulation on-load tap-changer



Model		SHZVG III 1300/72.5~252/B				SHZVG III 1300/72.5 ~ 252/C				SHZVG III 1300/72.5 ~ 252/D			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2482.5	2612.5	2742.5	2842.5	2657.5	2787.5	2917.5	3017.5	3110.5	3240.5	3370.5	3470.5
	i	1381	1511	1641	1741	1381	1511	1641	1741	1381	1511	1641	1741
	k	1101.5				1276.5				1729.5			
	n	147				172				237			
	o	210				260				390			
	t	105				130				195			
	m	252				302				432			
Oil volume (dm³)		150	170	210	230	150	170	210	230	150	170	210	230
Oil discharge (dm³)		225	245	265	285	250	270	290	310	315	330	345	360
Weight (kg)		340				360				380			

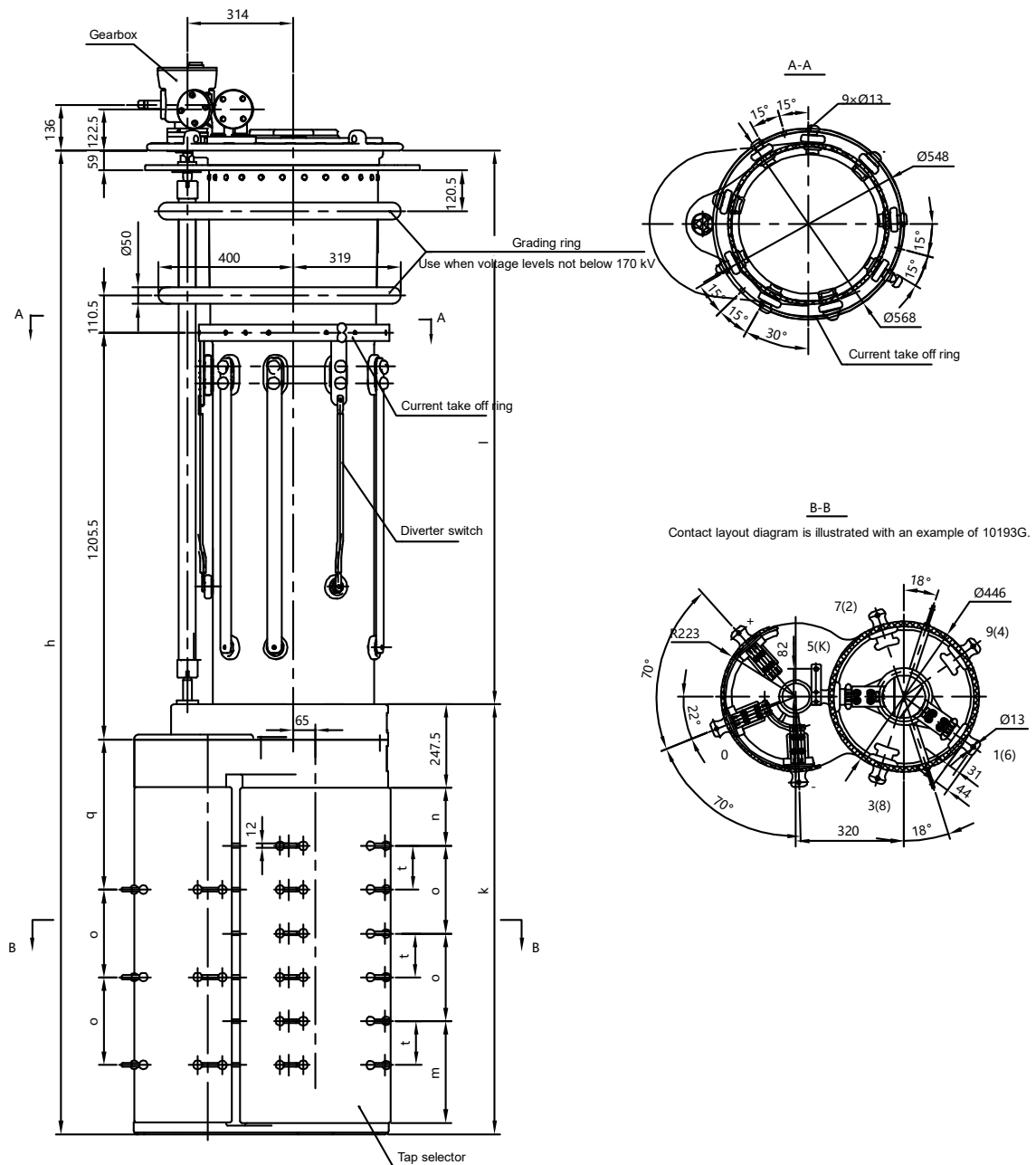
Unit: mm

9 Drawings Attached



Model		SHZVG III 1300/72.5 ~252/B				SHZVG III 1300/72.5 ~252/C				SHZVG III 1300/72.5 ~252/D			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2482.5	2612.5	2742.5	2842.5	2657.5	2787.5	2917.5	3017.5	3110.5	3240.5	3370.5	3470.5
	i	1381	1511	1641	1741	1381	1511	1641	1741	1381	1511	1641	1741
	k	1101.5				1276.5				1729.5			
	n	147				172				237			
	o	210				260				390			
	t	105				130				195			
	m	252				302				432			
	q	386				443				566			
Oil volume (dm³)		150	170	210	230	150	170	210	230	150	170	210	230
Oil discharge (dm³)		225	245	265	285	250	270	290	310	315	330	345	360
Weight (kg)		340				360				380			

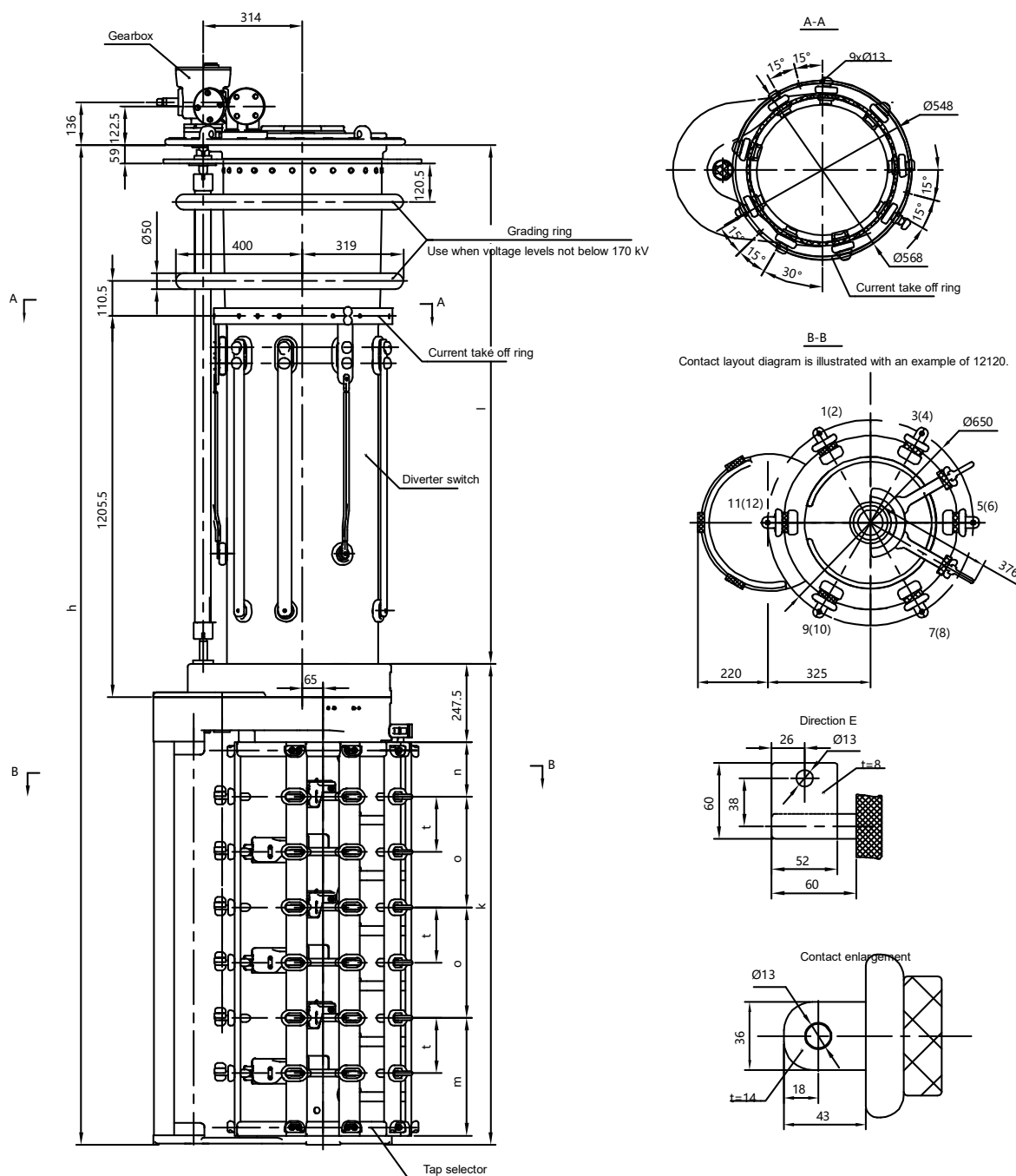
9.3 Outline dimension drawing of SHZVG III 1300A coarse & fine on-load tap-changer



Unit: mm

Model		SHZVG III1300/72.5 ~252/B				SHZVG III1300/72.5 ~252/C				SHZVG III1300/72.5 ~252/D			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2482.5	2612.5	2742.5	2842.5	2657.5	2787.5	2917.5	3017.5	3110.5	3240.5	3370.5	3470.5
	i	1381	1511	1641	1741	1381	1511	1641	1741	1381	1511	1641	1741
	k	1101.5				1276.5				1729.5			
	n	147				172				237			
	o	210				260				390			
	t	105				130				195			
	m	252				302				432			
	q	387				428				478			
Oil volume (dm³)		150	170	210	230	150	170	210	230	150	170	210	230
Oil discharge (dm³)		225	245	265	285	250	270	290	310	315	330	345	360
Weight (kg)		340				360				380			

9.4 Outline dimension drawing of SHZVG III 1500A linear regulation on-load tap-changer



Unit: mm

Model	SHZVG III1500/72.5 ~252/C				SHZVG III1500/72.5 ~252/D (DE)			
Highest equipment voltage (kV)	72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2904.5	3034.5	3164.5	3264.5	3209.5	3339.5	3469.5
	i	1381	1511	1641	1741	1381	1511	1641
	k	1523.5				1828.5		
	n	172				237		
	o	350				420		
	t	175				210		
	m	372				472		
Oil volume (dm ³)		170	210	230	250	170	210	230
Oil discharge (dm ³)		250	270	290	310	330	345	360
Weight (kg)		370				390		

9 Drawings Attached



Model		SHZVG III1500/72.5 ~252/C				SHZVG III1500/72.5 ~252/D (DE)			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2904.5	3034.5	3164.5	3264.5	3209.5	3339.5	3469.5	3569.5
	i	1381	1511	1641	1741	1381	1511	1641	1741
	k	1523.5				1828.5			
	n	172				237			
	o	350				420			
	t	175				210			
	m	372				472			
	q	478				578			
Oil volume (dm³)		170	210	230	250	170	210	230	250
Oil discharge (dm³)		250	270	290	310	330	345	360	375
Weight (kg)		370				390			

Technical drawing of a 170 kV oil-filled power transformer, showing front and side views with detailed dimensions and component labels.

Front View (Top):

- Dimensions:** 314, 136, 59, 122.5, 120.5, 400, 319, 110.5, 1205.5, 65, 247.5.
- Labels:** Gearbox, Grading ring (Use when voltage levels not below 170 kV), Current take off ring, Diverter switch.

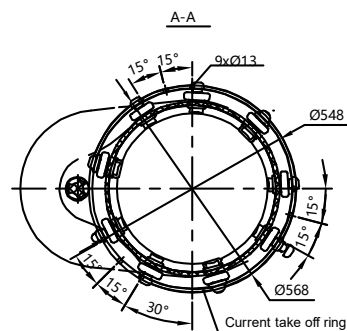
Side View (Bottom):

- Dimensions:** 65, 247.5, n, t, o, k, t, o, t, e, o.
- Labels:** Tap selector.

Other Labels: A, B, h, q.

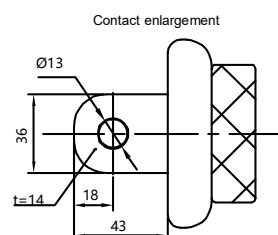
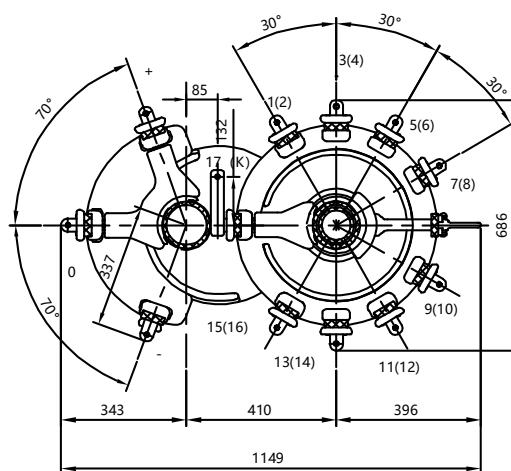
Notes:

- Grading ring Use when voltage levels not below 170 kV
- Current take off ring
- Diverter switch
- Tap selector



B-B

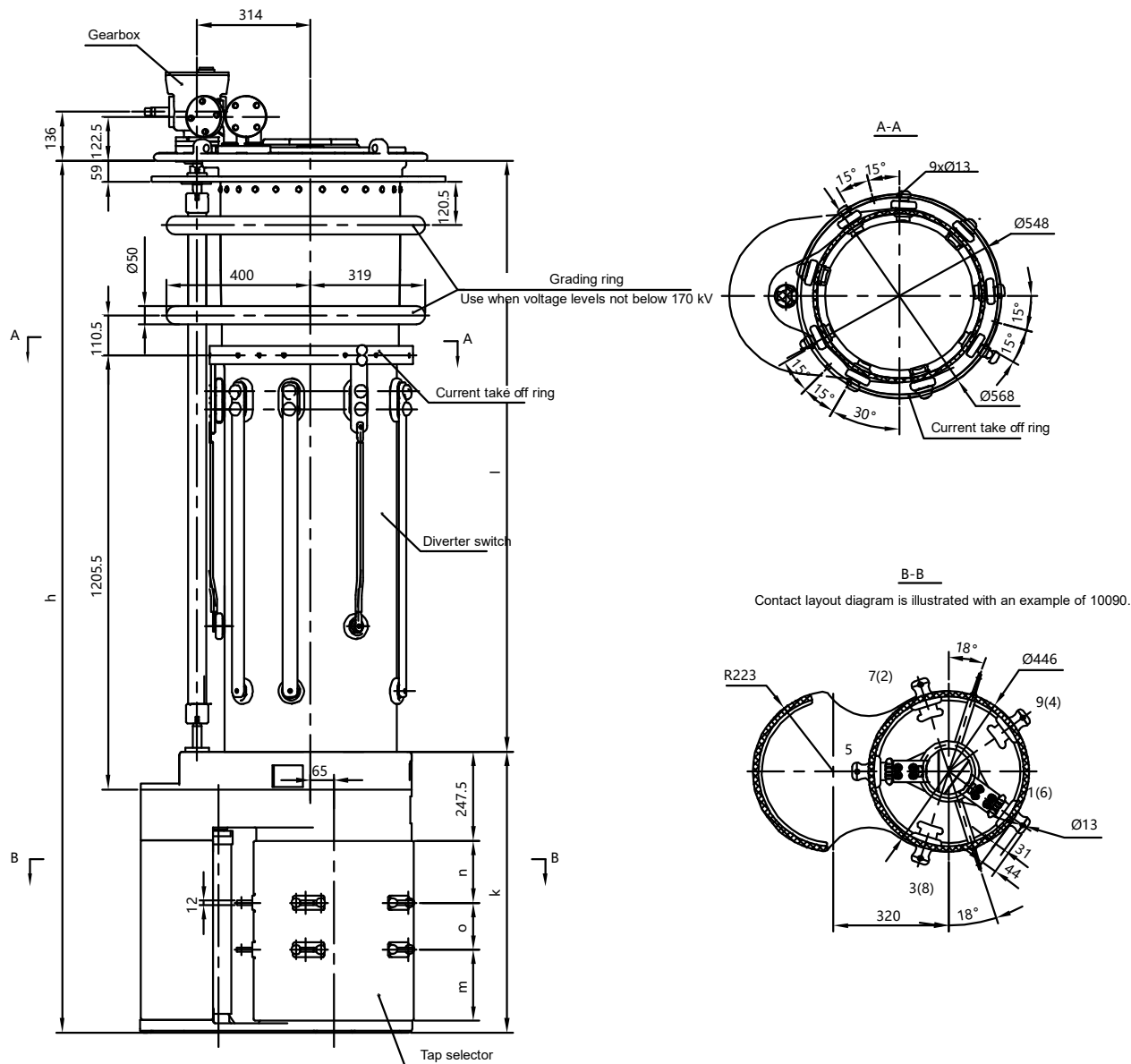
Contact layout diagram is illustrated with an example of 18353G.



Model		SHZVG III1500/72.5 ~252/C				SHZVG III1500/72.5 ~252/D (DE)			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	2904.5	3034.5	3164.5	3264.5	3209.5	3339.5	3469.5	3569.5
	i	1381	1511	1641	1741	1381	1511	1641	1741
	k	1523.5				1828.5			
	n	172				237			
	o	350				420			
	t	175				210			
	m	372				472			
	q	402.5				485			
Oil volume (dm³)		170	210	230	250	170	210	230	250
Oil discharge (dm³)		250	270	290	310	330	345	360	375
Weight (kg)		370				390			

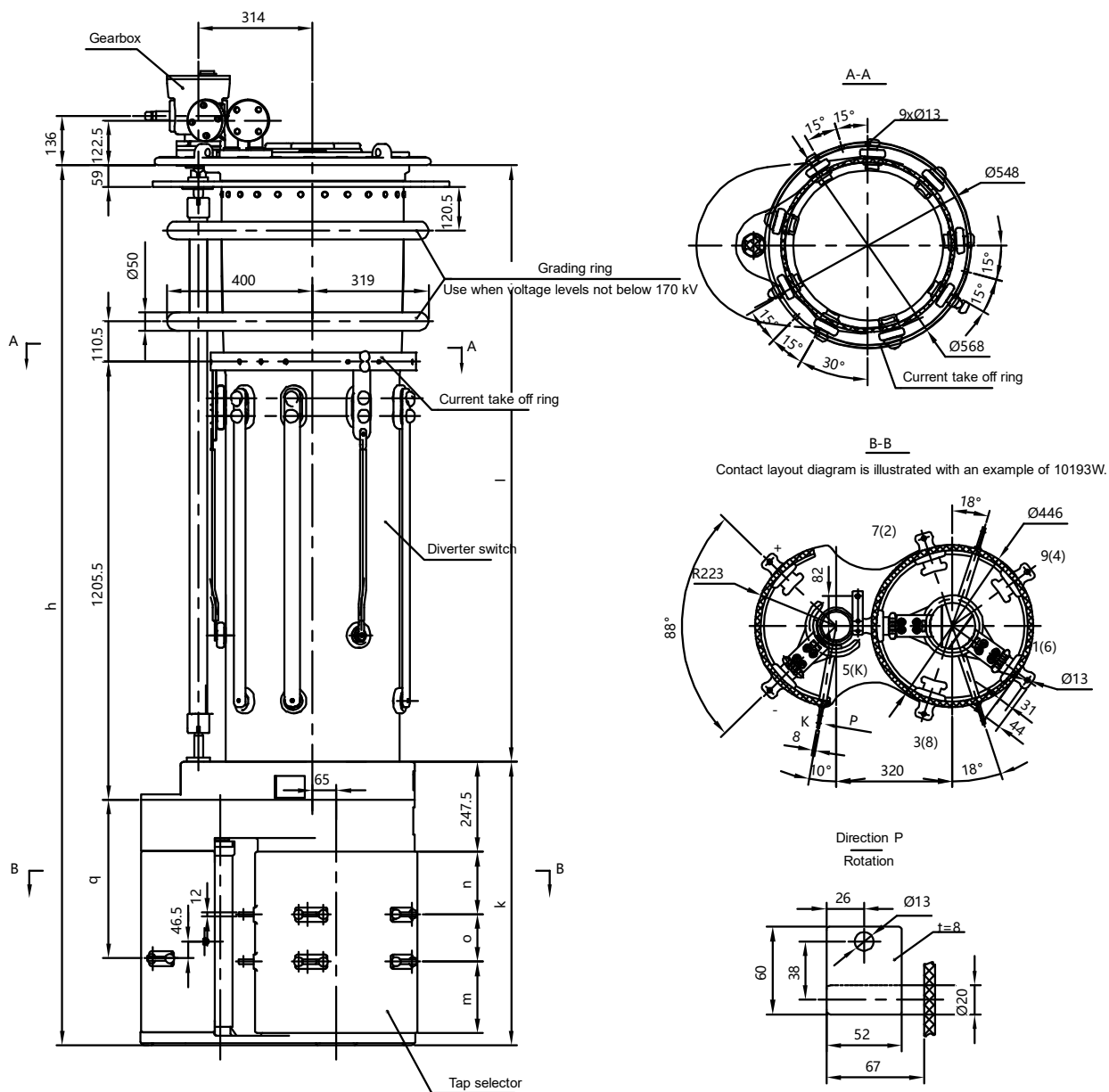
Unit: mm

9.7 Outline dimension drawing of SHZVG I 1300A linear regulation on-load tap-changer



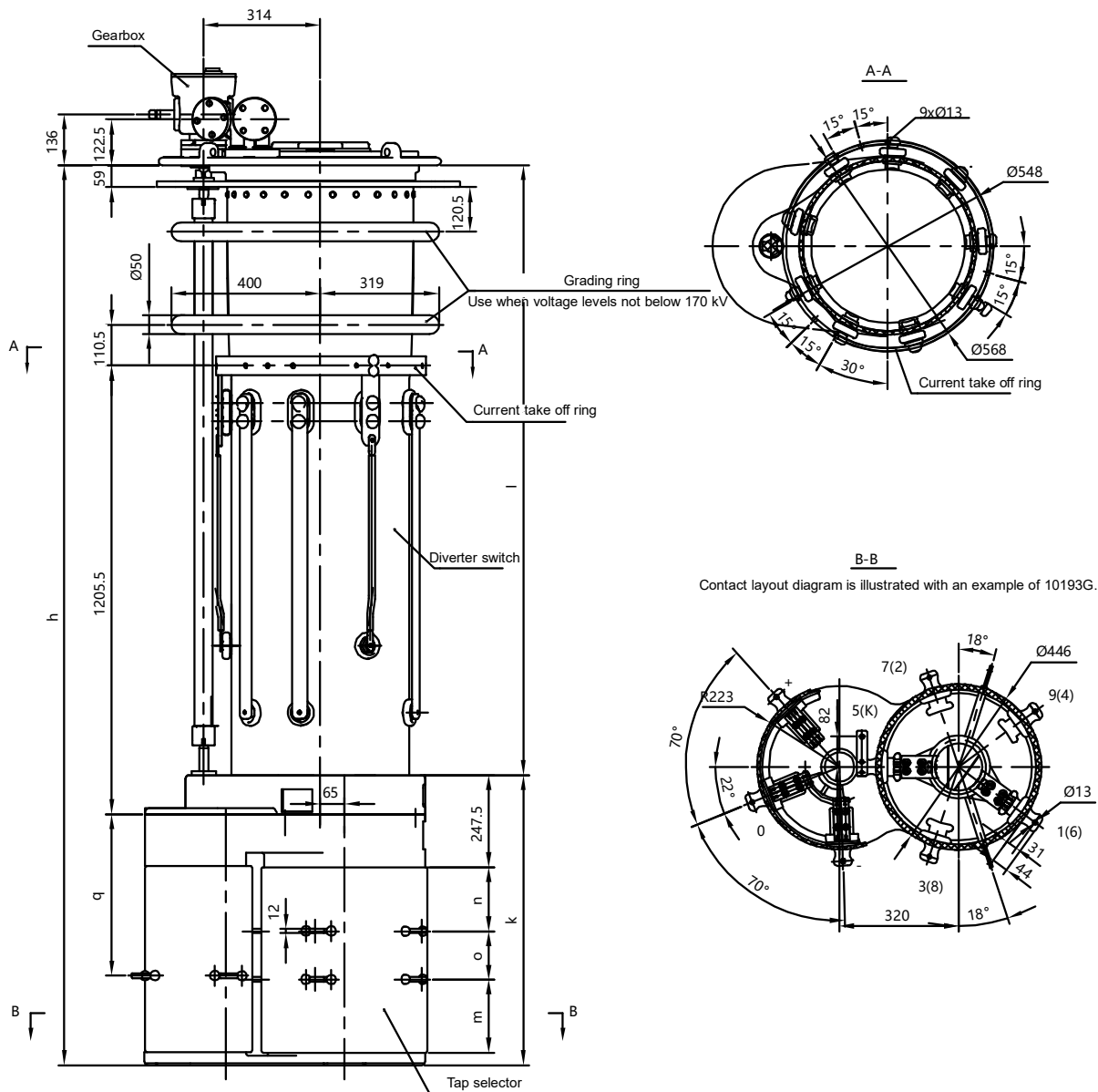
Model		SHZVG I 1300/72.5 ~ 363/C						SHZVG I 1300/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2160.5	2290.5	2420.5	2520.5	2570.5	2670.5	2330.5	2460.5	2590.5	2690.5	2740.5	2840.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	779.5						949.5					
	n	172						237					
	o	130						195					
	m	197						232					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		210	230	250	265	275	290	230	250	270	285	295	310
Weight (kg)		320						340					

Unit: mm



Model		SHZVG I 1300/72.5 ~ 363/C						SHZVG I 1300/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2160.5	2290.5	2420.5	2520.5	2570.5	2670.5	2330.5	2460.5	2590.5	2690.5	2740.5	2840.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	779.5						949.5					
	n	172						237					
	o	130						195					
	m	197						232					
	q	443						566					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		210	230	250	265	275	290	230	250	270	285	295	310
Weight (kg)		320						340					

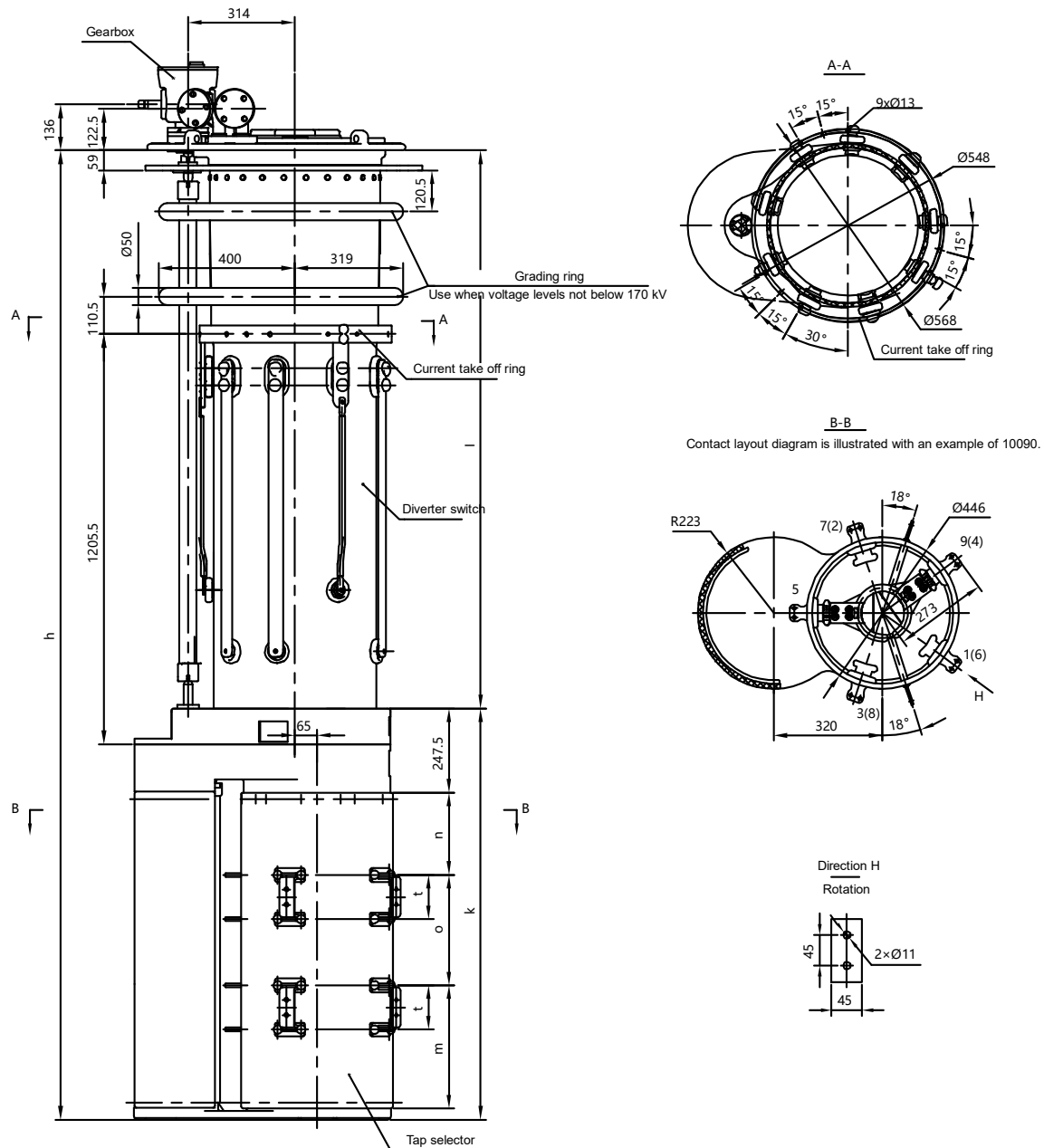
9.9 Outline dimension drawing of SHZVG I 1300A coarse & fine on-load tap-changer



Model		SHZVG I 1300/72.5 ~ 363/C						SHZVG I 1300/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2160.5	2290.5	2420.5	2520.5	2570.5	2670.5	2330.5	2460.5	2590.5	2690.5	2740.5	2840.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	779.5						949.5					
	n	172						237					
	o	130						195					
	m	197						232					
	q	428						478					
Oil volume (dm ³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm ³)		210	230	250	265	275	290	230	250	270	285	295	310
Weight (kg)		320						340					

Unit: mm

9.10 Outline dimension drawing of SHZVG I 1500A/2000A linear regulation on-load tap-changer



9 Drawings Attached

Model		SHZVG I 1500/2000/72.5 ~ 363/G						SHZVG I 1500/2000/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2420.5	2550.5	2680.5	2780.5	2830.5	2930.5	2590.5	2720.5	2850.5	2950.5	3000.5	3100.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	1039.5						1209.5					
	n	172						242					
	o	260						325					
	t	130						130					
	m	197						362					
	q	566						701					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		230	250	270	285	295	310	250	270	290	305	315	330
Weight (kg)		340						360					

9 Drawings Attached

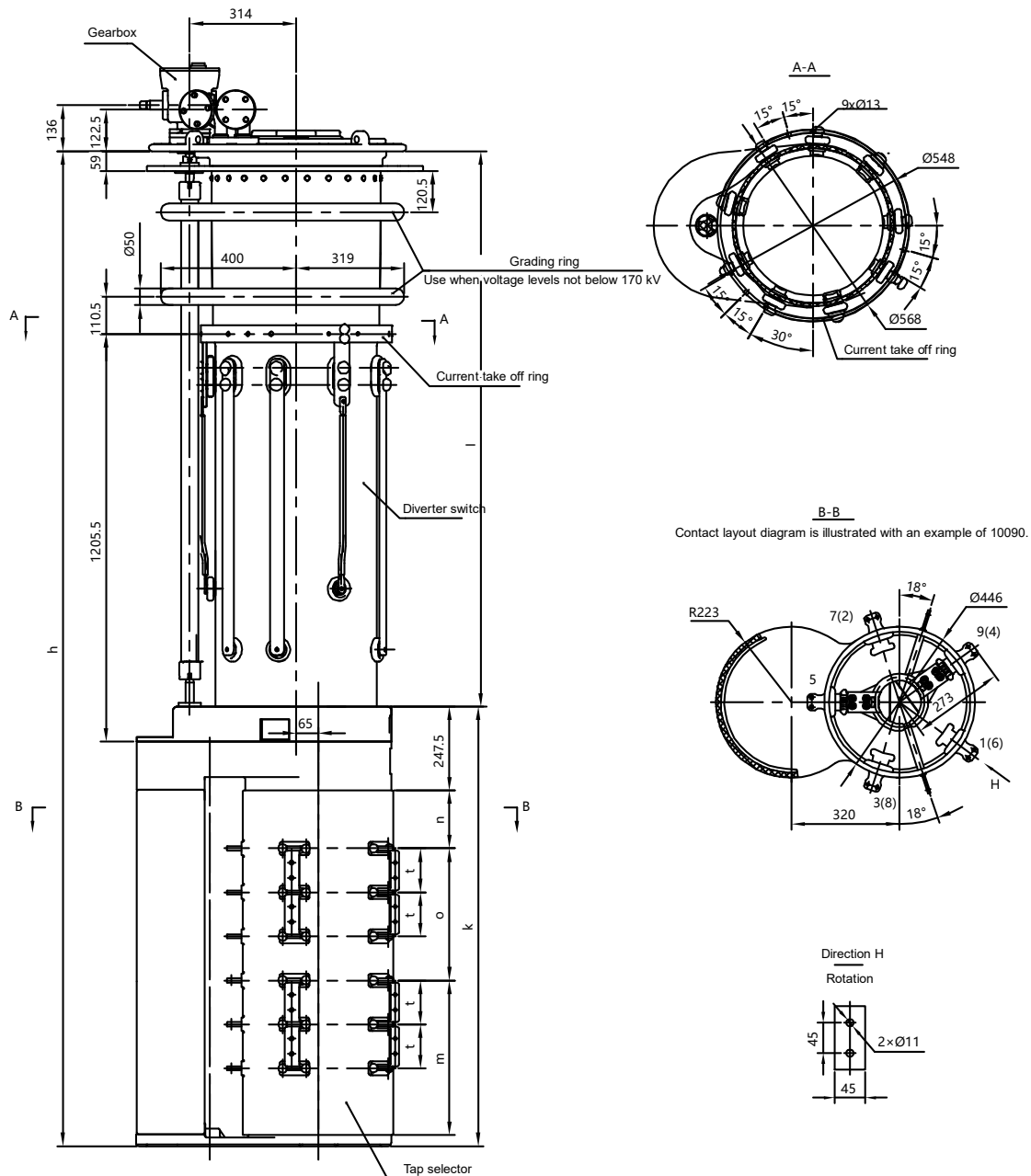
Unit: mm

[illegible]

9 Drawings Attached

Model		SHZVG I 1500/2000/72.5 ~ 363/C						SHZVG I 1500/2000/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2420.5	2550.5	2680.5	2780.5	2830.5	2930.5	2590.5	2720.5	2850.5	2950.5	3000.5	3100.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	1039.5						1209.5					
	n	172						242					
	o	260						325					
	t	130						130					
	m	197						362					
	q	669.5						804.5					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		230	250	270	285	295	310	250	270	290	305	315	330
Weight (kg)		340						360					

9.13 Outline dimension drawing of SHZVG I 2400A/3000A linear regulation on-load tap-changer



Model		SHZVG I 2400/3000/72.5 ~ 363/C						SHZVG I 2400/3000/72.5 ~ 363/D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2680.5	2810.5	2940.5	3040.5	3090.5	3190.5	2850.5	2980.5	3110.5	3210.5	3260.5	3360.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	1299.5						1469.5					
	n	172						242					
	o	390						455					
	t	130						130					
	m	457						492					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		270	290	310	325	335	350	290	310	330	345	355	370
Weight (kg)		370						390					

Unit: mm

9 Drawings Attached

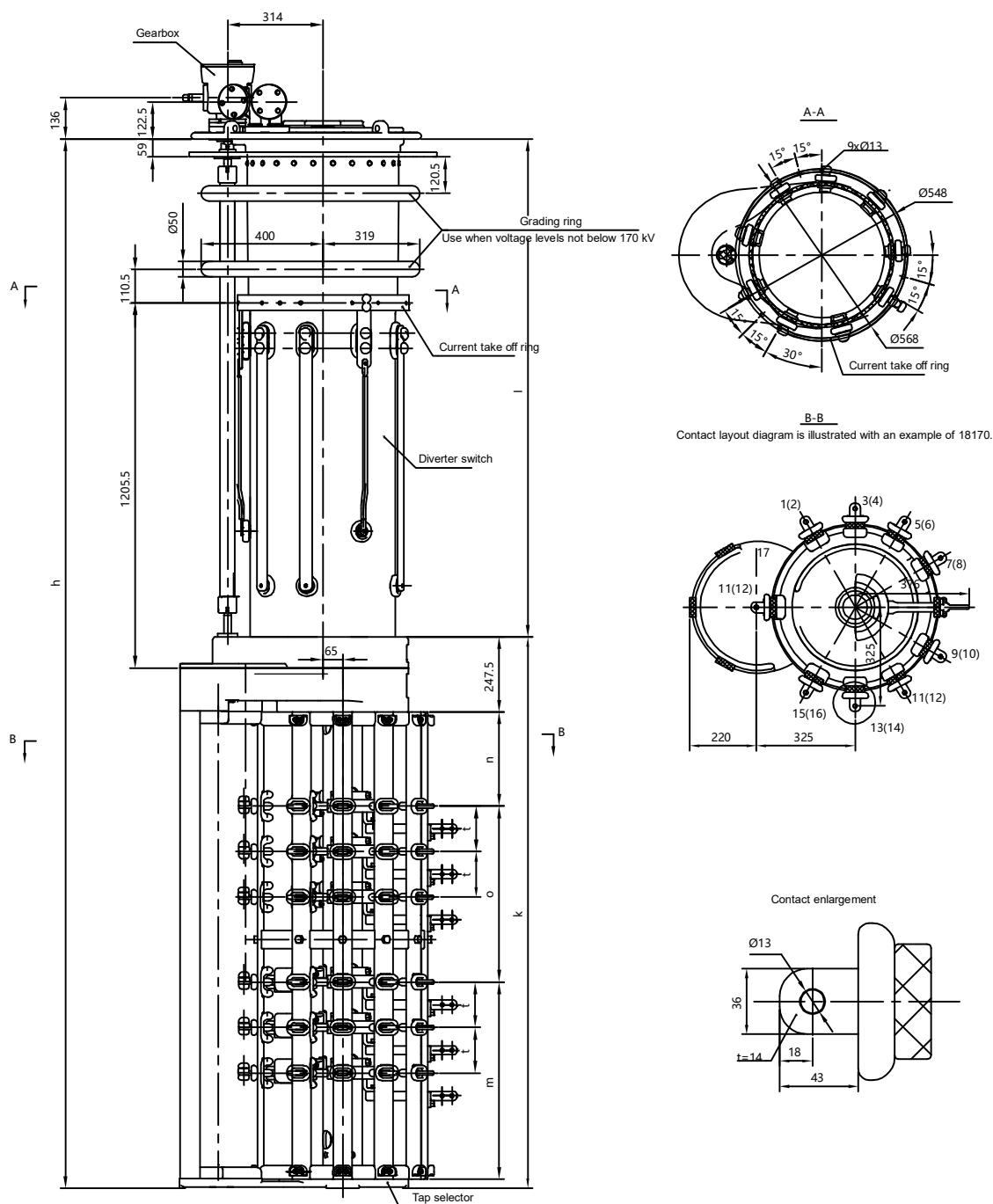


Model		SHZVG I 2400/3000/72.5 ~ 363/°C						SHZVG I 2400/3000/72.5 ~ 363/°D					
Highest equipment voltage (kV)		72.5	126	170	252	300	363	72.5	126	170	252	300	363
Dimension (mm)	h	2680.5	2810.5	2940.5	3040.5	3090.5	3190.5	2850.5	2980.5	3110.5	3210.5	3260.5	3360.5
	i	1381	1511	1641	1741	1791	1891	1381	1511	1641	1741	1791	1891
	k	1299.5						1469.5					
	n	172						242					
	o	390						455					
	t	130						130					
	m	457						492					
	q	696						701					
Oil volume (dm³)		170	210	230	250	260	275	170	210	230	250	260	275
Oil discharge (dm³)		270	290	310	325	335	350	290	310	330	345	355	370
Weight (kg)		370						390					

9 Drawings Attached



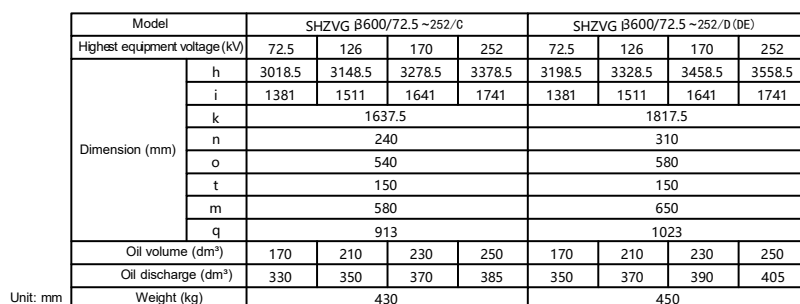
34

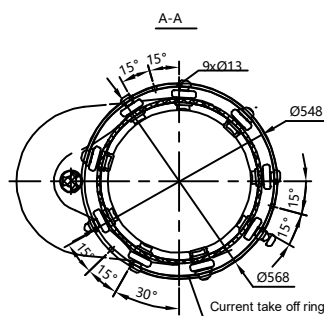


Model		SHZVG B600/72.5 ~252/c				SHZVG B600/72.5~252/d (DE)			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	3018.5	3148.5	3278.5	3378.5	3198.5	3328.5	3458.5	3558.5
	i	1381	1511	1641	1741	1381	1511	1641	1741
	k	1637.5				1817.5			
	n	240				310			
	o	540				580			
	t	150				150			
	m	580				650			
Oil volume (dm³)		170	210	230	250	170	210	230	250
Oil discharge (dm³)		330	350	370	385	350	370	390	405
Weight (kg)		430				450			

Unit: mm

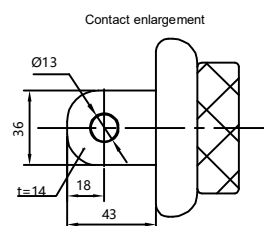
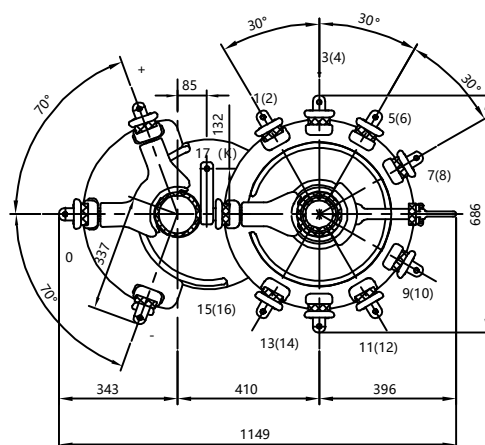
9 Drawings Attached



[illegible]

B-B

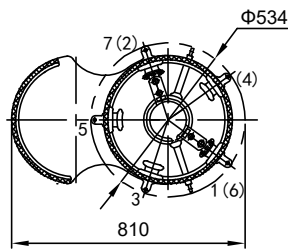
Contact layout diagram is illustrated with an example of 18353G.



						SHZVG B600/72.5 ~252/D (DE)			
Highest equipment voltage (kV)		72.5	126	170	252	72.5	126	170	252
Dimension (mm)	h	3018.5	3148.5	3278.5	3378.5	3198.5	3328.5	3458.5	3558.5
	i	1381	1511	1641	1741	1381	1511	1641	1741
	k	1637.5				1817.5			
	n	240				310			
	o	540				580			
	t	150				150			
	m	580				650			
	q	765				895			
Oil volume (dm³)		170	210	230	250	170	210	230	250
Oil discharge (dm³)		330	350	370	385	350	370	390	405
Weight (kg)		430				450			

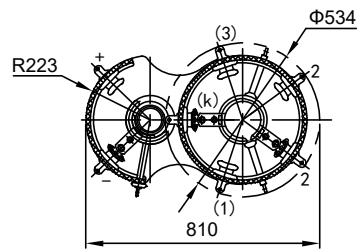
9.19 Layout drawing of SHZVG tap-changer tap selector contact

Linear regulation tap selector



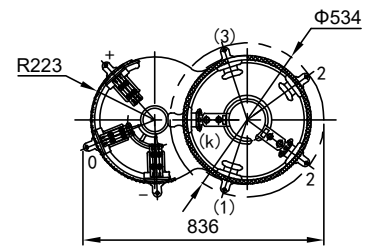
10070

Positive & reverse regulation tap selector

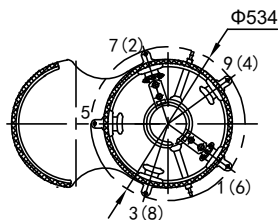


10051W

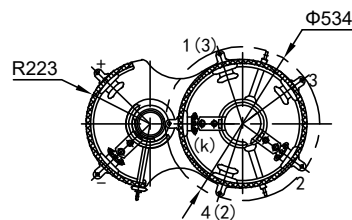
Coarse & fine tap selector



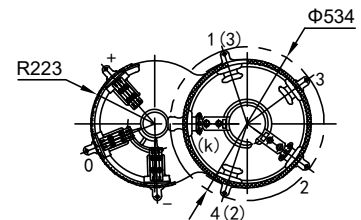
10051G



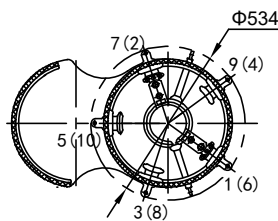
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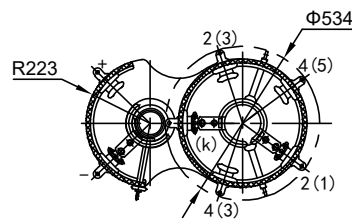
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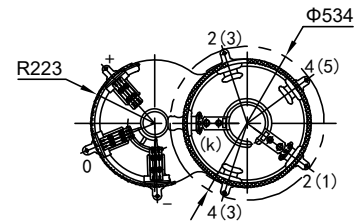
10071G



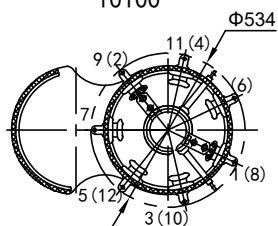
10100



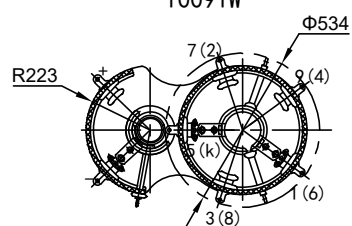
10091W



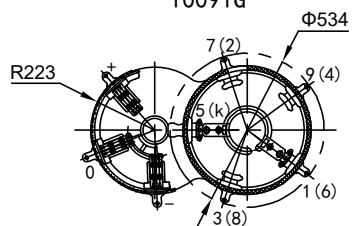
10091G



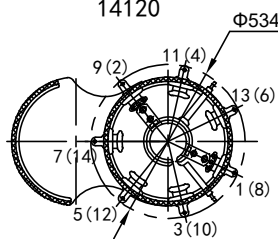
14120



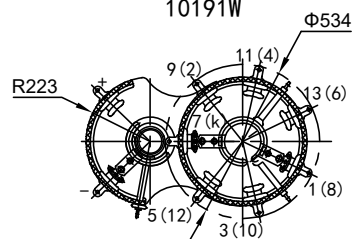
10191W



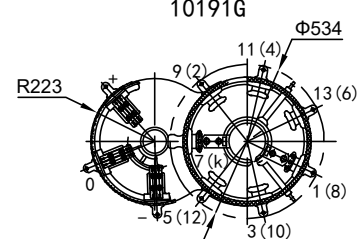
10191G



14140



14271W

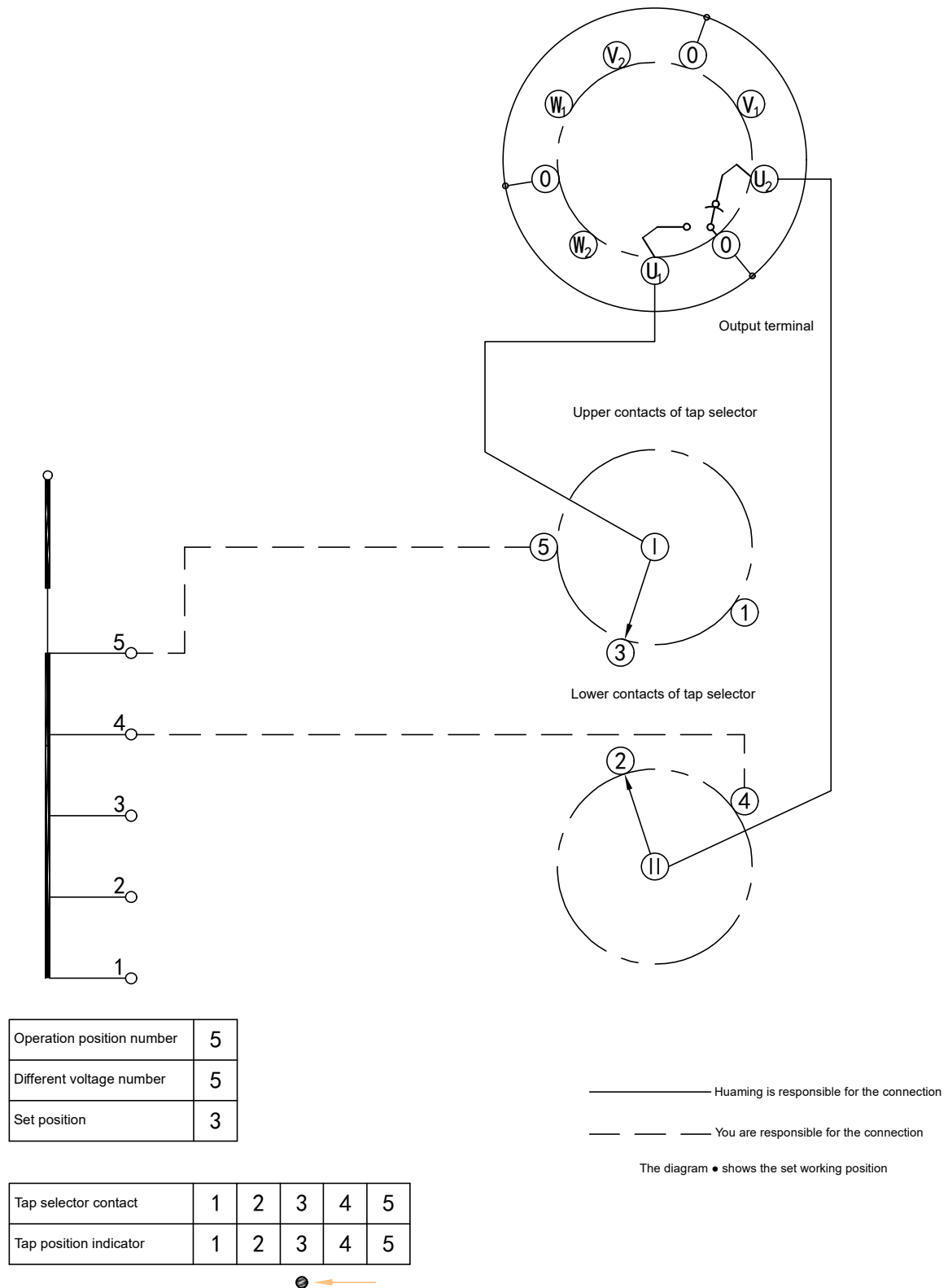


14271G

Note: The terminal code is subject to the relevant wiring diagram

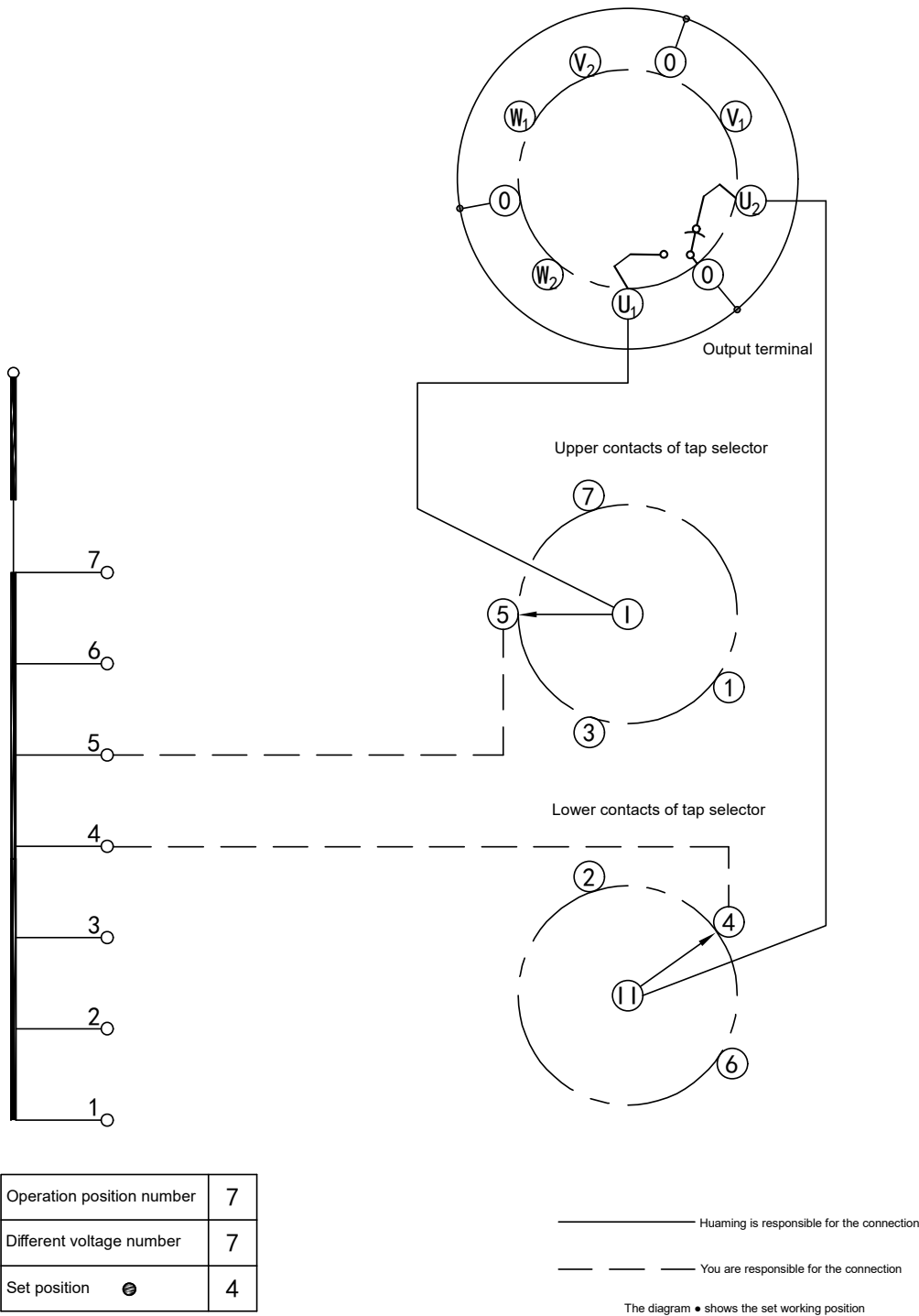
Tap selector terminal code:
X: indicates upper selector;
(X): indicates lower contact

9.20 Work position table and wiring diagram of SHZVG on-load changer (10050)



9.21 Work position table and wiring diagram of SHZVG on-load changer (10070)

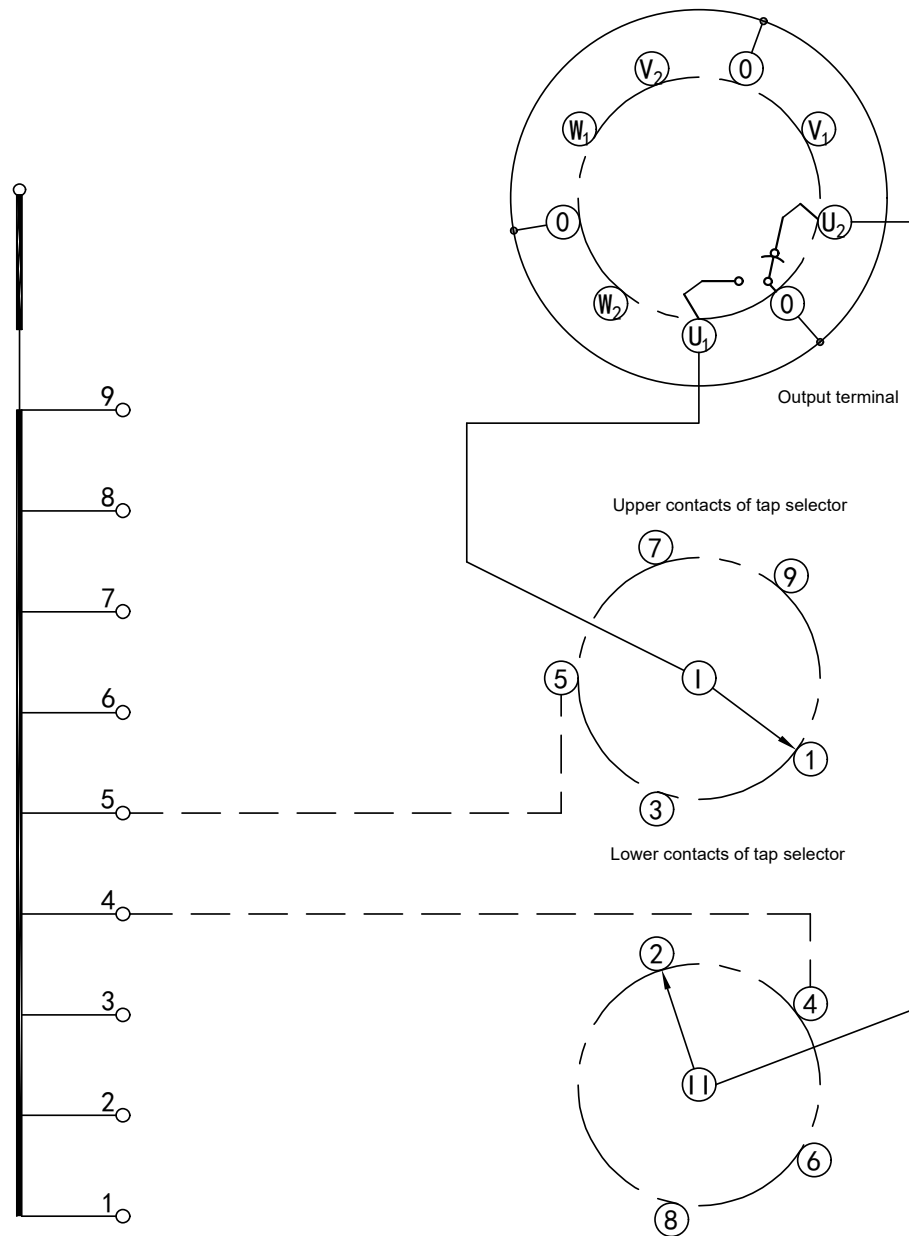
9 Drawings Attached



Tap selector contact	1	2	3	4	5	6	7
Tap position indicator	1	2	3	4	5	6	7



9.22 Work position table and wiring diagram of SHZVG on-load changer (10090)



Operation position number	9
Different voltage number	9
Set position	5

————— Huaming is responsible for the connection

- - - - - You are responsible for the connection

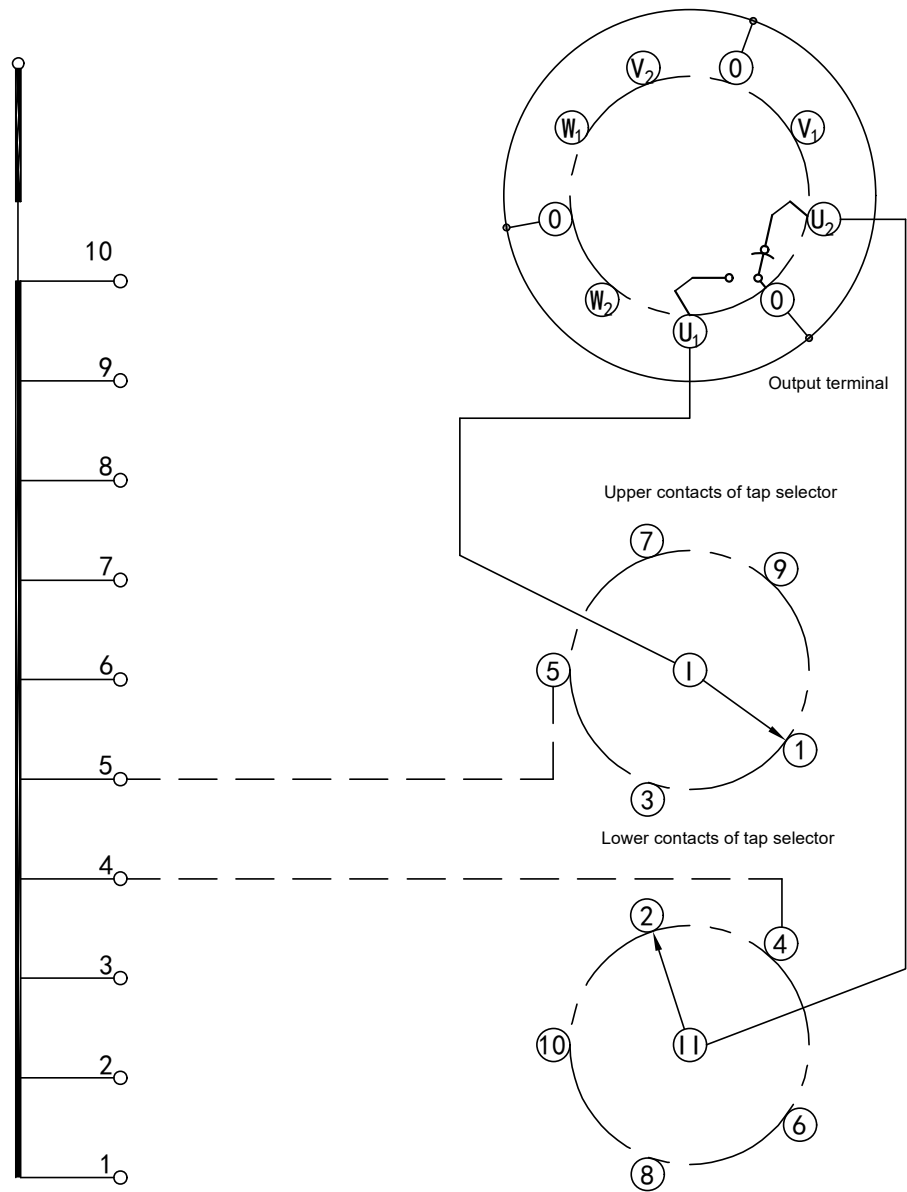
The diagram • shows the set working position

Tap selector contact	1	2	3	4	5	6	7	8	9
Tap position indicator	1	2	3	4	5	6	7	8	9



9.23 Work position table and wiring diagram of SHZVG on-load changer (10100)

9 Drawings Attached



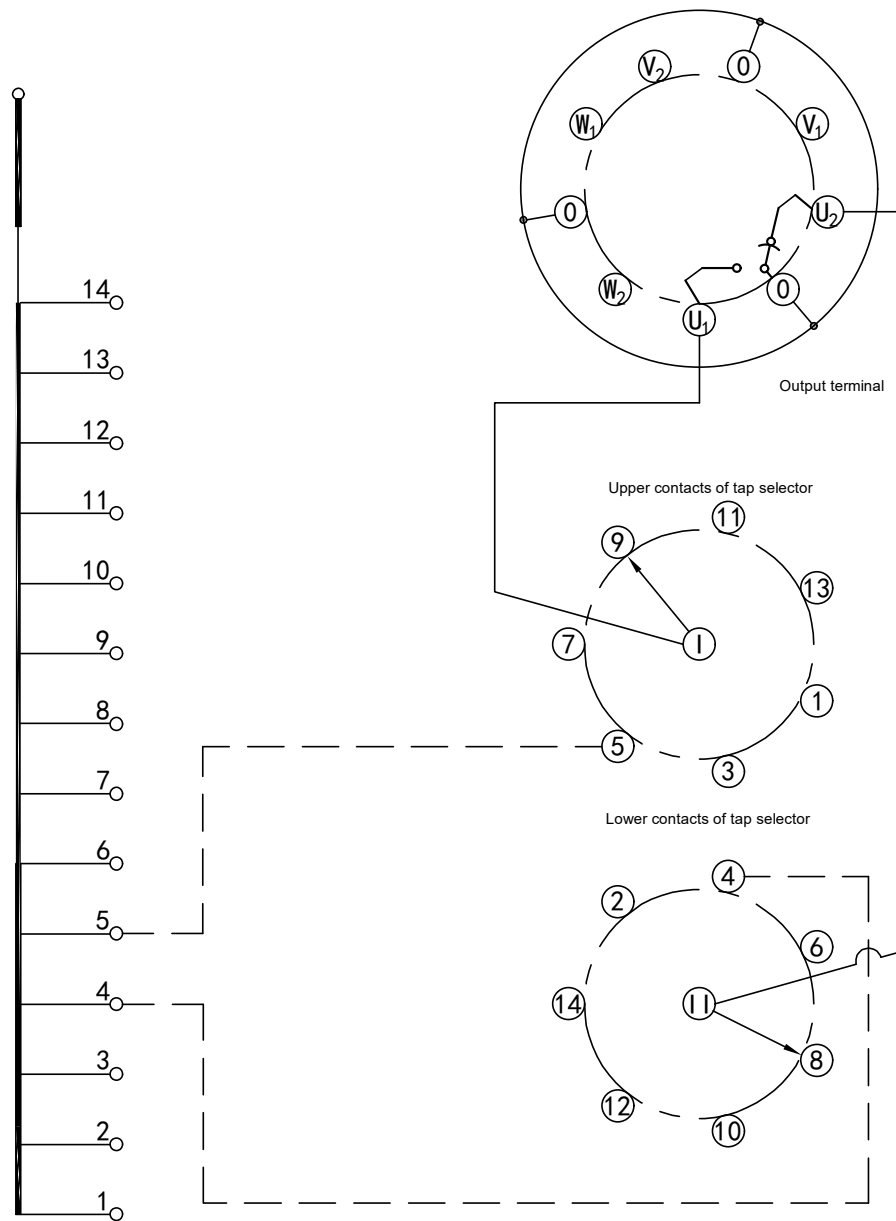
Operation position number	10
Different voltage number	10
Set position	6

————— Huaming is responsible for the connection
- - - - - You are responsible for the connection
The diagram • shows the set working position

Tap selector contact	1	2	3	4	5	6	7	8	9	10
Tap position indicator	1	2	3	4	5	6	7	8	9	10



9.24 Work position table and wiring diagram of SHZVG on-load changer (14140)



Operation position number	14
Different voltage number	14
Set position	8

————— Huaming is responsible for the connection

----- You are responsible for the connection

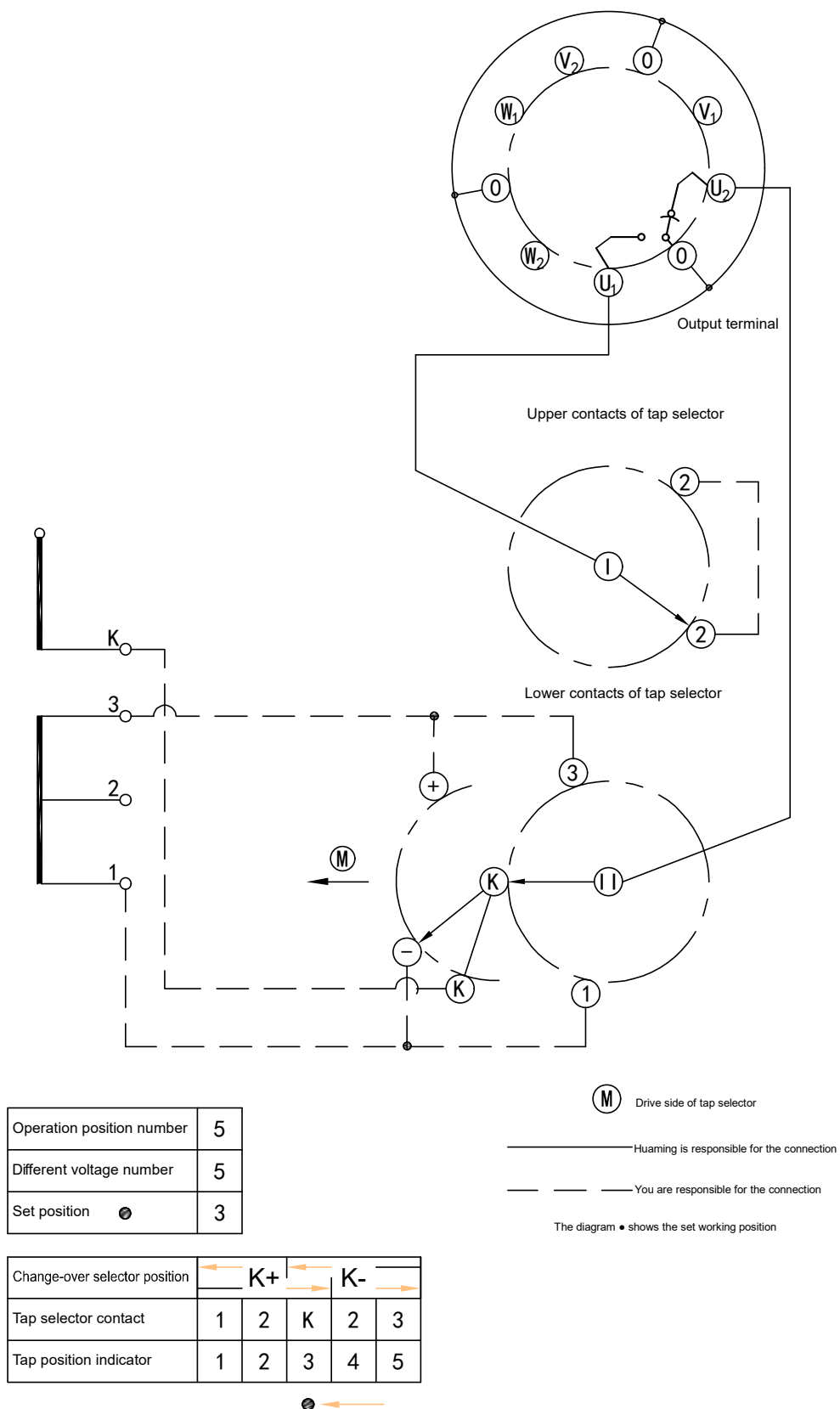
The diagram • shows the set working position

Tap selector contact	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Tap position indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14

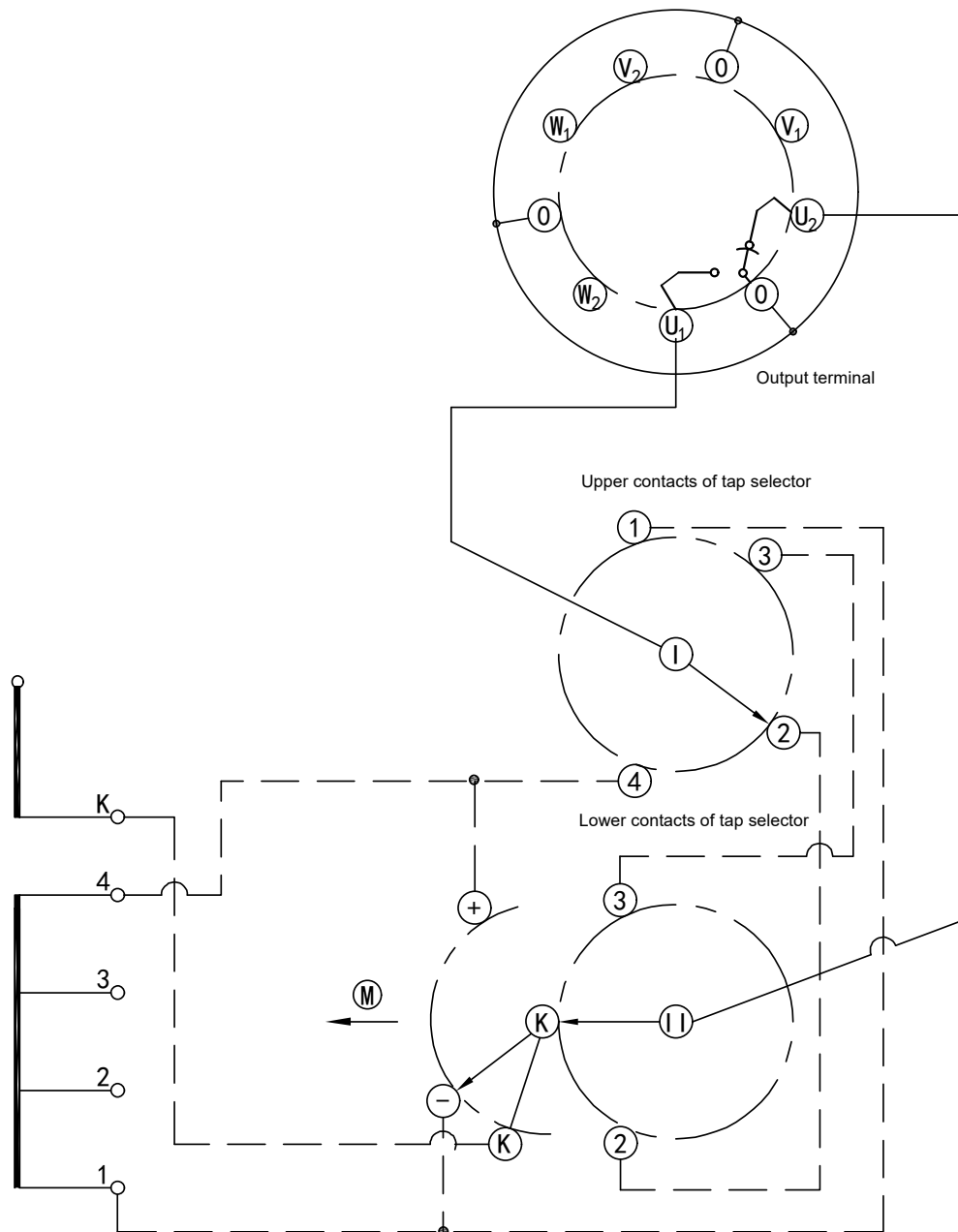


9.25 Work position table and wiring diagram of SHZVG on-load changer (10051W)

9 Drawings Attached



9.26 Work position table and wiring diagram of SHZVG on-load changer (10071W)



Operation position number	7
Different voltage number	7
Set position	4

(M) Drive side of tap selector

————— Huaming is responsible for the connection

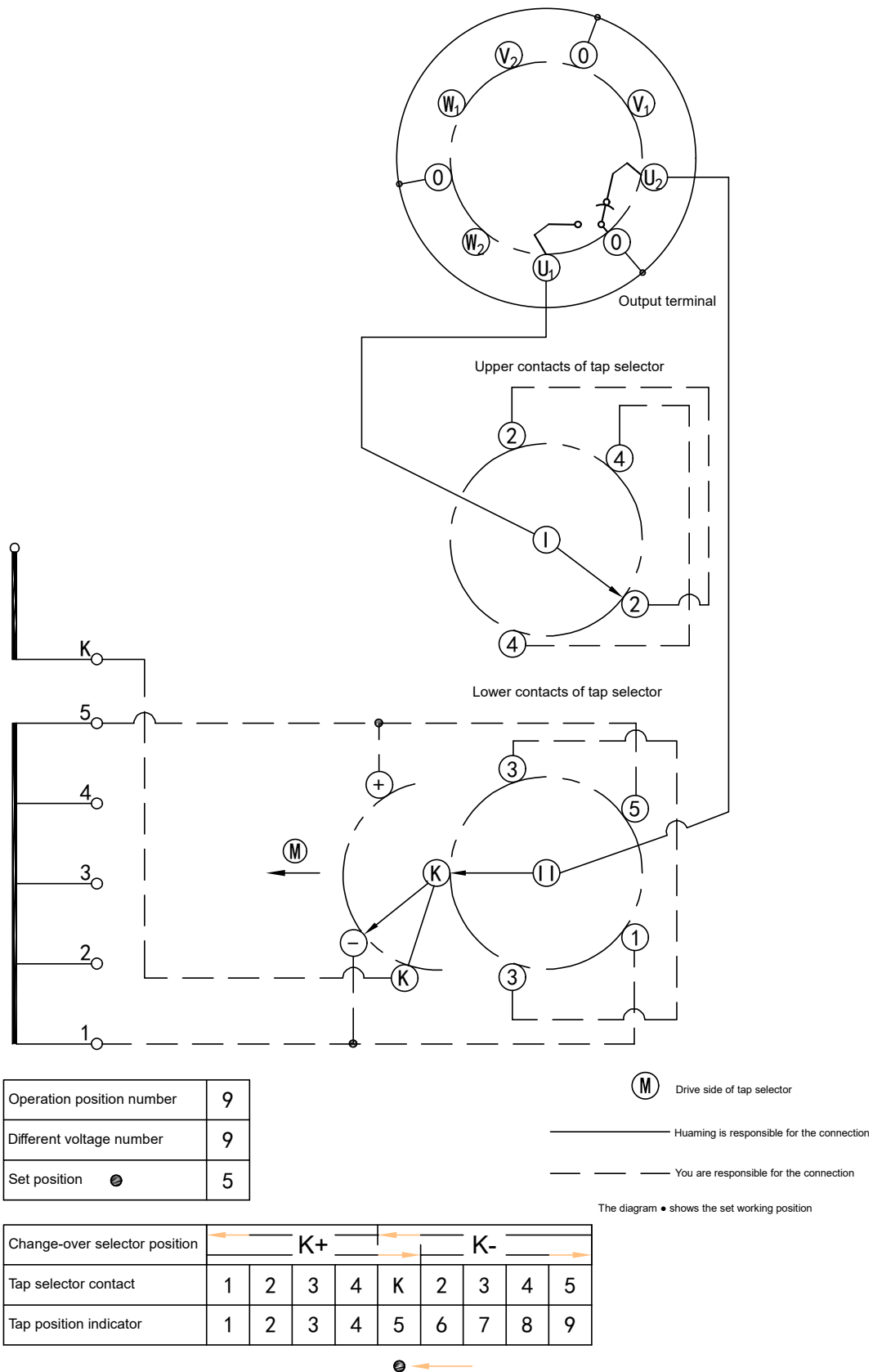
----- You are responsible for the connection

The diagram • shows the set working position

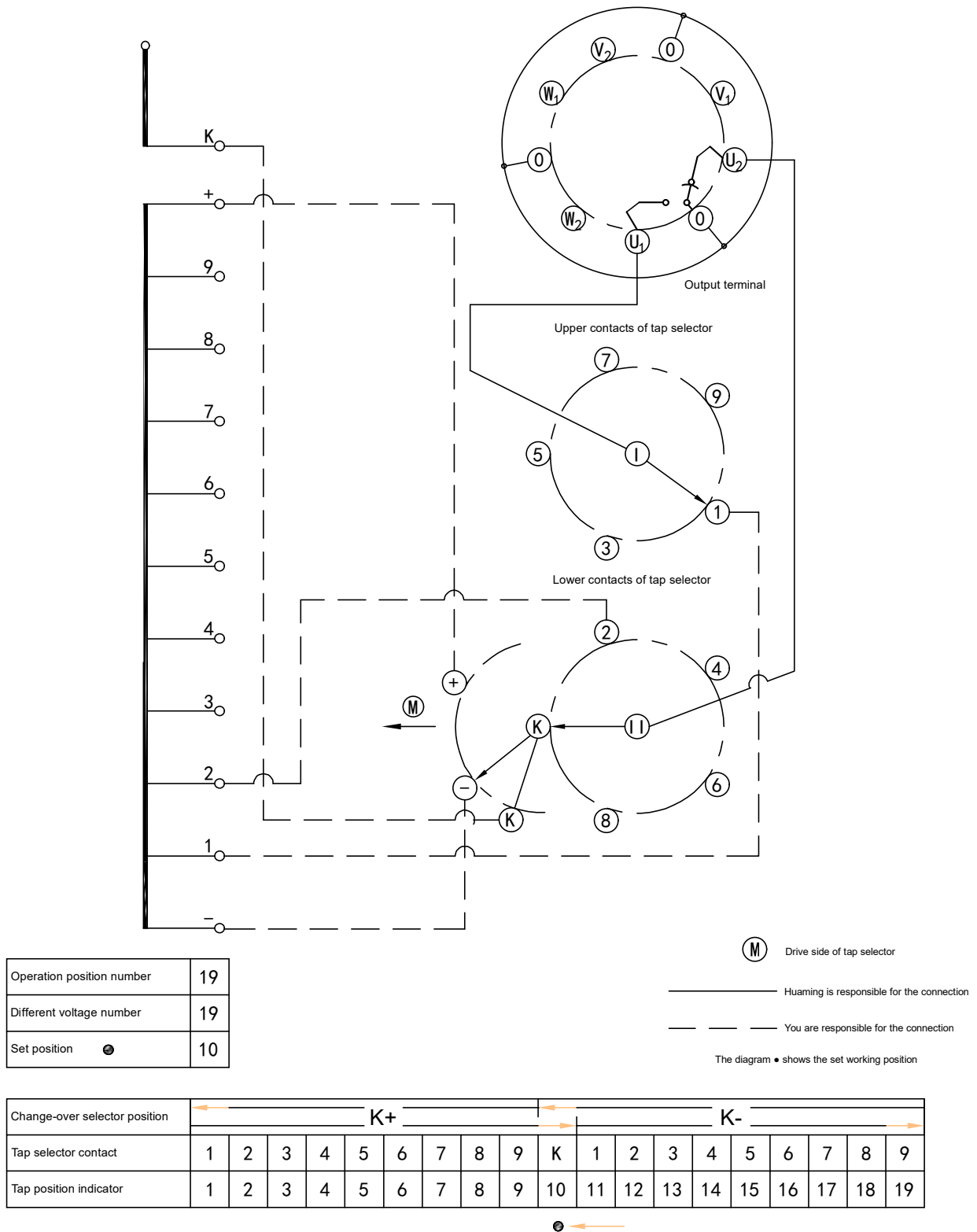
Change-over selector position							
Tap selector contact	1	2	3	K	2	3	4
Tap position indicator	1	2	3	4	5	6	7

• ———

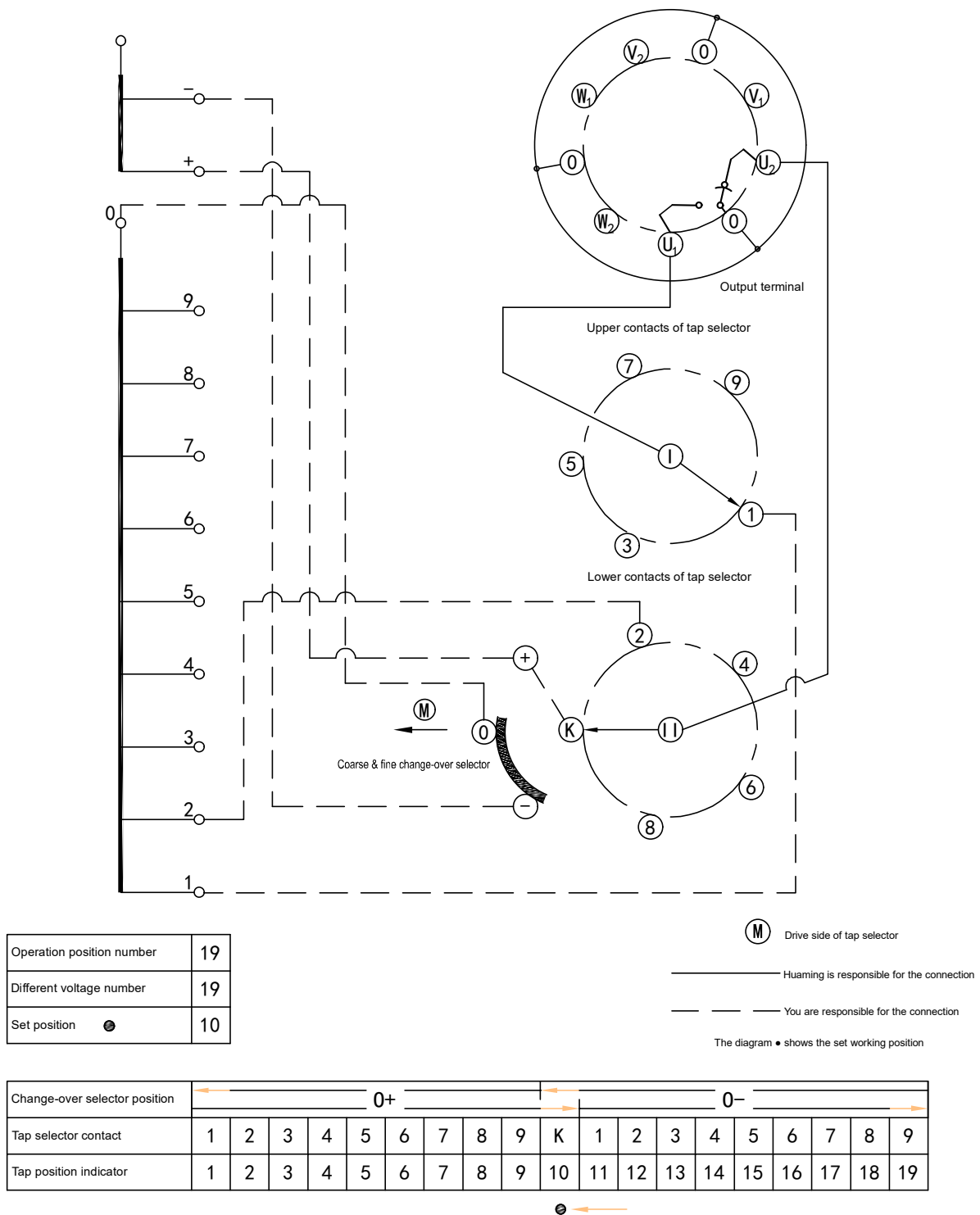
9.27 Work position table and wiring diagram of SHZVG on-load changer (10091W)



9.28 Work position table and wiring diagram of SHZVG on-load changer (10191W)

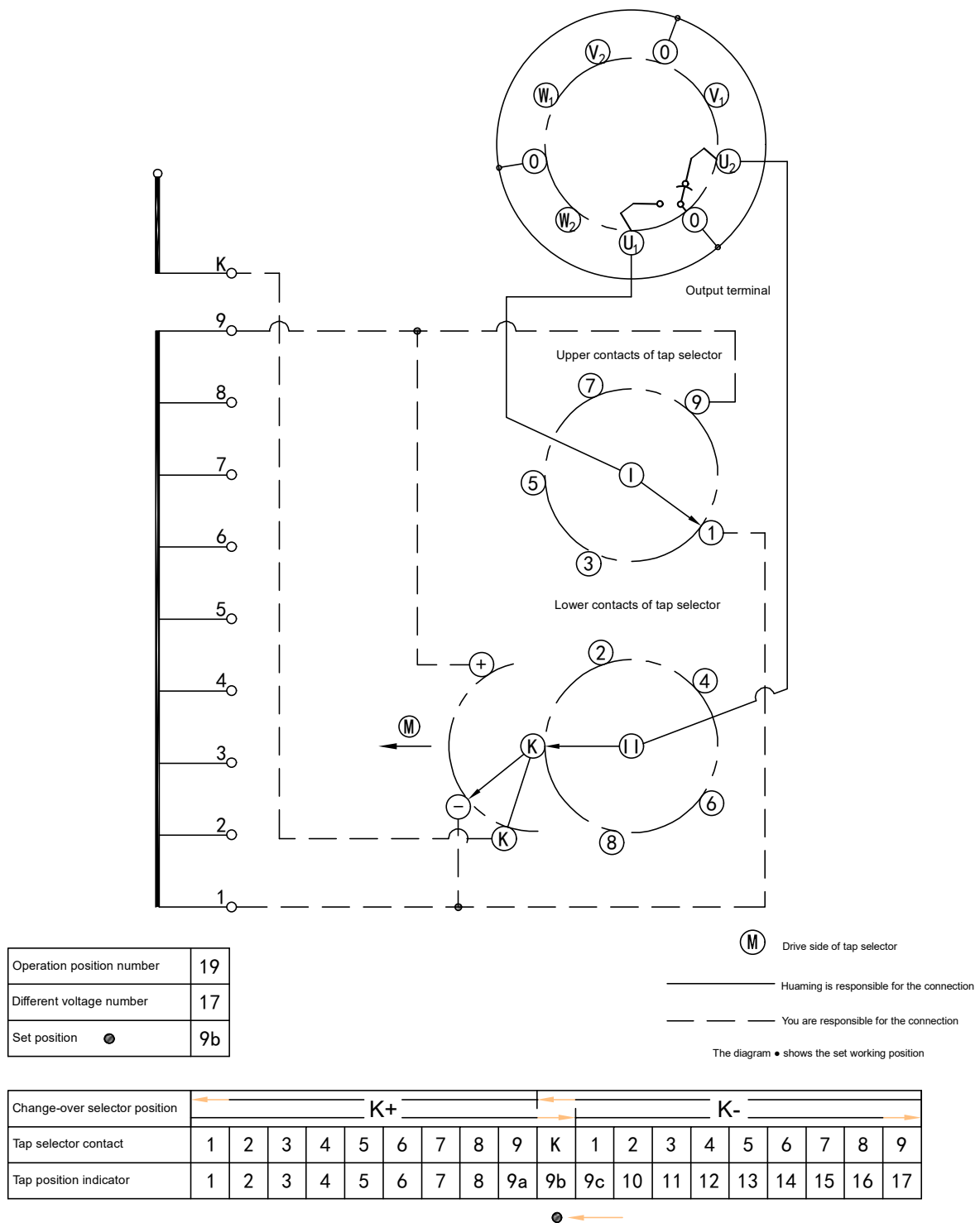


9.29 Work position table and wiring diagram of SHZVG on-load changer (10191G)

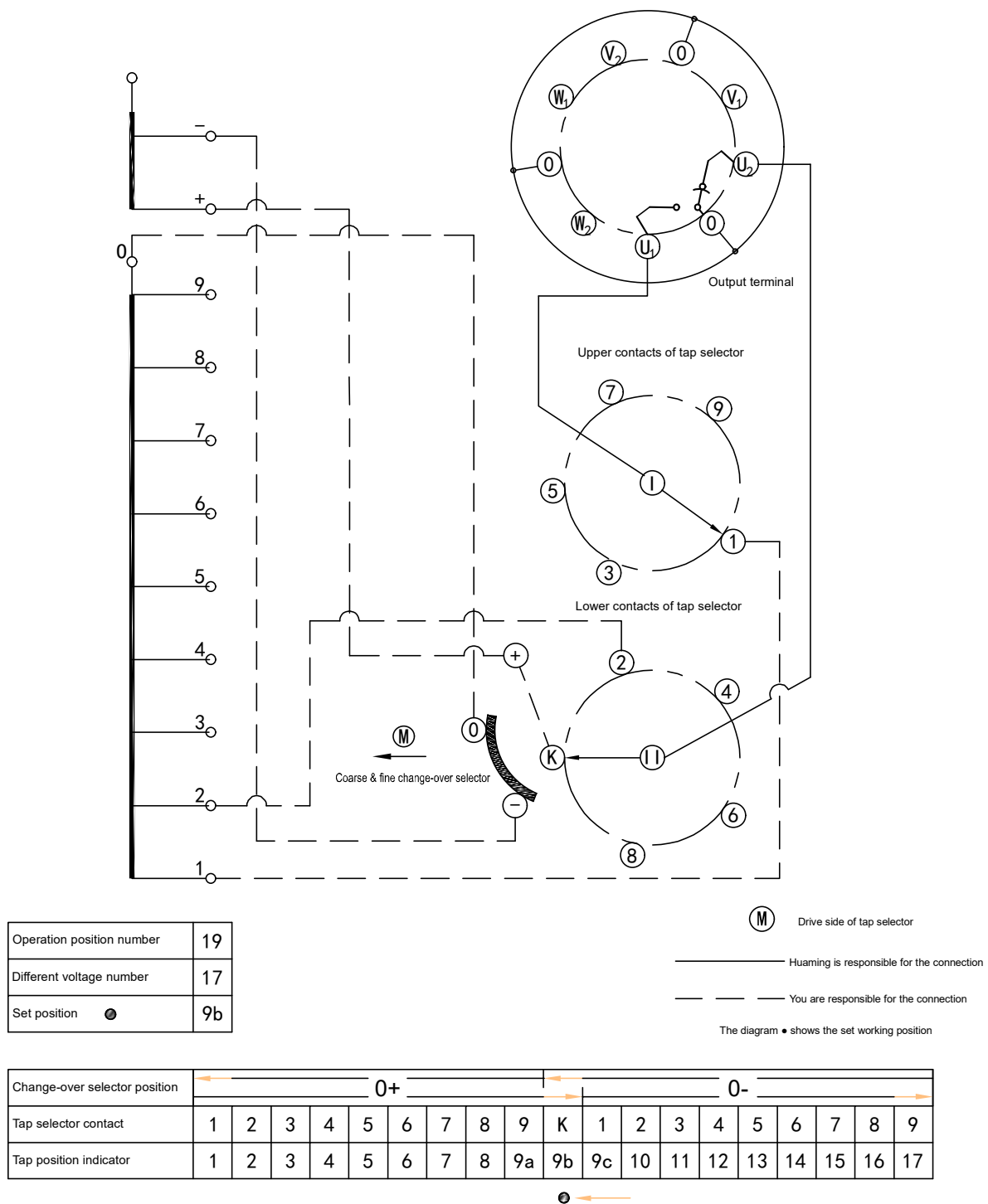


9 Drawings Attached

9.30 Work position table and wiring diagram of SHZVG on-load changer (10193W)

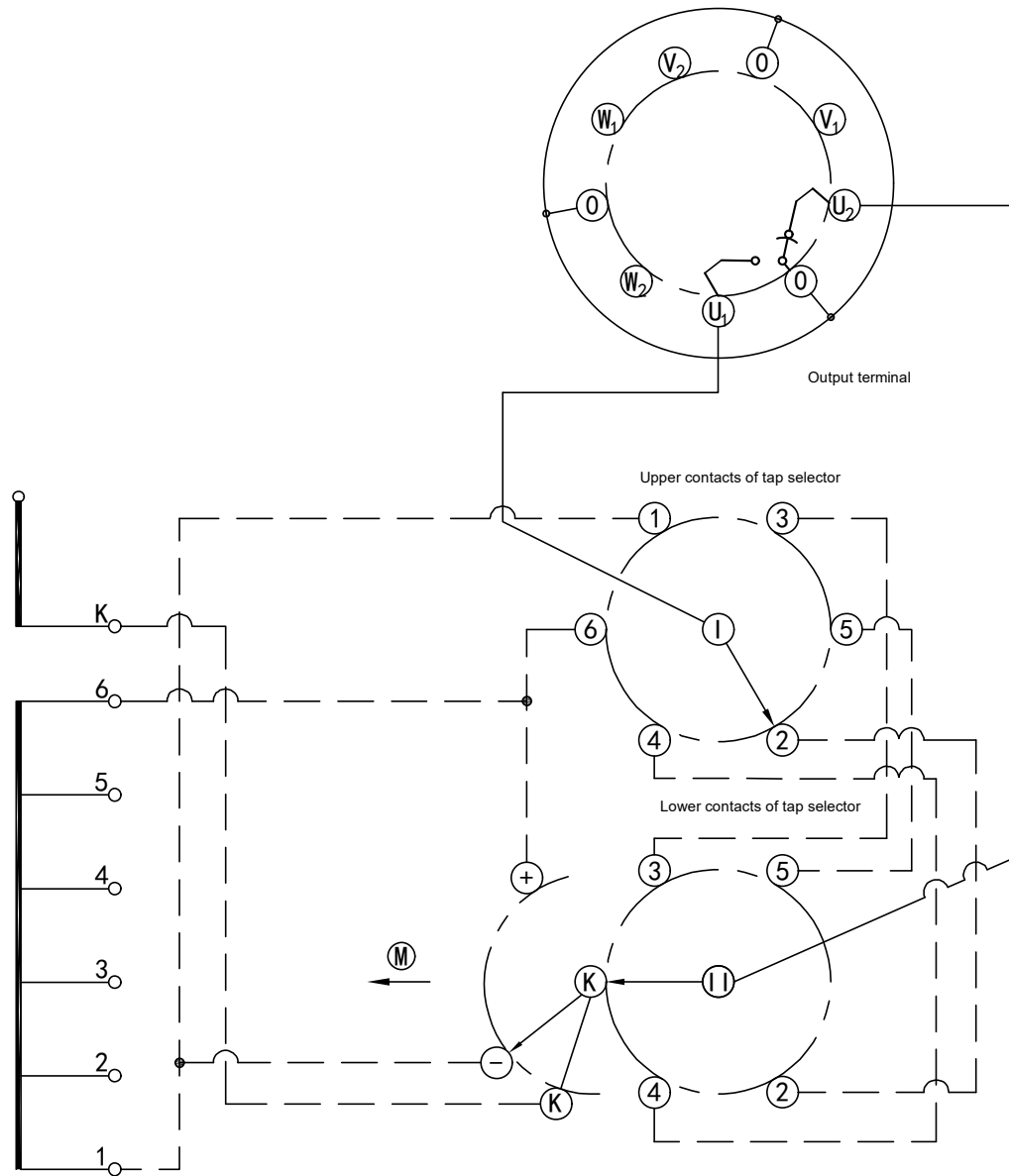


9.31 Work position table and wiring diagram of SHZVG on-load changer (10193G)



9 Drawings Attached

9.32 Work position table and wiring diagram of SHZVG on-load changer (12111W)



Operation position number	11
Different voltage number	11
Set position •	6

(M) Drive side of tap selector

————— Huaming is responsible for the connection

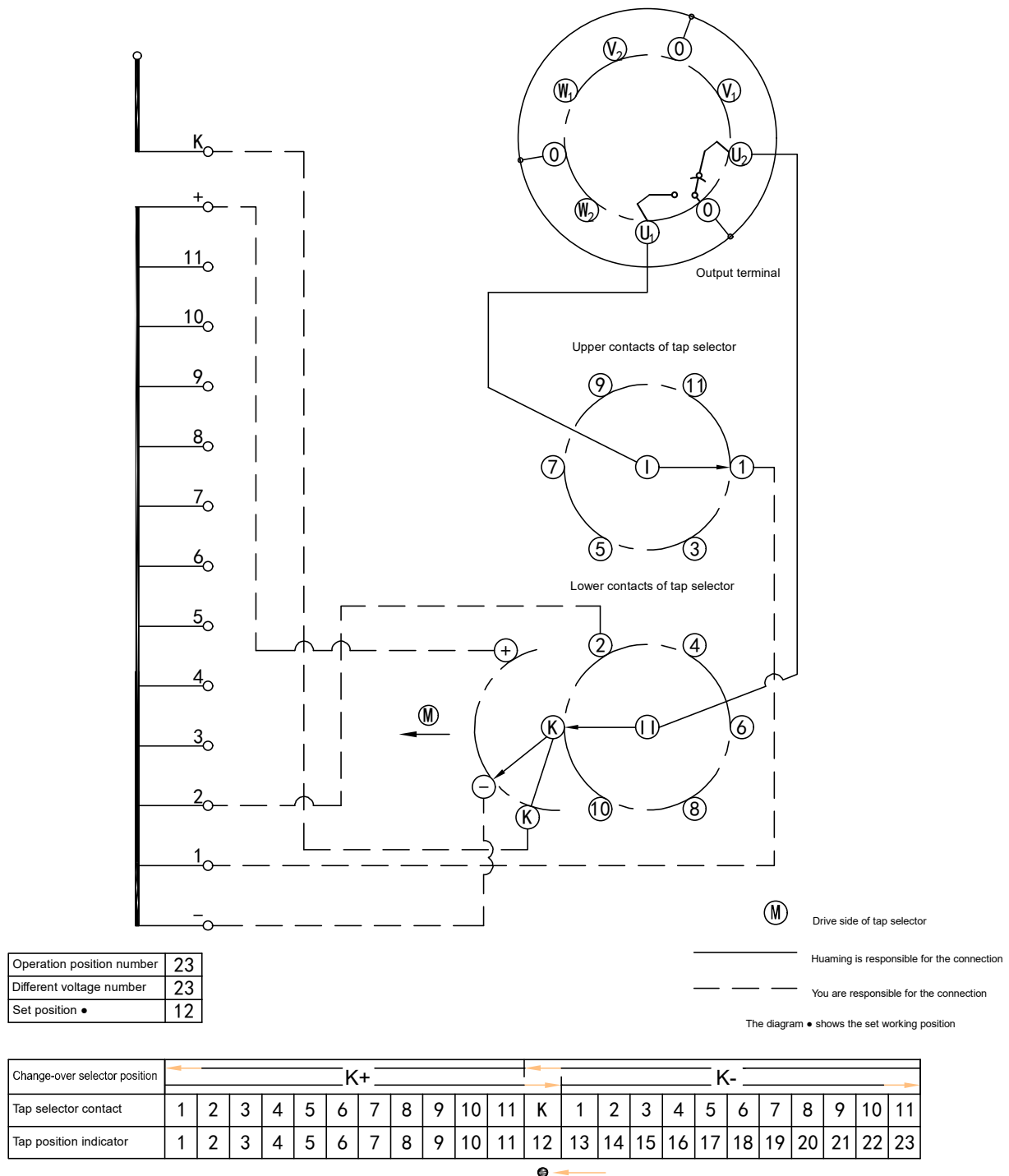
- - - - - You are responsible for the connection

The diagram • shows the set working position

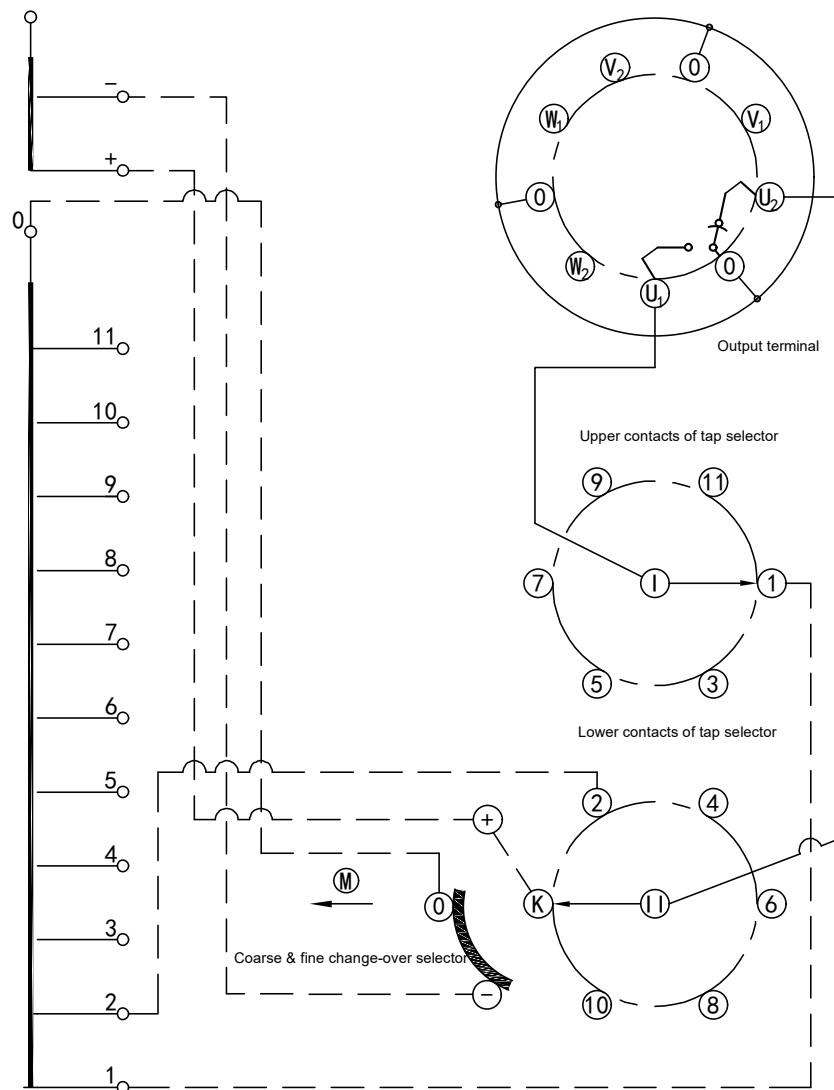
Change-over selector position	 K+					 K-					
Tap selector contact	1	2	3	4	5	K	2	3	4	5	6
Tap position indicator	1	2	3	4	5	6	7	8	9	10	11



9.33 Work position table and wiring diagram of SHZVG on-load changer (12231W)



9.34 Work position table and wiring diagram of SHZVG on-load changer (12231G)



Operation position number	23
Different voltage number	23
Set position	12

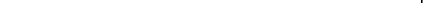
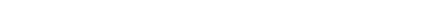


Drive side of tap selector

— Huaming is responsible for the connection

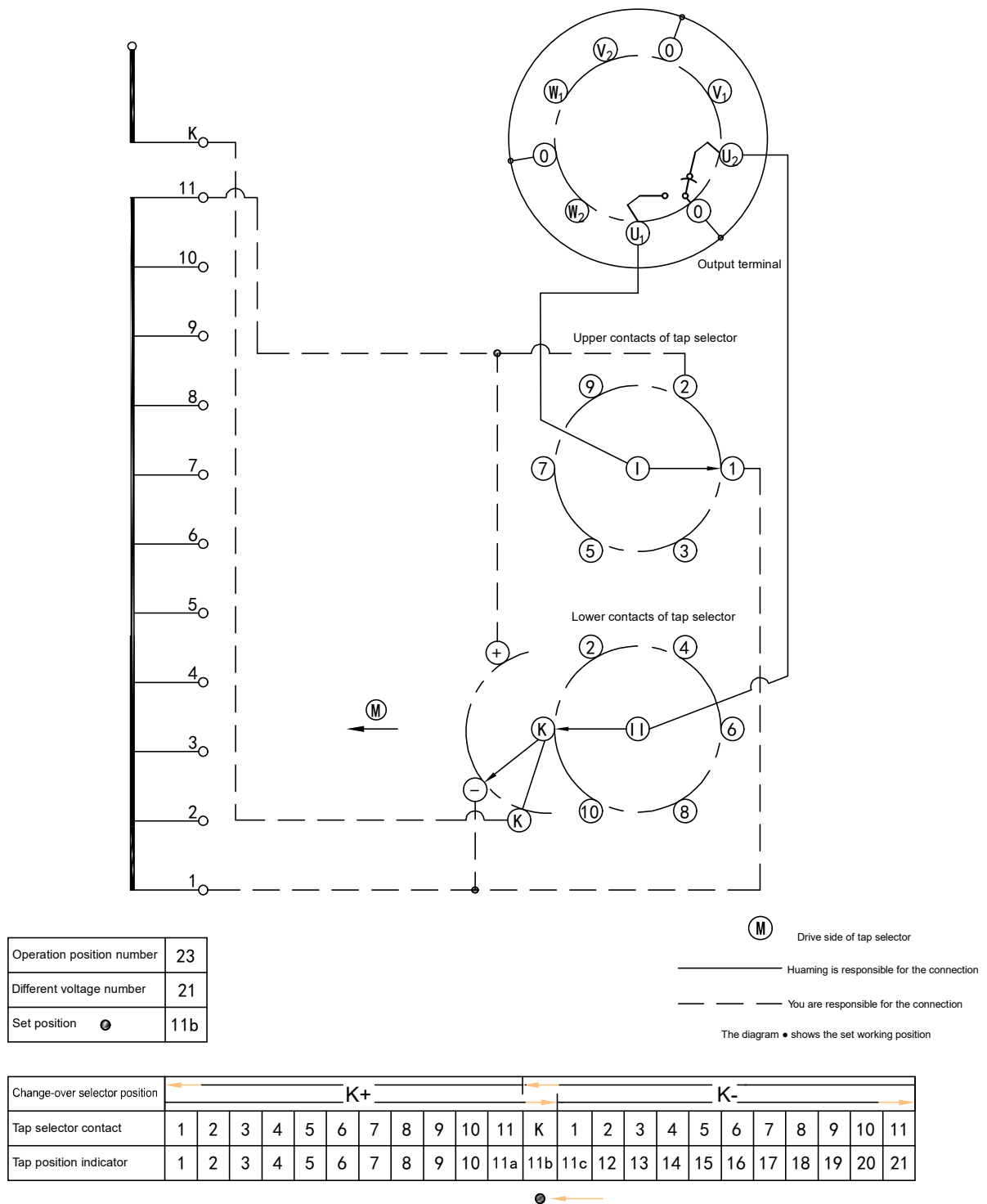
- - - You are responsible for the connection

The diagram ● shows the set working position

Change-over selector position	 0+												 0-											
Tap selector contact	1	2	3	4	5	6	7	8	9	10	11	K	1	2	3	4	5	6	7	8	9	10	11	
Tap position indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	

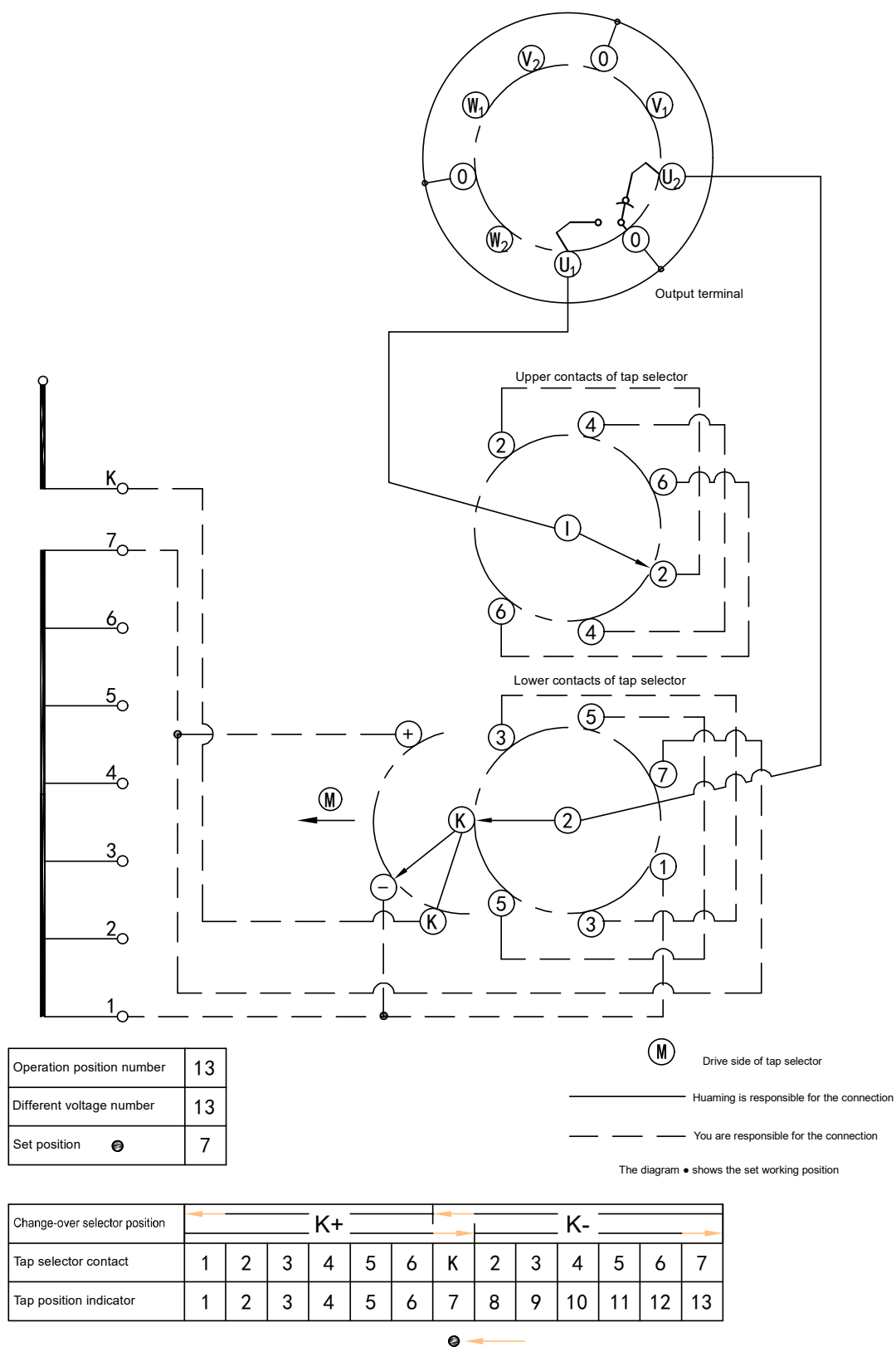


9.35 Work position table and wiring diagram of SHZVG on-load changer (12233W)

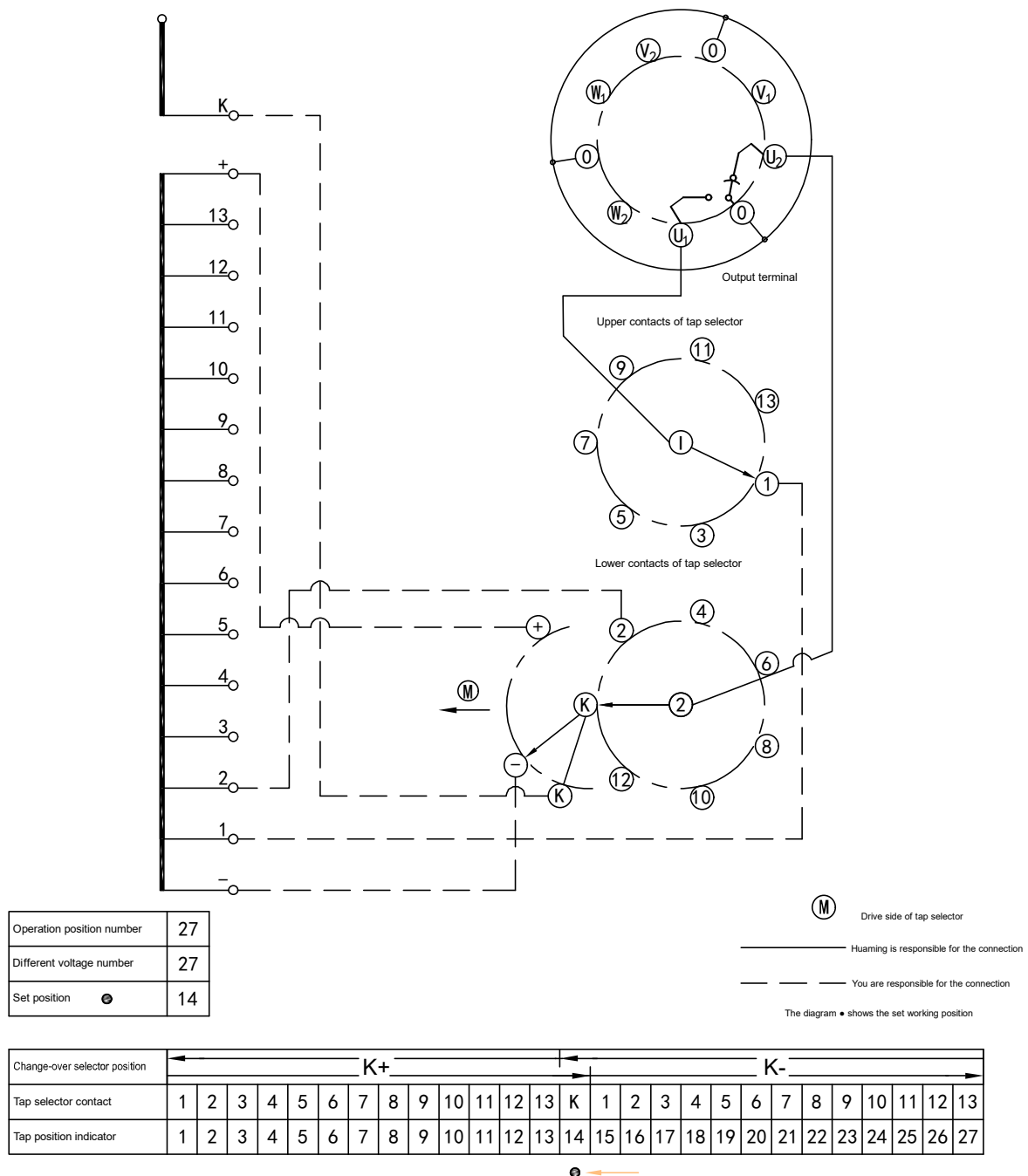


9 Drawings Attached

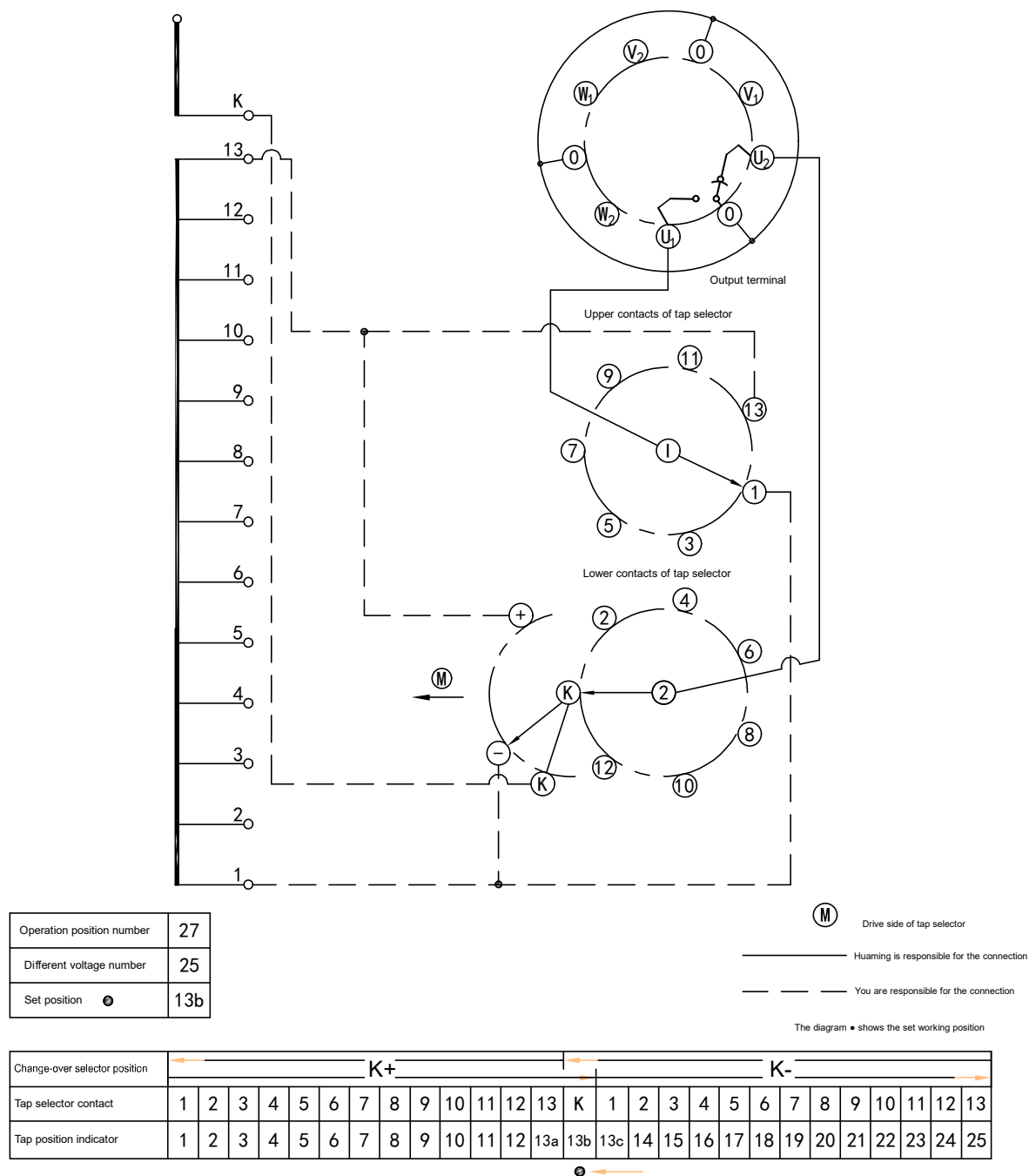
9.36 Work position table and wiring diagram of SHZVG on-load changer (14131W)



9.37 Work position table and wiring diagram of SHZVG on-load changer (14271W)

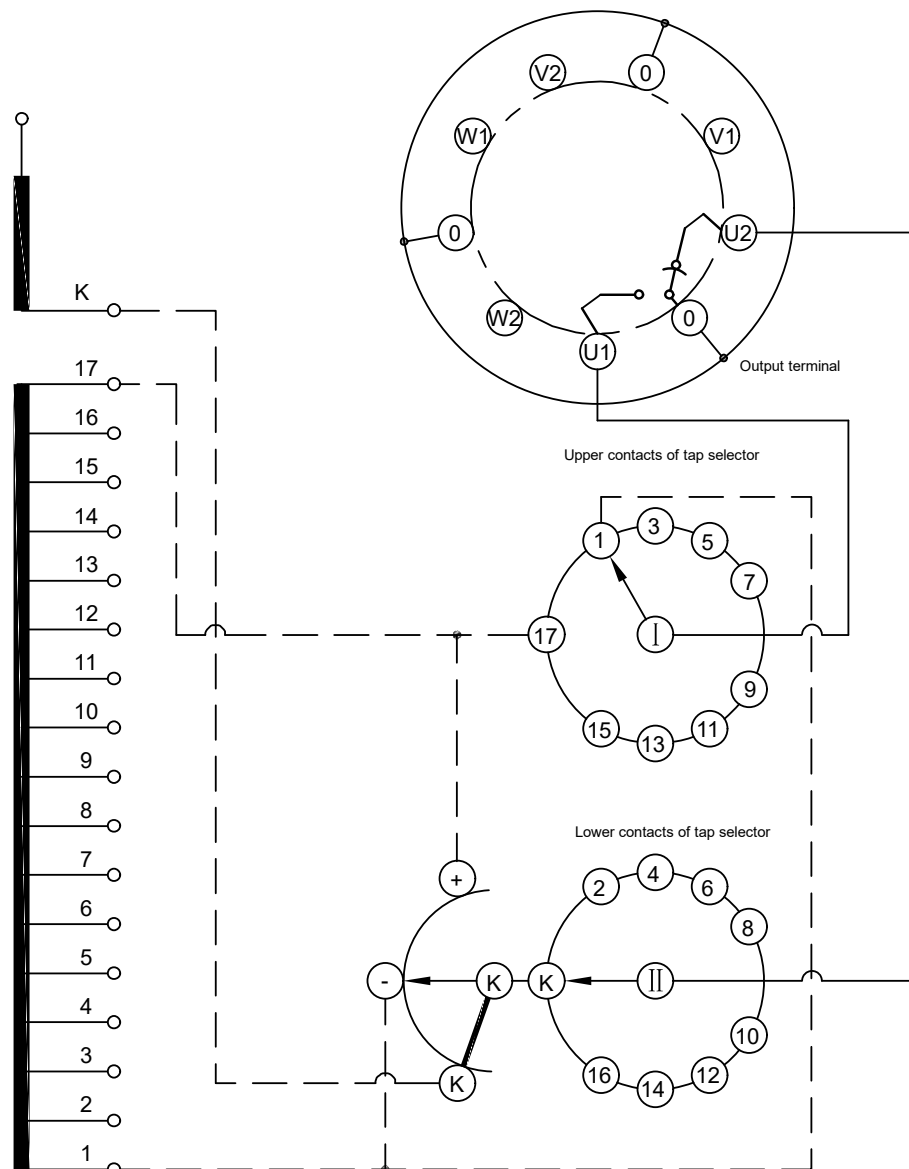


9.38 Work position table and wiring diagram of SHZVG on-load changer (14273W)



9.39 Work position table and wiring diagram of SHZVG on-load changer (18353W) (L)

9 Drawings Attached



Operation position number	35
Different voltage number	35
Set position	17b

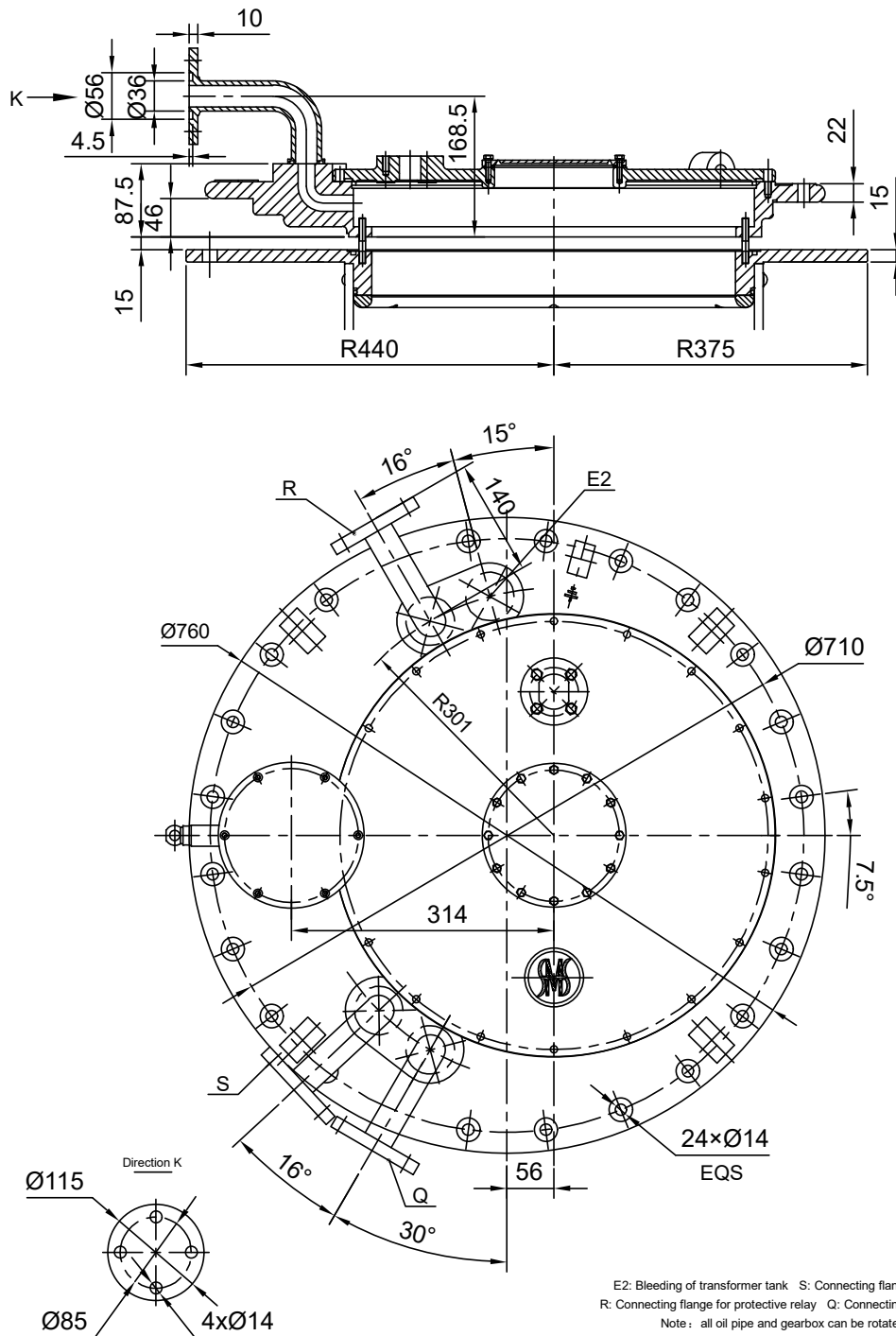
— Huaming is responsible for the connection

--- You are responsible for the connection

The diagram ● shows the set working position

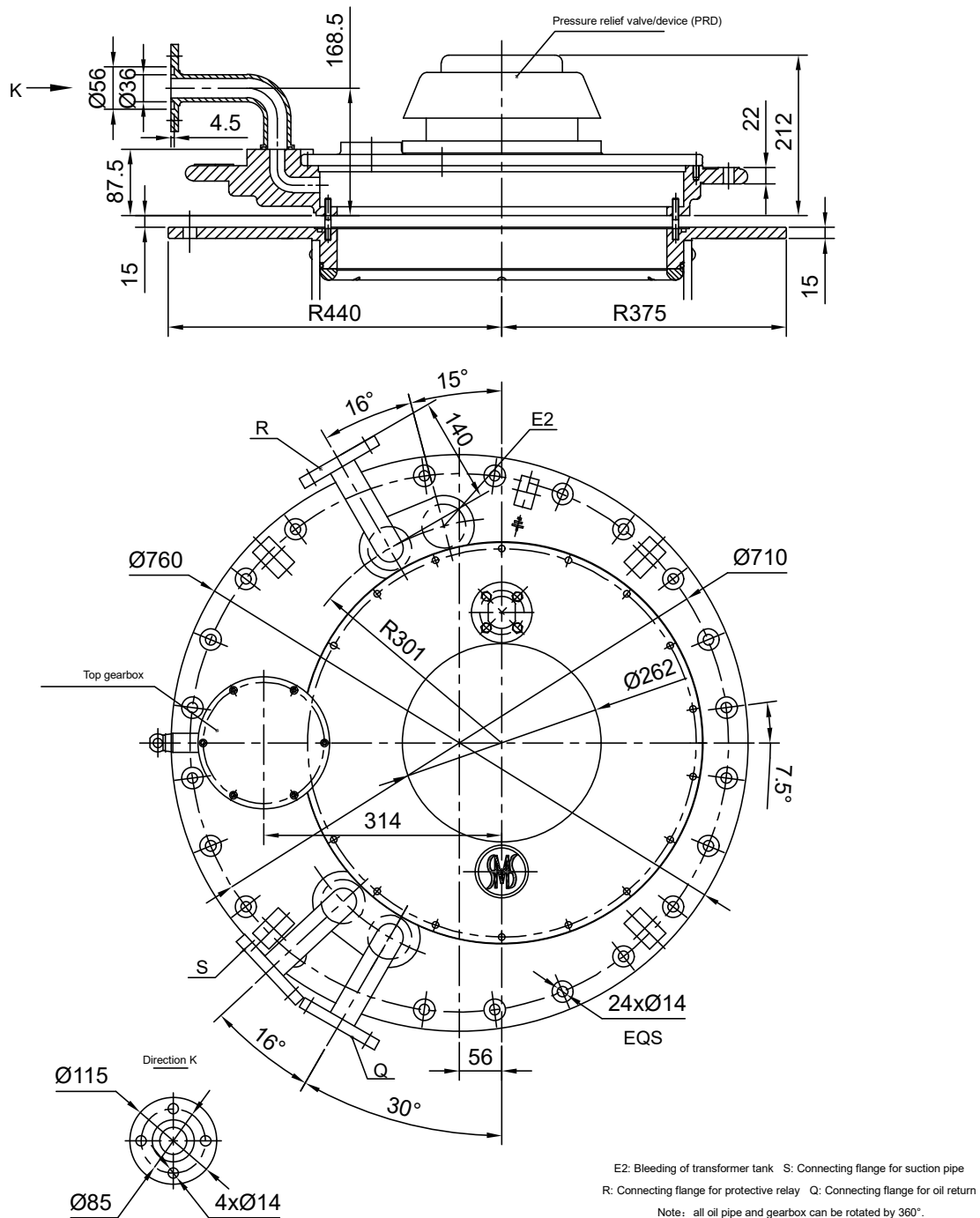
Change-over selector position	← K+ K- →																																		
Tap selector contact	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
Tap position indicator	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17a	17b	17c	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33

9.40 Dimension drawing of bell-type mounting flange for SHZVG on-load tap-changer

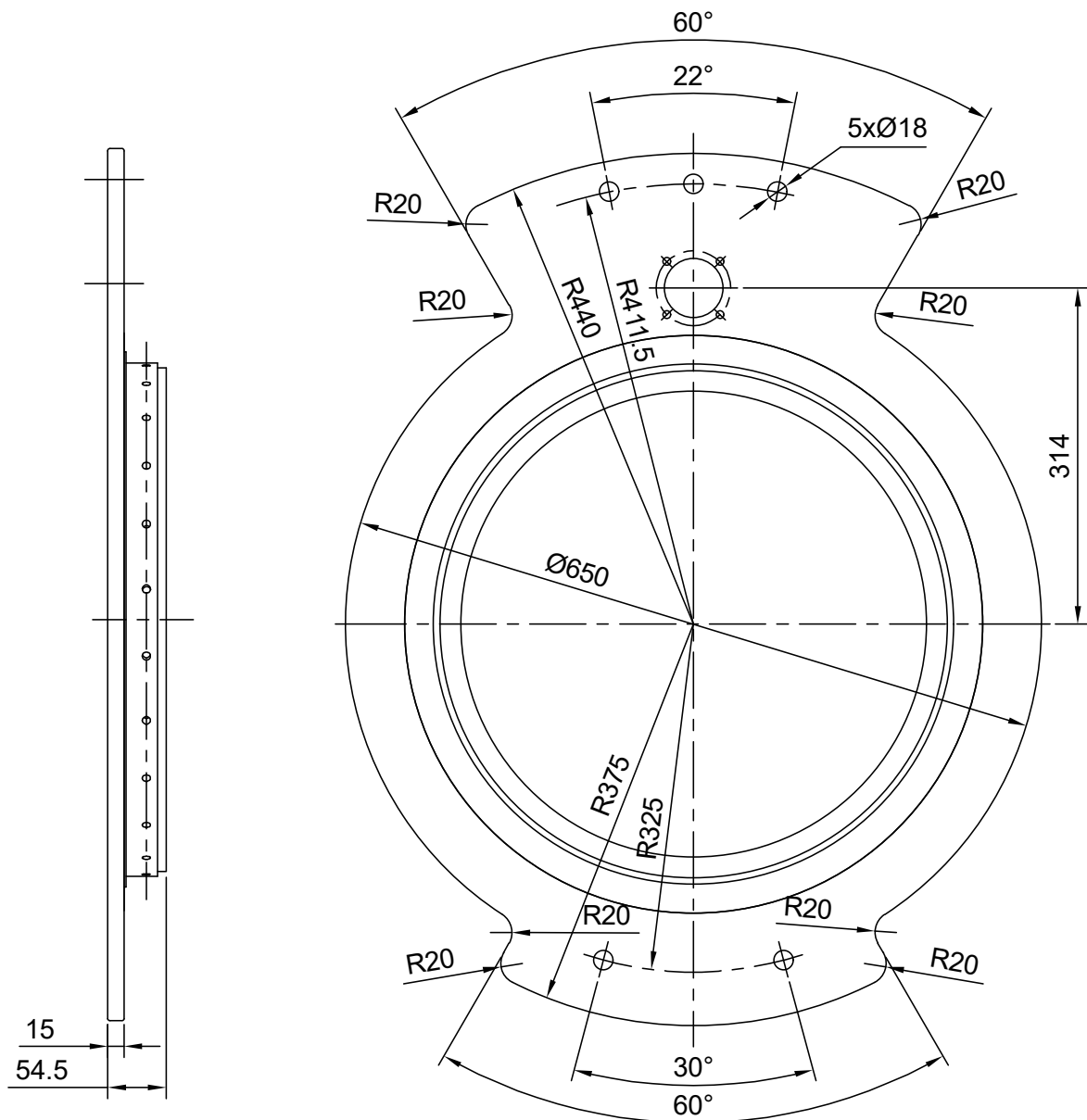


9 Drawings Attached

9.41 Dimension drawing of bell-type mounting flange for SHZVG on-load tap-changer with PRD

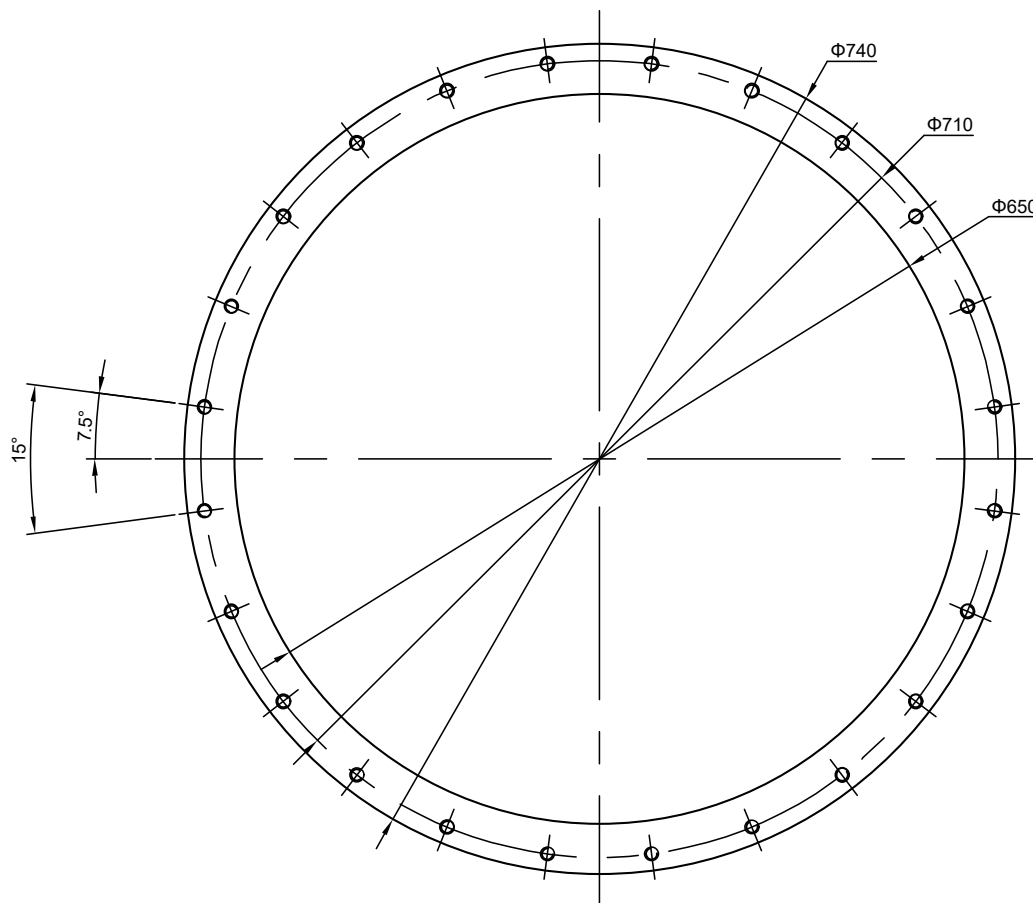
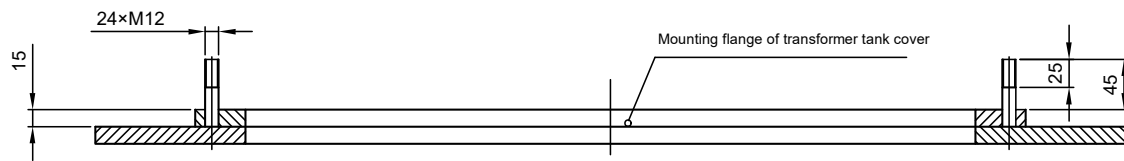


9.42 Dimension drawing of bell-type support flange for SHZVG on-load tap-changer



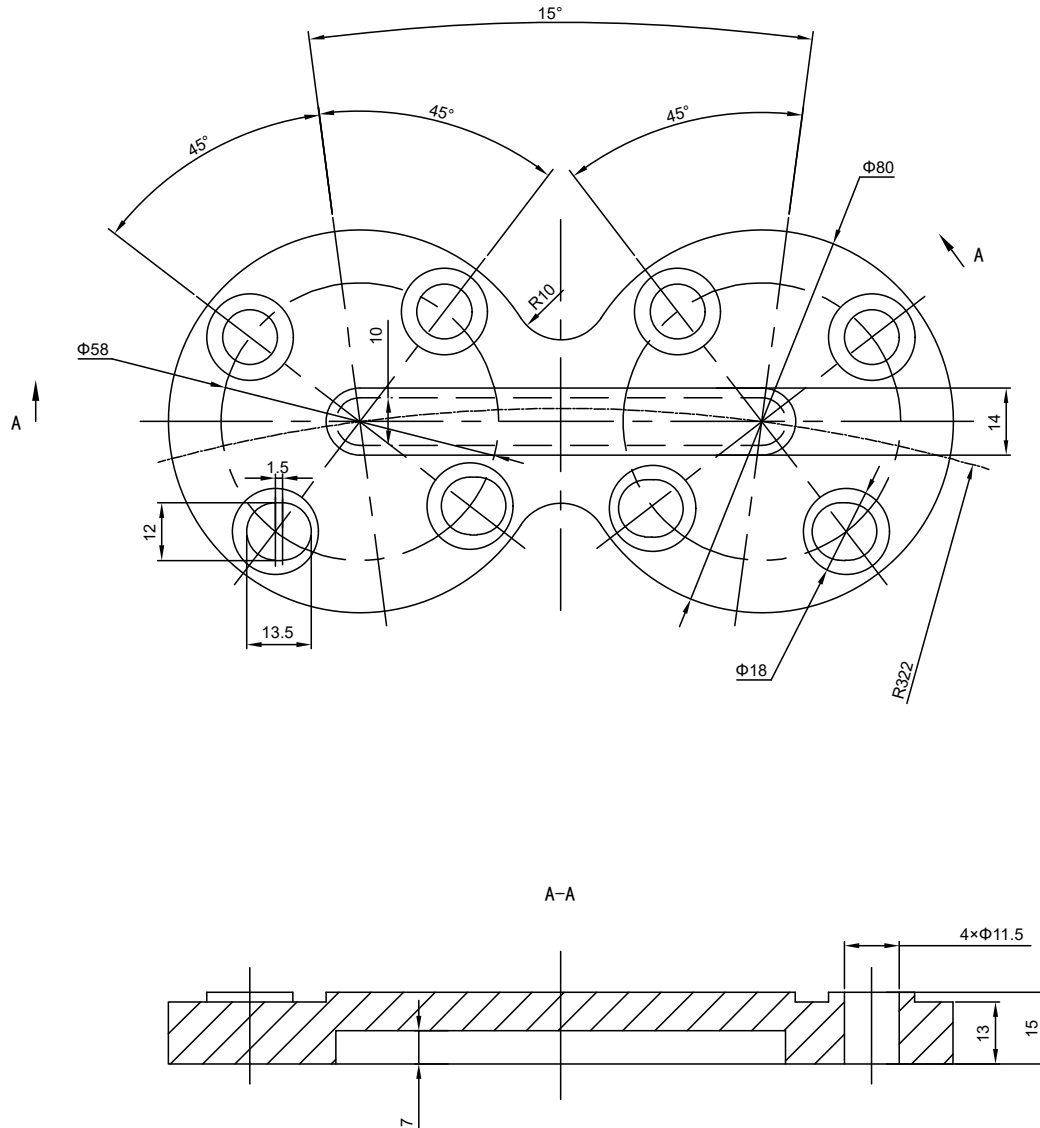
9 Drawings Attached

9.43 Dimension drawing of transformer cover mounting flange



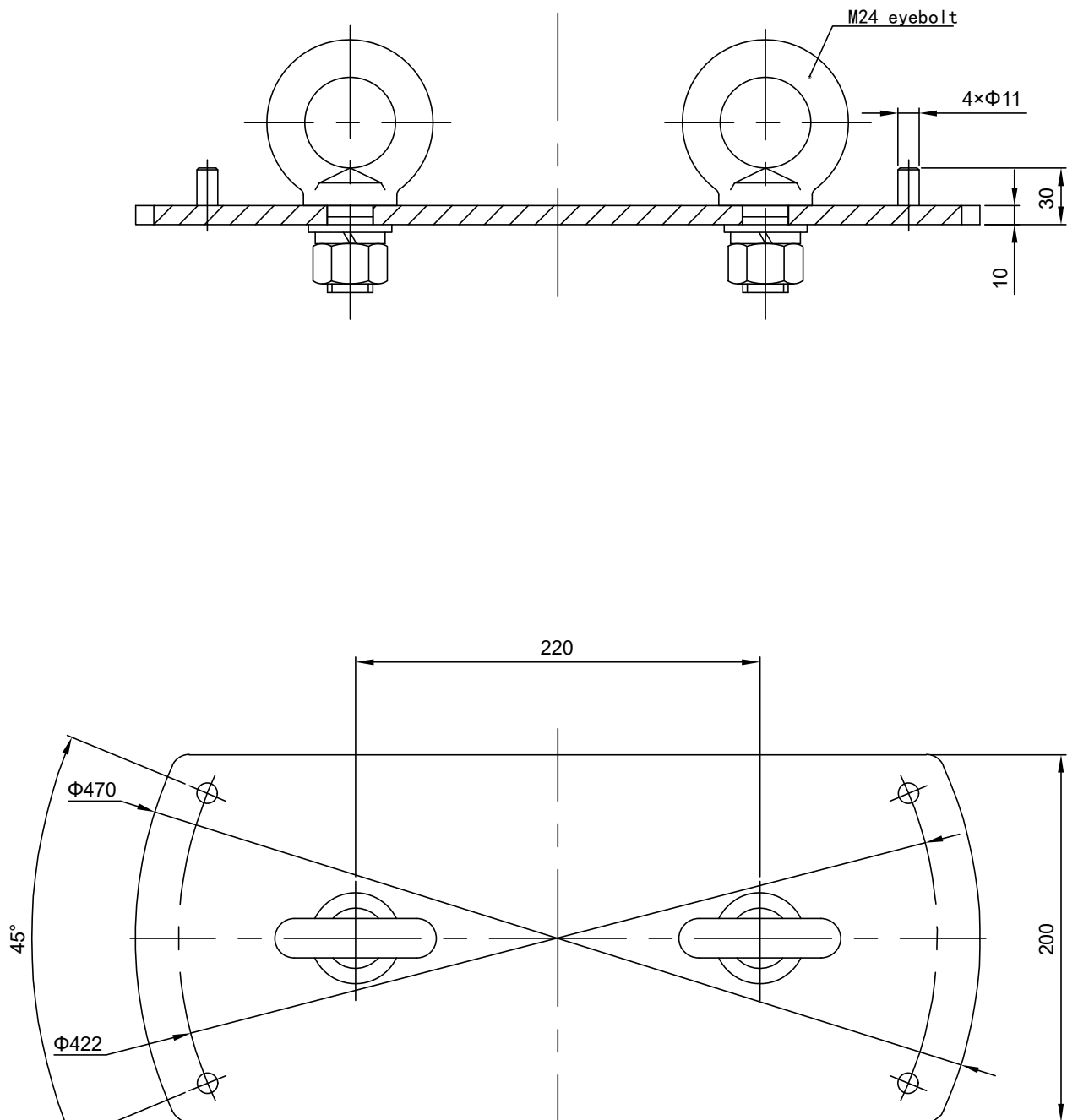
9 Drawings Attached

9.44 Dimension figure of bypass pipe



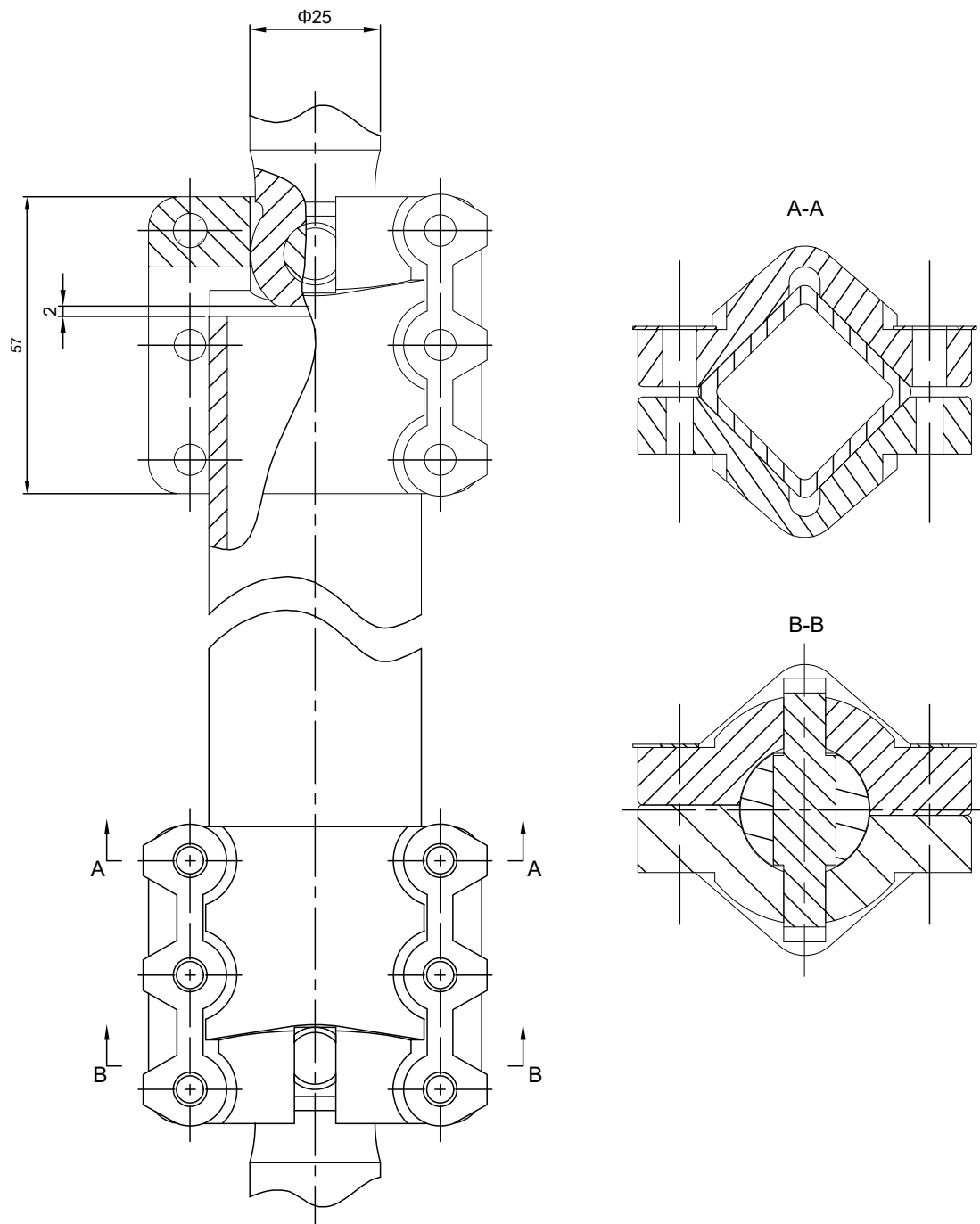
9 Drawings Attached

9.45 Dimension drawing of bell-type structural lifting plate



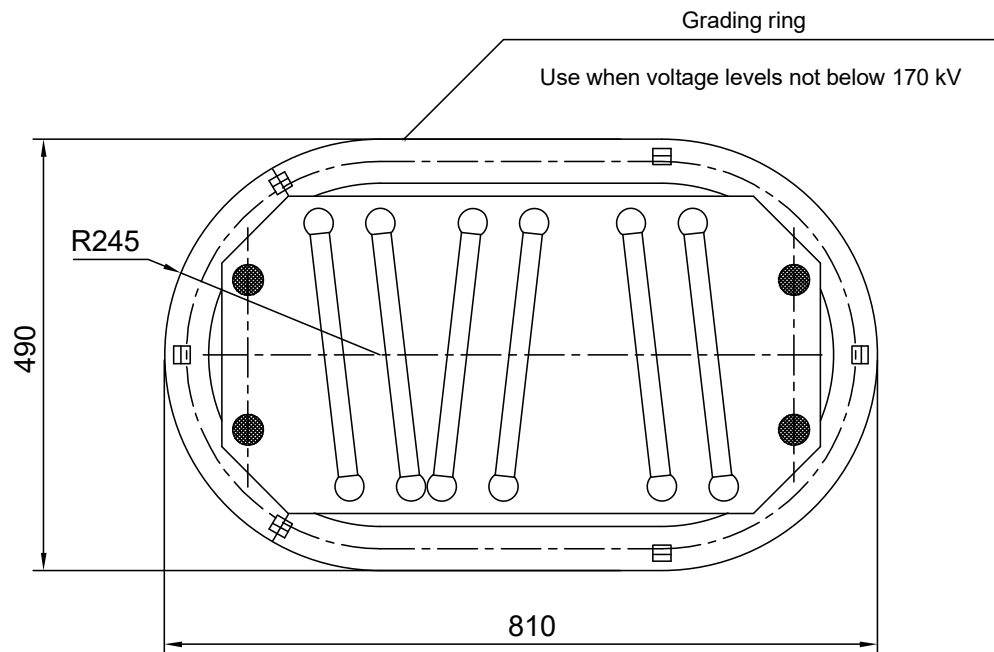
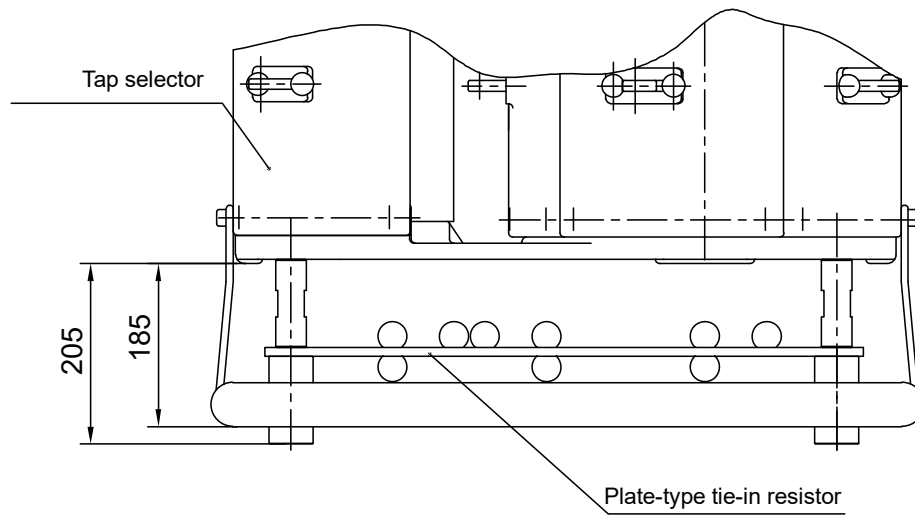
9 Drawings Attached

9.46 Schematic diagram of horizontal and vertical shaft transmission

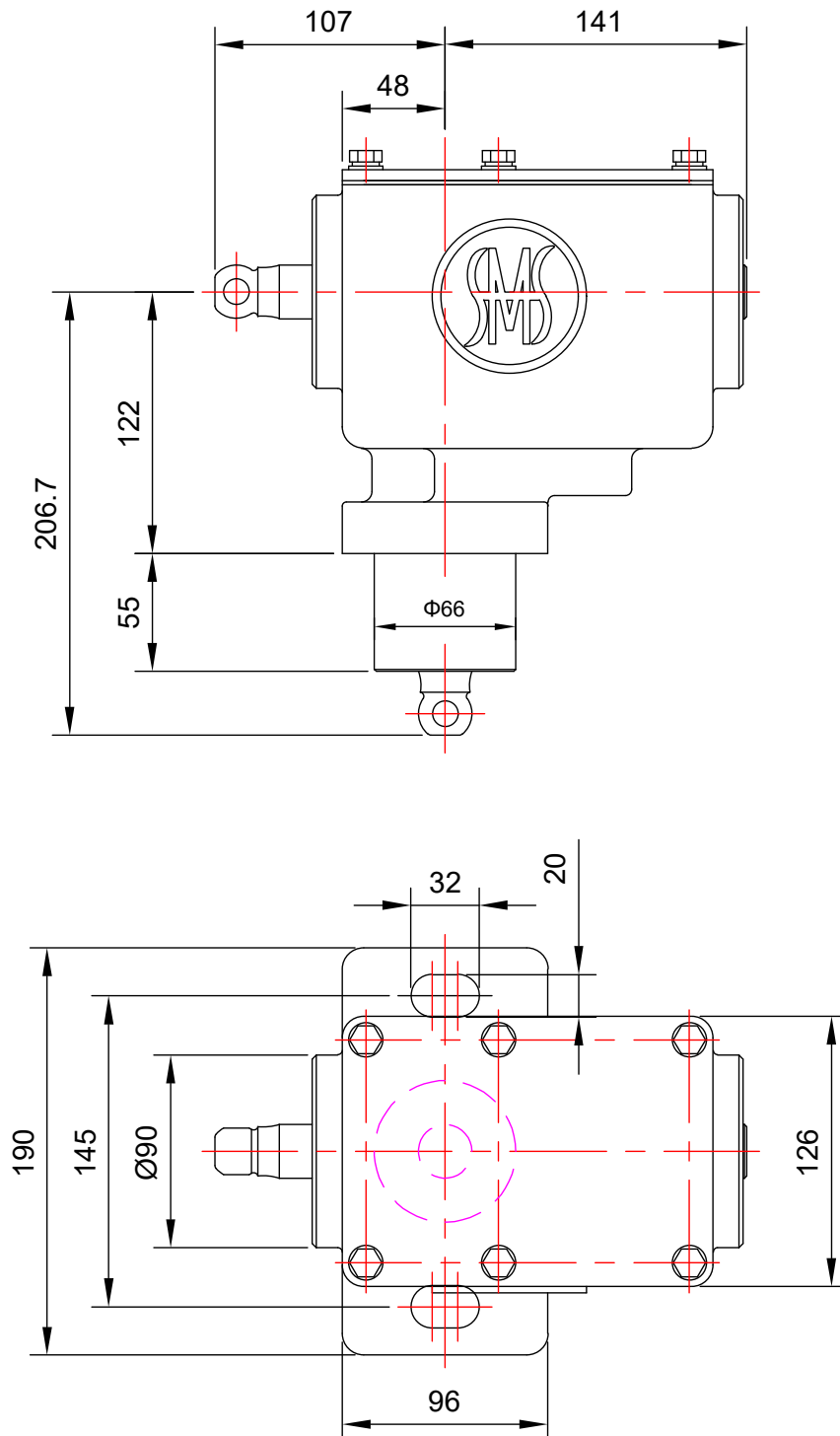


9 Drawings Attached

9.47 Installation dimension drawing of tie-in resistor

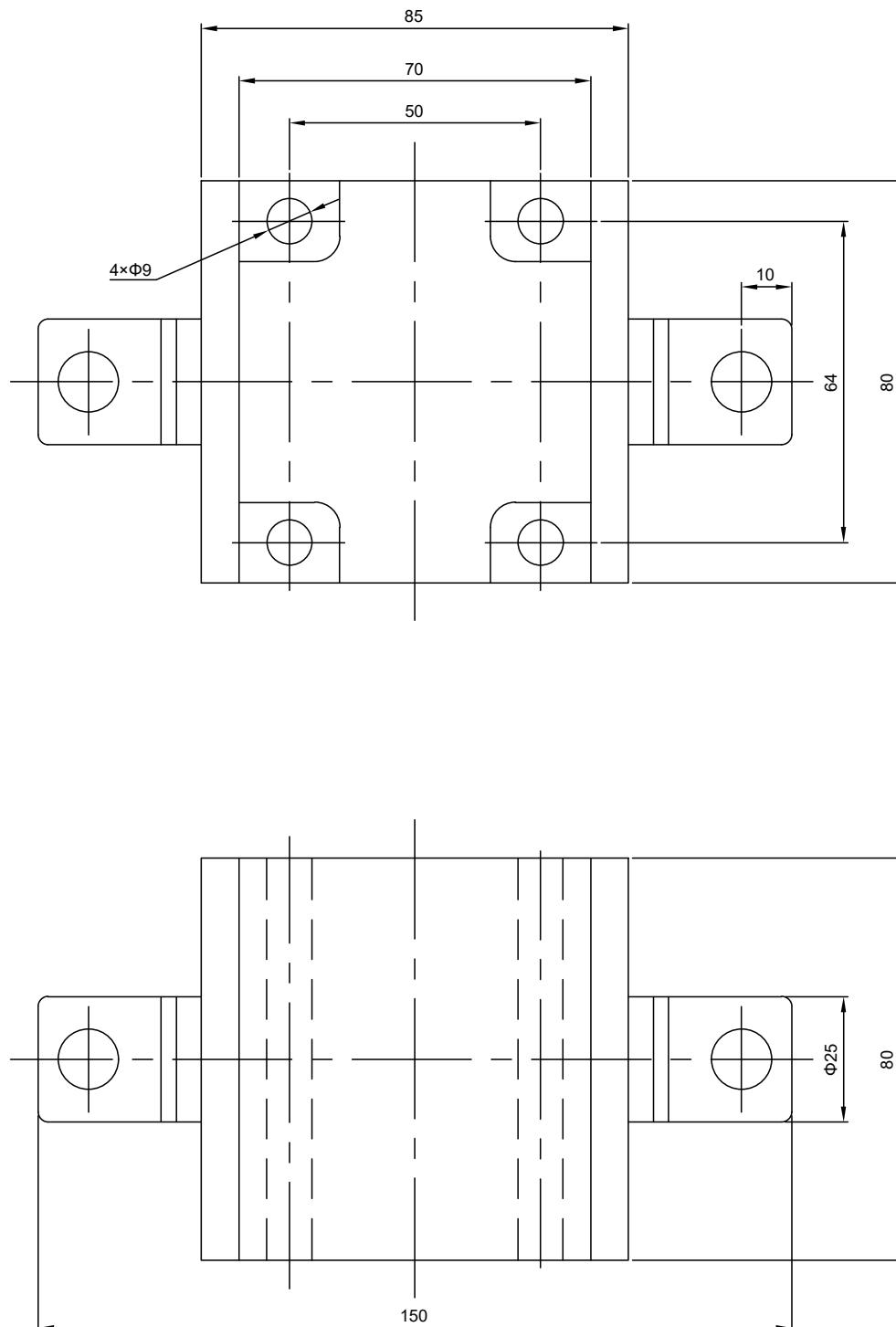


9.48 Dimension drawing of bevel gearbox



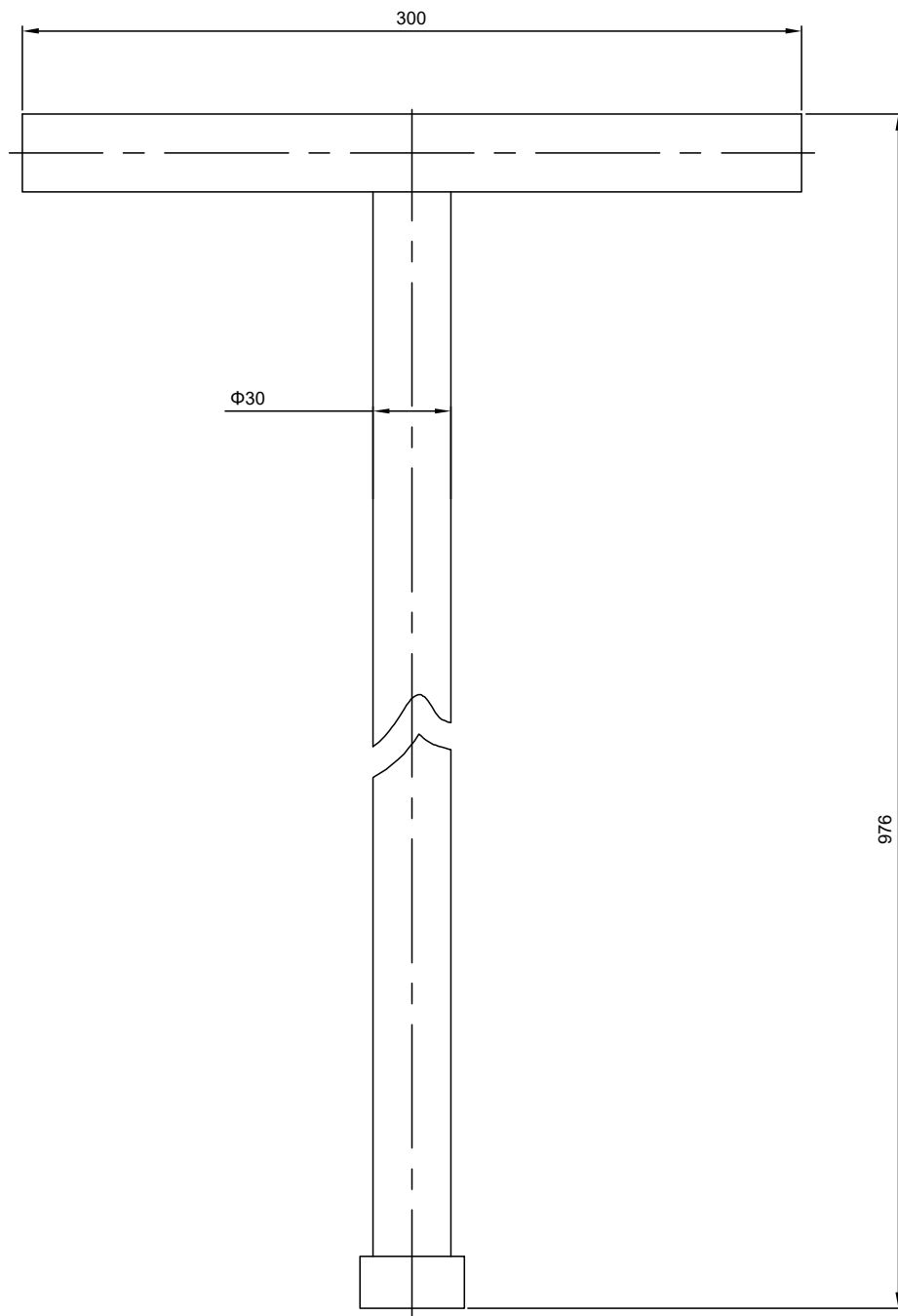
9 Drawings Attached

9.49 Intermediate support



9 Drawings Attached

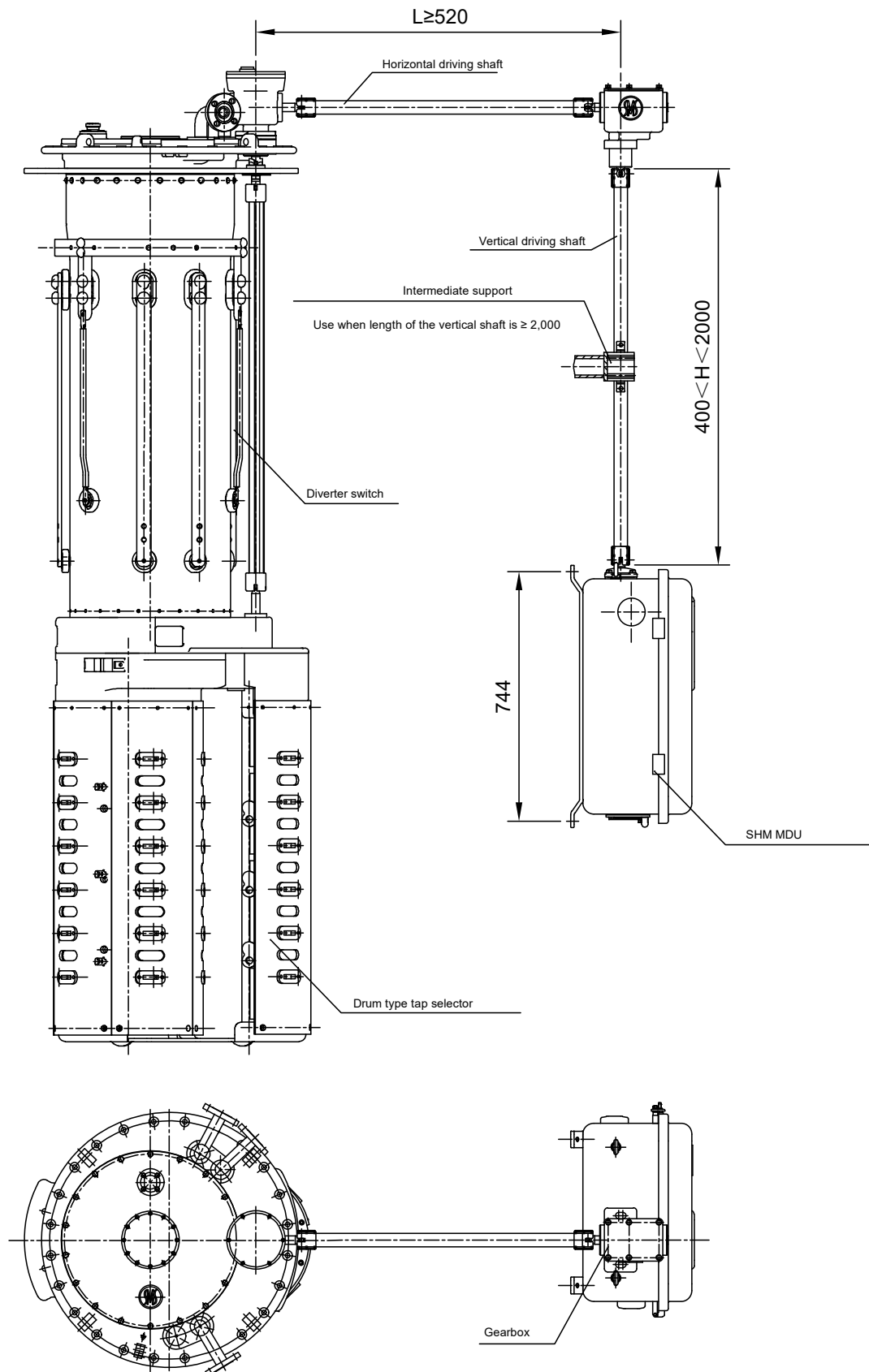
9.50 transformer oil PRD lever



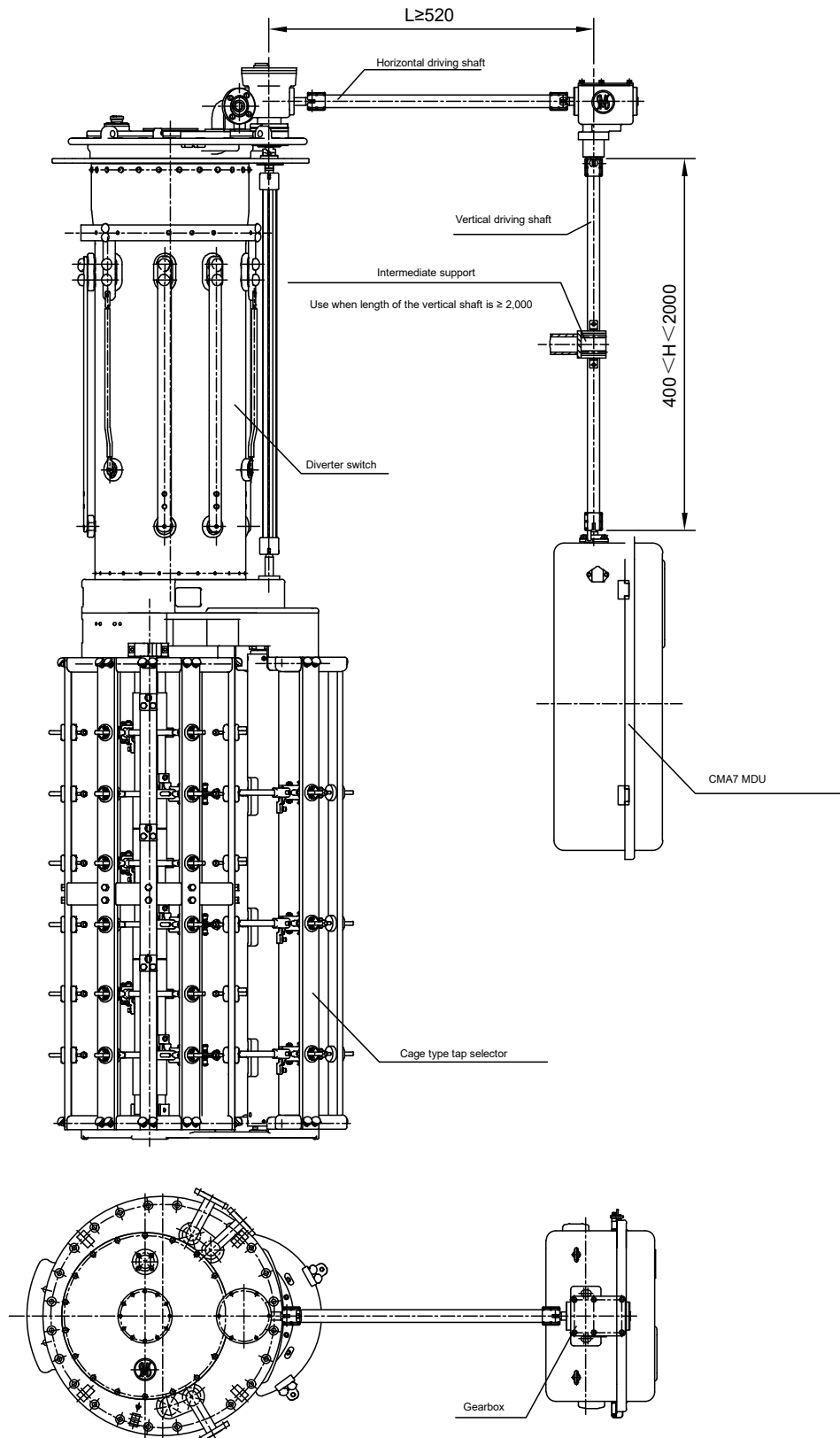
When it is shortened, it can be used as a wrench to operate the diverter switch

When it is extended, it can be used as a lever to operate the transformer oil release valve

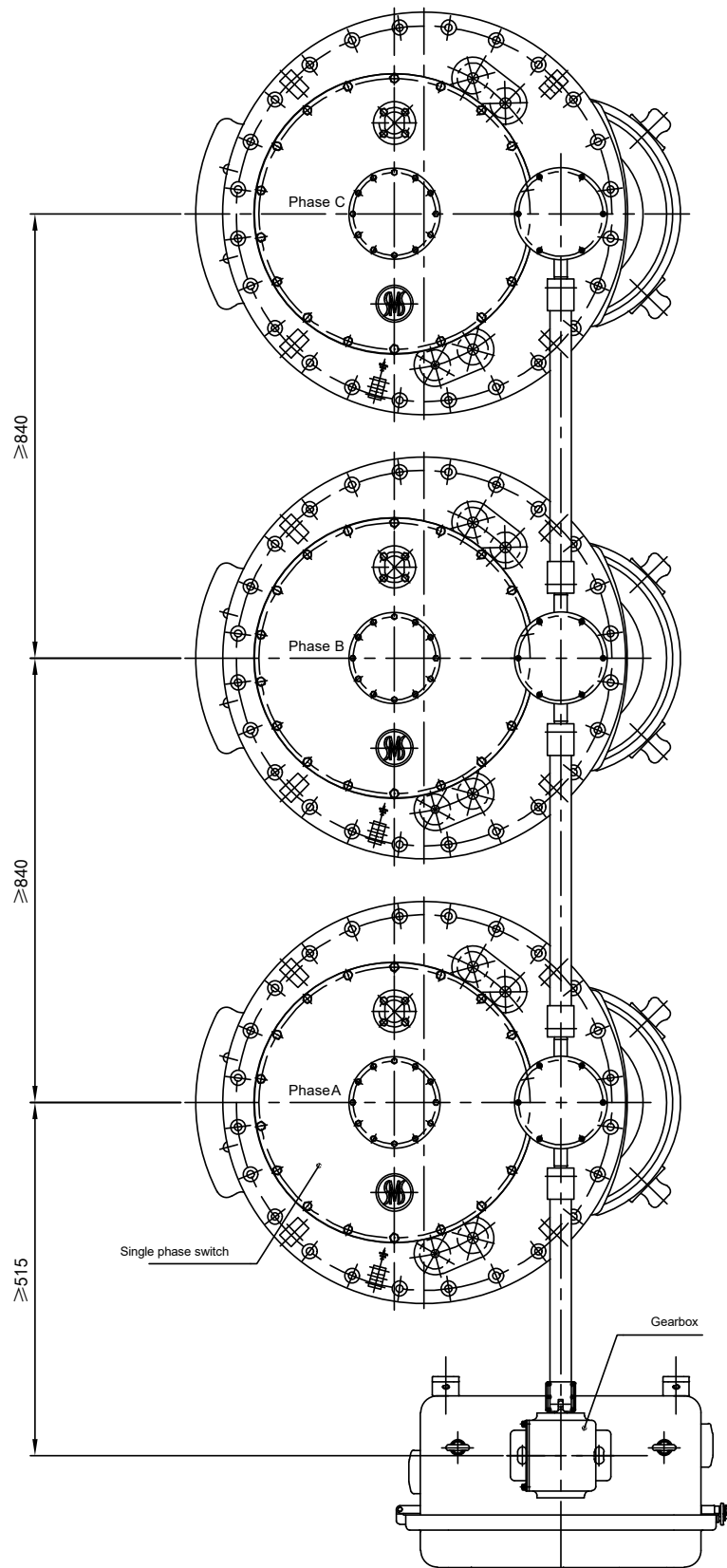
9.51 Layout dimension drawing of connection between SHZVG tap-changer and SHM



9.52 Layout dimension drawing of connection between SHZVG tap-changer and CMA7

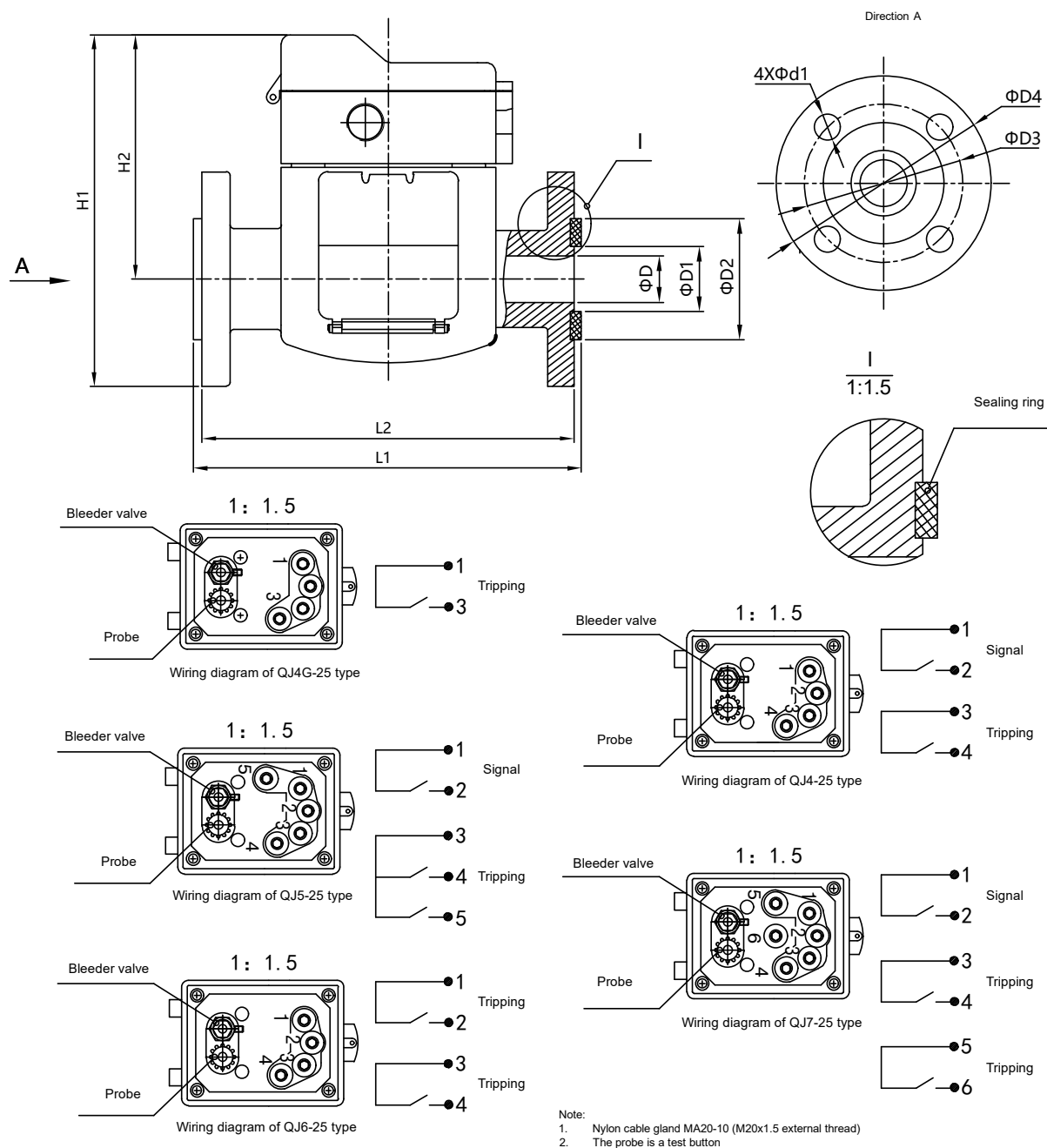


9.53 Layout dimension drawing of SHZVG three linked tap-changers



9 Drawings Attached

9.54 Outline dimension drawing of protective relay



Model	D	D1	D2	D3	D4	d1	H1	H2	L1	L2	Remarks
QJ4-25	25	35	65	85	115	14	215	158	204	200	Single signal, single jump turn
QJ4G-25											Single jump turn
QJ5-25											Single signal, with common point dual jump turns
QJ6-25											Dual independent jump turns
QJ7-25											Single signal, dual independent jump turns

Shanghai Huaming Power Equipment Co., Ltd.

Address: No.977 Tongpu Road, Putuo Postal Code: 200333
District, Shanghai, China

Tel.: +86 21 5270 8966 (Main line)

Fax: +86 21 5270 3385

Website: www.huaming.com

E-mail: Order@huaming.com